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PHYSICS

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Chapter 1

Physics and Measurement

UNITS

Measurement of any physical quantity involves its comparison with a certain basic, reference standard called unit.

Measurement = nu

Here, n is numerical value and u is unit. The numerical value is inversely proportional to the size of unit.

$n \times u$ = constant

$n \propto \frac{1}{u}$

DIMENSIONS OF PHYSICAL QUANTITIES

All the physical quantities represented by derived units can be expressed in terms of some combination of seven fundamental quantities. These seven fundamental quantities are called seven dimensions of the physical world. They are denoted with square brackets [ ].

S. No.	Physical quantity	Dimensional formula	Useful result
1.	Absolute permittivity (ε_0)	$M^{-1}L^{-3}T^4A^2$	$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$
2.	Absolute permeability (μ_0)	$MLT^{-2}A^{-2}$	$\frac{F}{l} = \frac{\mu_0 i_1 i_2}{2\pi r}$
3.	Resistance R	$MLT^{-2}A^{-2}$	$P = I^2R$
4.	Inductance L	$ML^2T^{-2}A^{-2}$	$U = \frac{1}{2}LI^2$
5.	Capacitance C	$M^{-1}L^{-2}T^4A^2$	$U = \frac{q^2}{2C}$
6.	L/R (time constant)	$[M^0L^0T^1]$	–
7.	RC (time constant)	$[M^0L^0T^1]$	–
8.	$\sqrt{LC}$	$M^0L^0T^1$	–
9.	Stress, Pressure, Energy density, $\frac{1}{2}\varepsilon_0E^2, B^2/2\mu_0$	$ML^{-1}T^{-2}$	–
10.	Heat capacity, Boltzmann constant	$ML^2T^{-2}K^{-1}$	–

DIMENSIONAL ANALYSIS AND ITS APPLICATIONS

Principle of Homogeneity of dimensions : It is based on the simple fact that length can be added to length. It states that in a correct equation, the dimensions of each term added or subtracted must be same. If two quantities are being added or subtracted, they must be of same dimensional formula. Every correct equation must have same dimensions on both sides of the equation.

> ***Note :*** *Although torque and work done by a force have same dimensional formula yet they cannot be added as their nature is different.*

Conversion of units : The numerical value of a physical quantity in a system of units can be changed to another system of units using the equation $n[u]$ = constant *i.e.*, $n_1[u_1] = n_2[u_2]$ where n is the numerical value and u is the unit.

$$n_2 = n_1\left[\frac{M_1}{M_2}\right]^a\left[\frac{L_1}{L_2}\right]^b\left[\frac{T_1}{T_2}\right]^c$$ where the dimensional formula of the physical quantity is $[M^aL^bT^c]$.

To find a relation among the physical quantities. If one knows the quantities on which a particular physical quantity depends and guesses that this dependence is of product type, method of dimensions are helpful in deducing their relation.

Suppose we want to find the relation between force, mass and acceleration. Let force depends on mass and acceleration as follows.

$F = Km^ba^c$ when K = dimensionless constant, b and c are powers of mass and acceleration.

According to principle of homogeneity,

$[F] = [K]\ [m]^b\ [a]^c$

$\Rightarrow$ $[MLT^{-2}] = [M^0L^0T^0]\ [M]^b\ [LT^{-2}]^c$

$\Rightarrow$ $[MLT^{-2}] = M^bL^c\,T^{-2c}$

Equating the dimension on both sides we get $1 = b$, $1 = c$, $-2c = -2$.

$\Rightarrow$ $b = 1$ and $c = 1$.

ACCURACY AND PRECISION

Accuracy

The closeness of the measured value to the true value of the physical quantity is known as the accuracy of the measurement.

Precision

It is the measure of the extent to which successive measurements of a physical quantity differ from one another.

Suppose the true value of a measurement is 35.75 and two measured values are 35.73 and 35.725. Here 35.73 is closest to 35.75, so its accuracy is more than 35.725 but 35.725 is more precise than 35.73 because 35.725 is measured upto 3 decimal places.

SIGNIFICANT FIGURES

The number of digits in the measured value about the correctness of which we are sure plus one more digit are called significant figures.

Rules for counting the significant figures

Rule I : All non-zero digits are significant.

Rule II : All zeros occurring between the non zero digits are significant. For example 230089 contains six significant figures.

Rule III : All zeros to the left of non zero digit are not significant. For example 0.0023 contains two significant figures.

Rule IV : If a number ends in zeros that are not to the right of a decimal, the zeros are not significant.

For example, number of significant figures in

1500 (Two)

1.5×10^3 (Two)

1.50×10^3 (Three)

1.500×10^3 (Four)

> ***Note :*** *Length of an object may be represented in many ways say 5 m, 5.0 m, 500 cm, 5.00 m, 5×10^2 cm. Here 5.00 m is most precise as it contains 3 significant figures.*

Rules for Arithmetic Operations with Significant Figures

Rule I : In addition or subtraction, the final result should retain as many decimal places as there are in the number with the least decimal places.

Rule II : In multiplication or division, the final result should retain as many significant figures as are there in the original number with the least significant figures.

Rounding Off of Uncertain Digits

Rule I : The preceding digit is raised by 1 if the insignificant digit to be removed is more than 5 and is left unchanged if the later is less than 5.

Rule II : When the insignificant digit to be removed is 5 and the uncertain digit is even, 5 is simply dropped and if it is odd, then the preceding digit is raised by 1.

ERRORS IN MEASUREMENT

1. **Mean Absolute Error :-** If a_1, a_2, a_3, ... a_n are n measurements then

$$a_m = \frac{a_1 + a_2 + ... + a_n}{n}$$ is taken as the true value of a quantity, if the same is not known.

$$a_1 = a_m - a_1$$
$$a_2 = a_m - a_2$$
....
$$a_n = a_m - a_n$$

Mean absolute error, $\overline{a} = \frac{| \ a_1 | + | \ a_2 | + ... + | \ a_n |}{n}$

Final result of measurement may be written as :

$$a = a_m \pm \ \overline{a}$$

2. **Relative Error or Fractional Error :** It is given by

$$\frac{\overline{a}}{a_m} = \frac{\text{Mean absolute Error}}{\text{Mean value of measurement}}$$

3. **Percentage Error** $= \frac{\overline{a}}{a_m} \times 100\%$

4. **Combination of Errors :**

   (i) In Sum : If $Z = A + B$, then maximum absolute error in Z is given by, $Z = \ A + \ B$, maximum fractional error in this case $\frac{Z}{Z} = \frac{A}{A+B} + \frac{B}{A+B}$ *i.e.,* when two physical quantities are added then the maximum absolute error in the result is the sum of the absolute errors of the individual quantities.

(ii) **In Difference :** If $Z = A - B$, then maximum absolute error is $Z = \ \ A + \ \ B$ and maximum fractional error in this case $\frac{Z}{Z} = \frac{A}{A-B} + \frac{B}{A-B}$.

(iii) **In Product :** If $Z = AB$, then the maximum fractional error, $\frac{Z}{Z} = \frac{A}{A} + \frac{B}{B}$ where $\frac{Z}{Z}$ is known as fractional error.

(iv) **In Division :** If $Z = A/B$, then maximum fractional error is $\frac{Z}{Z} = \frac{A}{A} + \frac{B}{B}$

(v) **In Power :** If $Z = A^n$ then $\frac{Z}{Z} = n\frac{A}{A}$

In more general form if $Z = \frac{A^p B^q}{C^r}$ then the maximum fractional error in Z is

$$\frac{Z}{Z} = p\frac{A}{A} + q\frac{B}{B} + r\frac{C}{C}$$

Applications :

1. For a simple pendulum, $T \propto l^{1/2}$ $\Rightarrow$ $\frac{T}{T} = \frac{1}{2}\frac{l}{l}$
2. For a sphere, surface area and volume are given by

$$A = 4\pi r^2, V = \frac{4}{3}\pi r^3 \quad \Rightarrow \quad \frac{A}{A} = 2.\frac{r}{r} \text{ and } \frac{V}{V} = 3.\frac{r}{r}$$

3. When two resistors R_1 and R_2 are connected

(a) In series

$$R_s = R_1 + R_2 \quad \Rightarrow \quad R_s = \ \ R_1 + \ \ R_2$$

$$\frac{\Delta R_s}{R_s} = \frac{R_1 + \ \ R_2}{R_1 + R_2}$$

(b) In parallel,

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} \quad \Rightarrow \quad \frac{R_p}{R_p^2} = \frac{R_1}{R_1^2} + \frac{R_2}{R_2^2}$$

Also, $\frac{R_p}{R_p} = \frac{R_1}{R_1} + \frac{R_2}{R_2} - \frac{R_1 + \ \ R_2}{R_1 + R_2}$

4. If $x = 2a - 3b$ then, $x = 2\ \ a + 3\ \ b$

LEAST COUNT OF MEASURING INSTRUMENTS

The smallest measurement that can be taken by an instrument is equal to least count of the instrument.

For example, a meter scale has smallest division 1 mm. This represents the least count (and also the absolute error) in the measurement.

Let a length measured by the meter scale = 56.0 cm

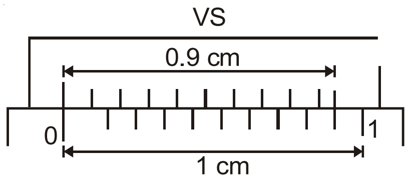
This implies that x = 56.0 cm

Absolute error x = 1 mm = 0.1 cm

Relative error = $\frac{x}{x} = \frac{0.1}{56.0}$

Vernier Callipers

It consists of two scales viz main scale and vernier scale. Vernier scale moves on the main scale. The least count of the instrument is the smallest distance between two consecutive divisions and it is equal to 1 MSD – 1 VSD.



In the figure shown, 1 MSD = 0.1 cm

1 VSD = 0.09 cm

Least count = 1 MSD – 1 VSD = 0.01 cm

Screw Gauge

It contains a main scale and a circular scale. The circular scale is divided into a number of divisions. In other words, the complete rotation of circular scale is divided into a number of parts. The least count of a screw gauge is pitch divided by no. of circular scale divisions.

$$\text{Least count of spherometer and Screw Gauge} = \frac{\text{Pitch}}{\text{No. of CSD}}$$

Total reading of screw gauge = Main scale reading + [(Circular scale reading) × Least count]

Table : SI Units and Dimensions of Some Important Physical Quantities

S.No.	Quantity	SI Unit	Dimensional Formula
1.	Volume	m^3	$[M^0L^3T^0]$
2.	Density	$kg\ m^{-3}$	$[M^1L^{-3}T^0]$
3.	Velocity	ms^{-1}	$[M^0L^1T^{-1}]$
4.	Acceleration	ms^{-2}	$[M^0L^1T^{-2}]$
5.	Angular Velocity	rad s^{-1}	$[M^0L^0T^{-1}]$
6.	Frequency	s^{-1} or hertz (Hz)	$[M^0L^0T^{-1}]$
7.	Momentum	$kg\ ms^{-1}$	$[M^1L^1T^{-1}]$
8.	Force	$kg\ ms^{-2}$ or newton (N)	$[M^1L^1T^{-2}]$
9.	Work, Energy	$kg\ m^2s^{-2}$ or joule (J)	$[M^1L^2T^{-2}]$
10.	Power	$kg\ m^2\ s^{-3}$ or Js^{-1} or watt (W)	$[M^1L^2T^{-3}]$
11.	Pressure, Stress	Nm^{-2} or pascal (Pa)	$[M^1L^{-1}T^{-2}]$
12.	Modulus of Elasticity	Nm^{-2}	$[M^1L^{-1}T^{-2}]$
13.	Moment of Inertia	$kg\ m^2$	$[M^1L^2T^0]$
14.	Torque	Nm	$[M^1L^2T^{-2}]$
15.	Angular Momentum	$kg\ m^2\ s^{-1}$ or J.s	$[M^1L^2T^{-1}]$
16.	Impulse	Ns	$[M^1L^1T^{-1}]$
17.	Coefficient of Viscosity	$kg\ m^{-1}\ s^{-1}$	$[M^1L^{-1}T^{-1}]$
18.	Surface Tension	Nm^{-1}	$[M^1L^0T^{-2}]$

19.	Universal Gravitational Constant	$Nm^2\ kg^{-2}$	$[M^{-1}L^3T^{-2}]$
20.	Latent Heat	$J\ kg^{-1}$	$[M^0L^2T^{-2}]$
21.	Specific Heat	$J\ kg^{-1}\ K^{-1}$	$[M^0L^2T^{-2}K^{-1}]$
22.	Thermal Conductivity	$J\ m^{-1}\ s^{-1}\ K^{-1}$	$[M^1L^1T^{-3}K^{-1}]$
23.	Electric Charge	Coulomb (C) or A.s	$[M^0L^0T^1A^1]$
24.	Electric Potential	JC^{-1} or volt (V)	$[M^1L^2T^{-3}A^{-1}]$
25.	Electric Resistance	VA^{-1} or ohm ( )	$[M^1L^2T^{-3}A^{-2}]$
26.	Electric Resistivity	m	$[M^1L^3T^{-3}A^{-2}]$
27.	Electric Conductance	$^{-1}$ or siemen (S)	$[M^{-1}L^{-2}T^3A^2]$
28.	Electric Conductivity	$^{-1}\ m^{-1}$ or $S\ m^{-1}$	$[M^{-1}L^{-3}T^3A^2]$
29.	Capacitance	CV^{-1} or farad (F)	$[M^{-1}L^{-2}T^4A^2]$
30.	Inductance	$Vs\ A^{-1}$ or henry (H)	$[M^1L^2T^{-2}A^{-2}]$
31.	Electric field	NC^{-1} or Vm^{-1}	$[M^1L^1T^{-3}A^{-1}]$
32.	Magnetic Induction	$NA^{-1}\ m^{-1}$ or tesla (T)	$[M^1L^0T^{-2}A^{-1}]$
33.	Magnetic Flux	Tm^2 or weber (Wb)	$[M^1L^2T^{-2}A^{-1}]$
34.	Permittivity	$C^2\ N^{-1}\ m^{-2}$	$[M^{-1}L^{-3}T^4A^2]$
35.	Permeability	$Tm\ A^{-1}$ or $Wb\ A^{-1}\ m^{-1}$	$[M^1L^1T^{-2}A^{-2}]$
36.	Planck's Constant	Js	$[M^1L^2T^{-1}]$
37.	Boltzman Constant	JK^{-1}	$[M^1L^2T^{-2}K^{-1}]$
38.	Stefan's Constant	$W\ m^{-2}K^{-4}$	$[M^1L^0T^{-3}K^{-4}]$

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Chapter 2

Kinematics

DISTANCE TRAVERSED AND SPEED

Distance traversed (Path length)

1. The total length of actual path traversed by the body between initial and final positions is called distance.
2. It has no direction and is always positive.
3. Distance covered by particle never decreases.
4. Its SI unit is metre (m) and dimensional formula is $[M^0L^1T^0]$.

EQUATIONS OF MOTION

General equations of motion :

$$v = \frac{dx}{dt} \Rightarrow dx = vdt \Rightarrow \int dx = \int vdt$$ = area enclosed by velocity-time graph

$$a = \frac{dv}{dt} \Rightarrow dv = adt \Rightarrow \int dv = \int adt$$ = area enclosed by acceleration-time graph

$$a = \frac{vdv}{dx} \Rightarrow vdv = adx \Rightarrow \int vdv = \int adx$$ = area enclosed by acceleration-position graph

Equations of motion of a particle moving with uniform acceleration in straight line :

1. $v = u + at$
2. $S = ut + \frac{1}{2}at^2 = \left(\frac{v+u}{2}\right)t = vt - \frac{1}{2}at^2$
3. $v^2 = u^2 + 2aS$
4. $S_{n^{th}} = u + \frac{1}{2}a(2n-1)$
5. $x = x_0 + ut + \frac{1}{2}at^2$

Here,

u = velocity of particle at $t = 0$

S = Displacement of particle between 0 to t

$= x - x_0$ (x_0 = position of particle at $t = 0$, x = position of particle at time t)

a = uniform acceleration

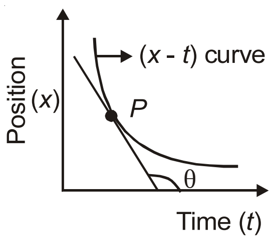
v = velocity of particle at time t

$S_{n^{th}}$ = Displacement of the particle in n^{th} second

GRAPHS

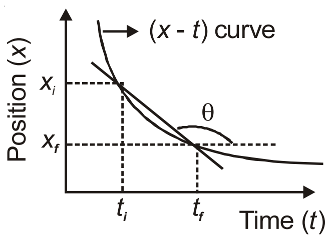
The important properties of various graphs are given below :

1. Slope of the tangent at a point on the position-time graph gives the instantaneous velocity at that point.



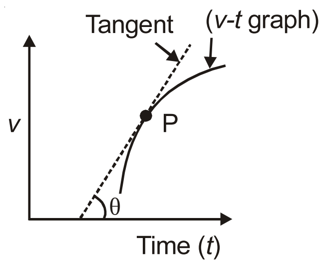
$\tan\theta = \dfrac{dx}{dt} = v$ (**Instantaneous Velocity** at point P)

2. Slope of a chord joining two points on the Position-time graph gives the average velocity during the time interval between those points.



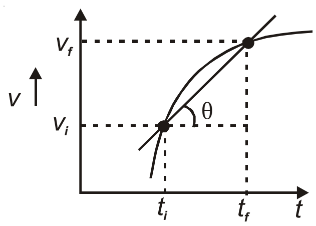
$\tan\theta = \dfrac{x_f - x_i}{t_f - t_i} = v_{\text{av}}$

3. Slope of the tangent at a point on the velocity-time graph gives the instantaneous acceleration at that point.



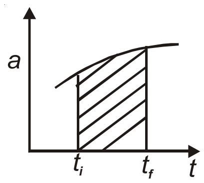
$\tan\theta = \dfrac{dv}{dt} = a$ (Instantaneous acceleration at P)

4. Slope of the chord joining two points on the velocity-time graph gives the average acceleration during the time interval between those points.



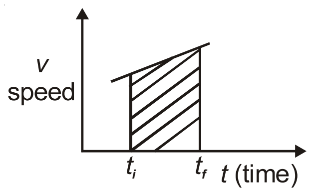
$\tan\theta = \dfrac{v_f - v_i}{t_f - t_i} = a_{\text{av}}$ = Average acceleration in time interval $t_f - t_i$

5. The area under the acceleration-time graph between t_i and t_f gives the change in velocity $(v_f - v_i)$ between the two instants.



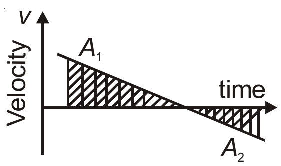
Shaded area = $v_f - v_i$ = change in velocity during interval t_i to t_f

6. The area under speed-time graph between t_i and t_f gives distance covered by particle in the interval $t_f - t_i$.



Shaded area = distance covered in time $(t_f - t_i)$

7. The area under the velocity-time graph between t_i and t_f gives the displacement $(x_f - x_i)$ between the two instants.

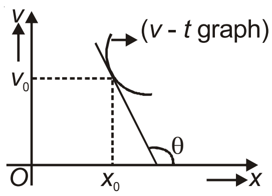


Shaded area $(A_1 - A_2)$ = Displacement in time $(t_f - t_i)$

Also, $A_1 + A_2$ = Distance covered in time $(t_f - t_i)$

8. In velocity-position graph, the acceleration of particle of any position x_0 is given as.

$$a = v_0 \tan\theta = v_0 \left|\frac{dv}{dx}\right|_{x=x_0}$$

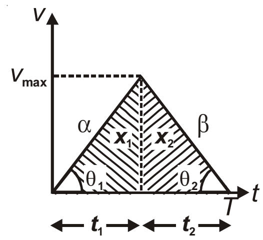


9. The position-time graph cannot be symmetric about the time-axis because at an instant a particle cannot have two displacements.
10. The distance-time graph is always an increasing curve for a moving body.
11. The displacement-time graph does not show the trajectory of the particle.

Applications

1. If a particle is moving with uniform acceleration on a straight line and have velocity v_A at A and v_B at B, then velocity of particle midway on line AB is $v = \sqrt{\dfrac{v_A^2 + v_B^2}{2}}$.

2. If a body starts from rest with acceleration α and then retards to rest with retardation β, such that total time of journey is T, then



(a) Maximum velocity during the trip $v_{max.} = \left(\frac{\alpha\beta}{\alpha+\beta}\right).T$

(b) Length of the journey $L = \frac{1}{2}\left(\frac{\alpha\beta}{\alpha+\beta}\right)T^2$

(c) Average velocity of the trip = $\frac{v_{max.}}{2} = \frac{\alpha\beta T}{2(\alpha+\beta)}$

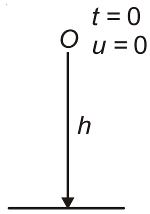
(d) $\frac{x_1}{x_2} = \frac{\beta}{\alpha} = \frac{t_1}{t_2}$.

MOTION UNDER GRAVITY

If height of object is very small as compared to radius of earth, motion of object will be uniformly accelerated. Equation of motion can be applied with proper sign convention.

Following are the important cases of interest.

1. Object is released from a height *h.*



Time taken to reach ground

$$-h = 0 - \frac{1}{2}gT^2 \text{ (taking up as positive)}$$

$$\Rightarrow T = \sqrt{\frac{2h}{g}}$$

Velocity of ball when it reaches ground

$$v = 0 - gT = -g\sqrt{\frac{2h'}{g}} = -\sqrt{2gh}$$

'–' sign indicate that velocity will be in downward direction.

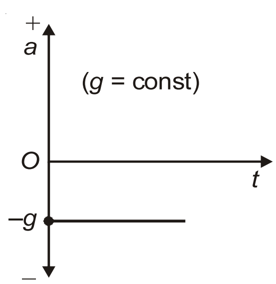
2. A particle is projected from ground with velocity *u* in vertically upward direction then

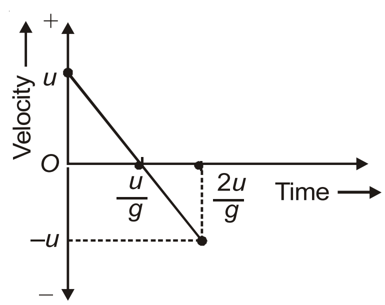
(a) Time of ascent = Time of descent = $\frac{\text{Time of flight}}{2} = \frac{T}{2} = \frac{u}{g}$

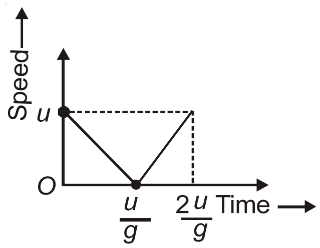
(b) Maximum height attained = $\frac{u^2}{2g}$

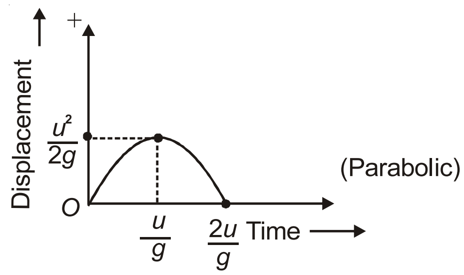
(c) Speed of particle when it hits the ground = *u*

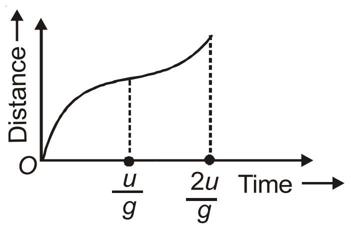
(d) Graphs











(e) Displacement of particle in complete journey = zero $\Rightarrow$ average velocity $v_{av} = 0$

(f) Distance covered by particle in complete journey = $\dfrac{u^2}{g}$

$\Rightarrow$ Average speed in complete journey = $\dfrac{u}{2}$

3. A body is thrown upward such that it takes t seconds to reach its highest point.
   (a) Distance travelled in $(t)^{th}$ second = distance travelled in $(t + 1)^{th}$ second.
   (b) Distance travelled in $(t - 1)^{th}$ second = distance travelled in $(t + 2)^{th}$ second.
   (c) Distance travelled in $(t - r)^{th}$ second = distance travelled in $(t + r + 1)^{th}$ second.
4. A body is projected upward from certain height h with initial speed u.
   (a) Its speed when it acquires the same level is u.
   (b) Its speed at the ground level is

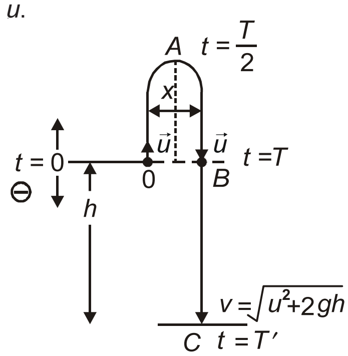
   $$v = \sqrt{u^2 + 2gh}$$

   (c) The time required to attain same level is

   $$T = \frac{2u}{g}$$

   (d) Total time of flight (T') is obtained by solving

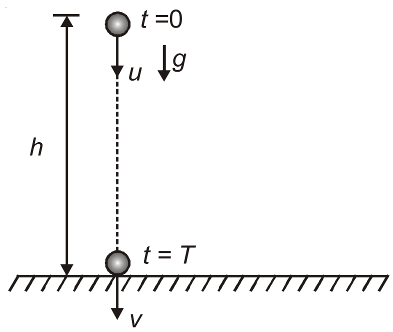
   $$\Rightarrow -h = uT' - \frac{1}{2}gT'^2 \text{ or } T' = \frac{u + \sqrt{u^2 + 2gh}}{g}$$



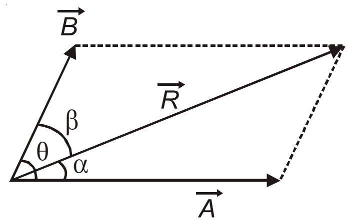
Here $x = 0$ particle follows same path during ascent and descent

5. A body is projected from a certain height h with initial speed u downward.
   (a) Its speed at ground level is $v = \sqrt{u^2 + 2gh}$
   (b) Time of flight (T)

   $$T = \frac{-u + \sqrt{u^2 + 2gh}}{g}$$



Parallelogram Law of Vector Addition : If two vectors having common origin are represented both in magnitude and direction as the two adjacent sides of a parallelogram, then the diagonal which originates from the common origin represents the resultant of these two vectors. The results are listed below



(a) $\vec{R} = \vec{A} + \vec{B}$

(b) $|\vec{R}| = (A^2 + B^2 + 2AB\cos\theta)^{1/2}$

(c) $\tan\alpha = \dfrac{B\sin\theta}{A + B\cos\theta}$, $\tan\beta = \dfrac{A\sin\theta}{B + A\cos\theta}$

(d) If $|\vec{A}| = |\vec{B}| = x\,(\text{say})$, then $R = x\sqrt{2(1+\cos\theta)} = 2x\cos\dfrac{\theta}{2}$ and $\alpha = \beta = \dfrac{\theta}{2}$ *i.e.,* resultant bisect angle between $\vec{A}$ and $\vec{B}$.

(e) If $|\vec{A}|\ \ |\vec{B}|$ then $\alpha < \beta$

(f) $R_{max} = A + B$, when $\theta = 0$ and $R_{min} = |A - B|$ when $\theta = 180°$.

(g) $R^2 = A^2 + B^2$, if $\theta = 90°$ *i.e.,* $\vec{A}$ and $\vec{B}$ are perpendicular.

(h) If $|\vec{A}| = |\vec{B}| = |\vec{R}|$, then $\theta = 120°$.

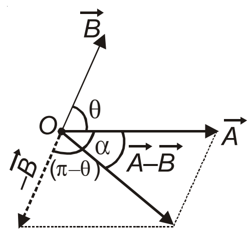
(i) If $\vec{R}$ is perpendicular to $\vec{A}$, then $\cos\theta = -\dfrac{A}{B}$ and $A^2 + R^2 = B^2$.

(j) For n coplanar vectors of same magnitude acting at a point such that angle between consecutive vectors are equal $\left(\dfrac{360}{n}\right)$, the resultant is zero.

VECTOR SUBTRACTION

Subtraction of vector $\vec{B}$ from vector $\vec{A}$ is simply addition of vector $-\vec{B}$ with $\vec{A}$ *i.e.,* $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$

Using parallelogram law,



Result : $R = |\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB\cos\theta}$, $\tan\alpha = \dfrac{B\sin\ \pi-\theta}{A + B\cos(\pi-\theta)} = \dfrac{B\sin\theta}{A - B\cos\theta}$

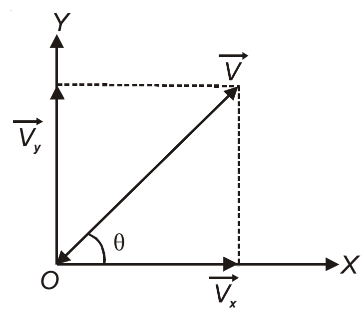
> ***Note :*** *If* $|\vec{A}| = |\vec{B}| = x\,(\text{say})$, *then* $R = x\sqrt{2(1-\cos\theta)} = 2x\sin\dfrac{\theta}{2}$.

RESOLUTION OF VECTORS

Any vector $\vec{V}$ can be represented as a sum of two vectors $\vec{P}$ and $\vec{Q}$ which are in same plane as $\vec{V} = \lambda\vec{P} + \mu\vec{Q}$, where λ and μ are two real numbers. We say that $\vec{V}$ has been resolved in two component vector $\lambda\vec{P}$ and $\mu\vec{Q}$ along $\vec{P}$ and $\vec{Q}$ respectively.

Rectangular components in two dimensions :

$\vec{V} = \vec{V}_x + \vec{V}_y, \vec{V} = V_x\hat{i} + V_y\hat{j},\ V = \sqrt{V_x^2 + V_y^2}$



$\vec{V}_x$ and $\vec{V}_y$ are rectangular component of vector in 2-dimension.

$V_x = V\cos\theta$

$V_y = V\sin\theta = V\cos(90 - \theta)$

V_z = zero.

$\vec{V} = V\cos\theta\,\hat{i} + V\sin\theta\,\hat{j}$

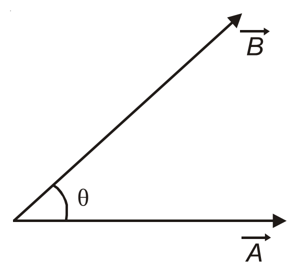
Note : *Unit vector along* $\vec{V}$ is $\cos\theta\,\hat{i} + \sin\theta\,\hat{j}$

SCALAR AND VECTOR PRODUCTS

Scalar (dot) Product of Two Vectors : The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is defined as

$\vec{A}.\vec{B} = AB\cos\theta$

$\cos\theta = \dfrac{\vec{A}.\vec{B}}{AB}$



If $\vec{A}$ and $\vec{B}$ are perpendicular, then $\vec{A}.\vec{B} = 0$

If $\theta < 90°$, then $\vec{A}.\vec{B}\quad 0$ and if $\theta > 90°$ then $\vec{A}.\vec{B}\quad 0$.

Projection of vector $\vec{A}$ on $\vec{B}$ is $(\vec{A}.\vec{B})\dfrac{\vec{B}}{B^2}$.

$A^2 = \vec{A}.\vec{A}$

$\hat{i}.\hat{i} = \hat{j}.\hat{j} = \hat{k}.\hat{k} = 1.$

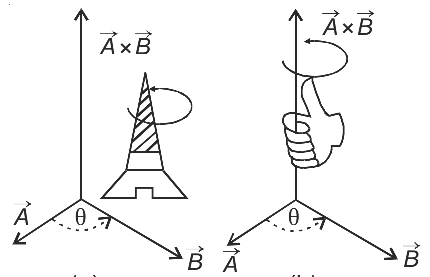
Scalar product is commutative *i.e.*, $\vec{A}.\vec{B} = \vec{B}.\vec{A}$.

Scalar product is distributive *i.e.*, $\vec{A}\ (\vec{B}\quad \vec{C}) = \vec{A}\ \vec{B} + \vec{A}\ \vec{C}$

Vector Product of two Vectors :

Mathematically, if θ is the angle between vectors A and B, then

$A \times B = AB\sin\theta\,\hat{n}$ ...(i)



(a) (b)

The direction of vector $A\times B$ is the same as that of unit vector $\hat{n}$. It is decided by any of the following two rules :

(a) **Right handed screw rule :** Rotate a right handed screw from vector A to B through the smaller angle between them; then the direction of motion of screw gives the direction of vector $A\times B$ (Fig. a).

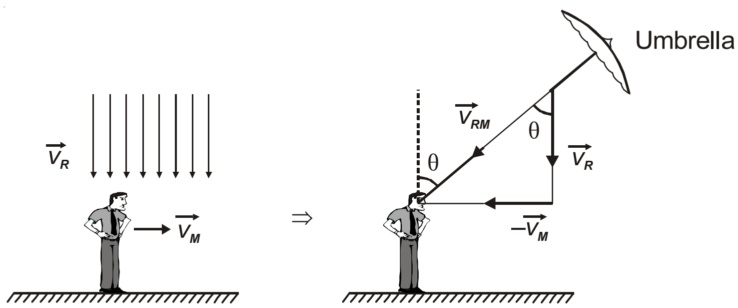
(b) **Right hand thumb rule :** Bend the finger of the right hand in such a way that they point in the direction of rotation from vector A to B through the smaller angle between them; then the thumb points in the direction of vector $A\times B$ (Fig. b).

RELATIVE MOTION IN TWO DIMENSIONS

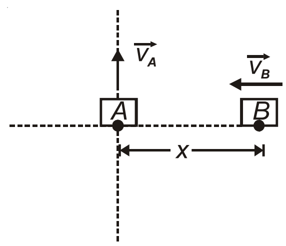
Relative velocity :

Velocity of object A w.r.t. object B is $\vec{v}_{AB}=\vec{v}_A-\vec{v}_B, \vec{v}_{BA}=\vec{v}_B-\vec{v}_A$

1. **Direction of Umbrella :** A person moving on straight road has to hold his umbrella opposite to direction of relative velocity of rain. The angle θ is given by $\tan\theta=\dfrac{v_M}{v_R}$ with vertical in forward direction.



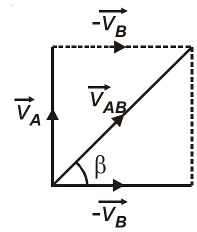
2. **Closest approach :** Two objects A and B having velocities $\vec{v}_A$ and $\vec{v}_B$ at separation x are shown in figure



The relative velocity of A with respect to B is given by

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$

$$\tan\beta = \frac{v_A}{v_B}$$



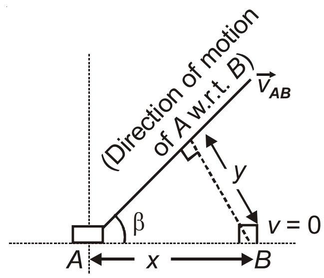
The above situation is similar to figure given below.

y is the distance of closest approach.

Now, $\sin\beta = \dfrac{y}{x}$

$\Rightarrow\ y = x\sin\beta$

$$y = \frac{x\tan\beta}{\sqrt{1+\tan^2\beta}} = \frac{xV_A}{\sqrt{V_B^2+V_A^2}}$$



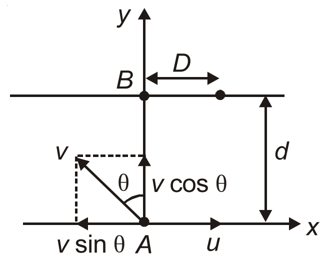
3. Crossing a river :

v = velocity of the man in still water.

θ = angle at which man swims w.r.t. normal to bank such that

$v_x = -v\sin\theta$, $v_y = v\cos\theta$

Time taken to cross the river is given by



$$t = \frac{d}{v_y} = \frac{d}{v\cos\theta}$$

Velocity along the river

$$v_x' = u - v\sin\theta$$

Distance drifted along the river $D = t\,v_x'$

$$D = \frac{d}{v\cos\theta}(u - v\sin\theta)$$

Case I : (Shortest time)

The Minimum time to cross the river is given by

$${}_{min} = \frac{d}{v} \qquad \text{(when } \cos\theta = 1,\ \theta = 0°,\ u \quad v\text{)}$$

Distance drifted is given by

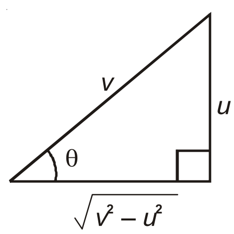
$$D = \frac{d}{v}\times u$$

Case II : (Shortest path)

To cross the river straight

drift $D = 0 \quad\Rightarrow\quad u - v\sin\theta = 0$

$\sin\theta = \dfrac{u}{v} \quad\Rightarrow\quad$ provided $v > u$



Time to cross the river straight across is given by

$$t = \frac{d}{v\cos\theta} = \frac{d}{\sqrt{v^2-u^2}}$$

PROJECTILE MOTION

An object moving in space under the influence of gravity is called projectile. Two important cases of interest are discussed below :

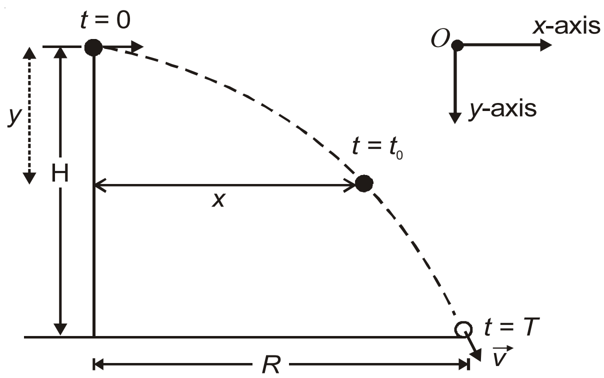
1. **Horizontal projection :**

A body of mass m is projected horizontally with a speed u from a height h at the moment $t = 0$. The path followed by it is a parabola.

It hits the ground at the moment $t = T$, with a velocity $\vec{v}$ such that

$$T = \sqrt{\frac{2H}{g}}$$

$$|\vec{v}| = \sqrt{u^2 + 2gH} = |u\hat{i} + gT\hat{j}|$$



The position at any instant t_0 is given by

$$x = ut_0$$

$$y = \frac{1}{2}gt_0^2$$

$$y = \frac{gx^2}{2u^2} \text{ (trajectory of particle)}$$

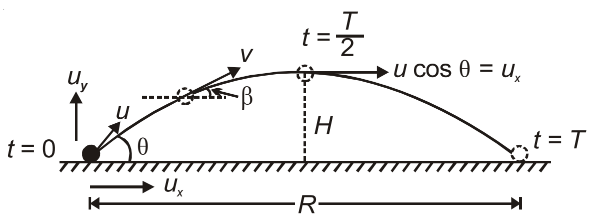
The velocity at any instant t_0 is given by

$$\overrightarrow{v_0} = u\hat{i} + gt_0\hat{j}$$

The range R will be given by $\boxed{R = u\sqrt{\frac{2H}{g}}}$

2. **Oblique projection :** A body of mass m is projected from ground with speed u at an angle θ above horizontal at the moment $t = 0$.

It hits the ground at a horizontal distance R at the moment $t = T$.



1. Time of flight $T = \frac{2u_y}{g} = \frac{2u\sin\theta}{g}$

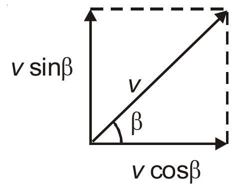
2. Maximum height $H = \frac{u_y^2}{2g} = \frac{u^2\sin^2\theta}{2g}$

3. Horizontal range $R = u_x \times T = \frac{2u_x u_y}{g} = \frac{u^2\sin 2\theta}{g}$

4. Equation of trajectory: $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

or $y = x\tan\theta\left(1 - \frac{x}{R}\right)$

5. Instantaneous velocity $v = \sqrt{u^2 + (gt)^2 - 2u(gt)\sin\theta}$



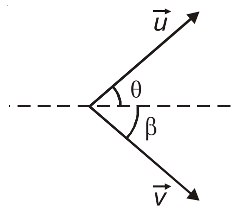
and direction of motion is such that, $\tan\beta = \dfrac{u\sin\theta - gt}{u\cos\theta}$

(a) $v = \dfrac{u\cos\theta}{\cos\beta}$ [$\because$ Horizontal component is same everywhere]

(b) $v\sin\beta = u\sin\theta - gt$

(c) When $\vec{v}$ (velocity at any instant '*t*') is perpendicular to $\vec{u}$ (initial velocity)

$\Rightarrow$ $\beta = -(90° - \theta)$



(i) $v = \dfrac{u\cos\theta}{\cos(90 - \theta)} = u\cot\theta$

(ii) $t = \dfrac{u}{g\sin\theta}$

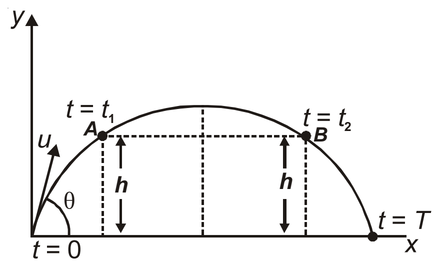
Applications :

1. The height attained by the particle is largest when $\theta = 90°$. In this situation, time of flight is maximum and range is minimum (zero).
2. When R is range, T is time of flight and H is maximum height, then

(a) $\tan\theta = \dfrac{gT^2}{2R}$

(b) $\tan\theta = \dfrac{4H}{R}$

3. When horizontal range is maximum, $H = \dfrac{R_{max}}{4}$
4. The horizontal range is same for complimentary angles like $(\theta, 90° - \theta)$ or $(45° + \theta, 45° - \theta)$. It is maximum for $\theta = 45°$.
5. If A and B are two points at same level such that the object passes A at $t = t_1$ and B at $T = t_2$, then



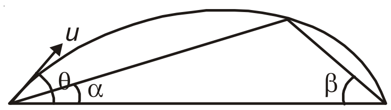
(i) $T = \dfrac{2u\sin\theta}{g} = t_1 + t_2$

(ii) $h = \dfrac{1}{2}gt_1t_2$

(iii) Average velocity in the interval AB is

$v_{av} = u\cos\theta$ [$\because$ vertical displacement is zero]

6. If a projectile is projected from one vertex of a triangle such that it grazes second vertex and finally fall down on 3rd vertex of the triangle on the same horizontal level, then $\boxed{\tan\theta = \tan\alpha + \tan\beta}$.



7. A projectile has same range for two angle of projection. If time of flight in two cases are T_1 and T_2, maximum height is H_1 and H_2 and the horizontal range is R. Then

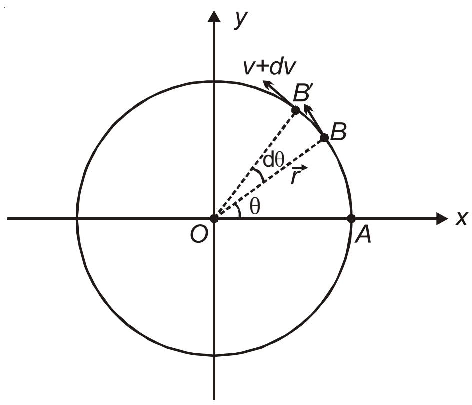
(i) Range of projectile is $R = \frac{1}{2}gT_1T_2$

(ii) Velocity of projection of projectile is $u = \frac{1}{2}g\left[T_1^2 + T_2^2\right]^{1/2}$

(iii) $R = 4\sqrt{H_1H_2}$

CIRCULAR MOTION

An object of mass m is moving on a circular track of radius r. At t = 0, it was at A. At any moment of time 't', it has moved to B, such that $AOB = \theta$. Let its speed at this instant be v and direction is along the tangent. In a small time dt, it moves to B' such that $B'OB = d\theta$.



The angular displacement vector is $\overrightarrow{d\theta} = d\theta\hat{k}$

The angular velocity vector is $\vec{\ } = \frac{d\theta}{dt}\hat{k}$.

At B', the speed of the object has become $v + dv$.

The tangential acceleration is $a_t = \frac{dv}{dt} = r\alpha$

The radial (centripetal acceleration) is $a_c = \frac{v^2}{r} = \ ^2r$

The angular acceleration is $\alpha = \frac{d}{dt}$

Relations among various quantities.

1. $\vec{v} = \vec{\ } \times \vec{r}$

2. $\vec{a} = \frac{d\vec{v}}{dt} = \vec{\ } \times \frac{d\vec{r}}{dt} + \frac{d\vec{\ }}{dt} \times \vec{r} = \overrightarrow{a_c} + \overrightarrow{a_t}$

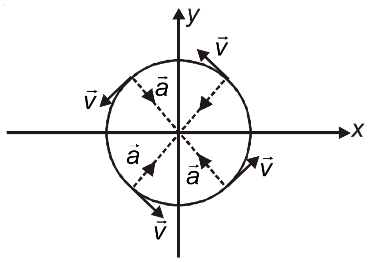
3. $\overrightarrow{a_c} = \vec{\ } \times \vec{v}$

4. $\overrightarrow{a_t} = \vec{\alpha} \times \vec{r}$

Uniform Circular Motion :

1. In uniform circular motion, the speed (v) of particle is remain constant (= constant)

2. $a_T = 0$ and $\alpha = \frac{d}{dt} = 0$



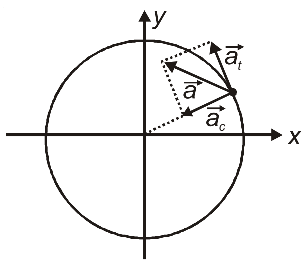
3. Only centripetal acceleration (also called normal acceleration) exists in uniform circular motion

$$a_c = r\ \ ^2 = \frac{v^2}{r}$$

4. In uniform circular motion $\vec{v}$ $\vec{a}$

Nonuniform Circular Motion :

1. In nonuniform circular motion the speed (v) and angular velocity ( ) change w.r.t. time.
2. Net acceleration of particle in non-uniform circular motion.



$$a = \sqrt{a_c^2 + a_t^2} = \sqrt{\left(\frac{v^2}{r}\right)^2 + \ r\alpha\ \ ^2}$$

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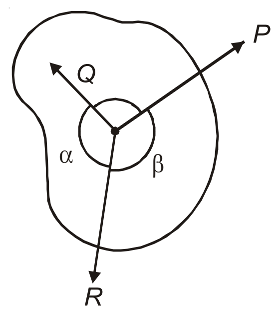
Chapter 3

Laws of Motion

Equilibrium of Concurrent Forces

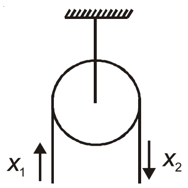
If three forces $\vec{P}$, $\vec{Q}$ and $\vec{R}$ are acting on an object such that forces are concurrent and the object is in equilibrium then $\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin}$.



APPLICATIONS OF NEWTON'S LAWS OF MOTION

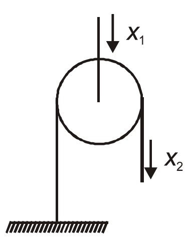
The strings connected to pulley are considered as ideal. Their length is fixed, so the ends of string follow a fixed relation between displacement velocity and acceleration. These relations are called constraint relation.

Case I**:** When the middle end is fixed

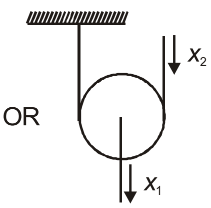


$x_1 = x_2$
$v_1 = v_2$
$a_1 = a_2$

Case II**:** When the side end is fixed



OR



$x_2 = 2x_1$
$v_2 = 2v_1$
$a_2 = 2a_1$

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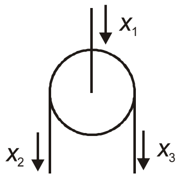
Case III **:** When all the three ends are free to move



$$x_1 = \frac{x_2 + x_3}{2}$$

$$v_1 = \frac{v_2 + v_3}{2}$$

$$a_1 = \frac{a_2 + a_3}{2}$$

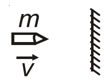
> ***Note :*** *In all the above relations downward direction is taken as positive. If any of the direction is upward in any case then –ve sign must be incorporated in the corresponding equation.*

1. A machine gun fires n bullet per second with speed u and mass of each bullet is m.

The Force required to keep the gun stationary is

$\vec{F} = nm\vec{v}$

2. Bullets moving with a speed v hit a wall normally.



(i) If the bullets come to rest in wall

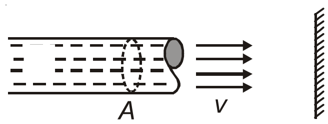
Force on wall $F_{wall} = nmv$

(Here n is number of bullets hitting the wall in one second)

(ii) If the bullets rebound elastically,

$F_{wall} = 2nmv$

3. Liquid jet of area A moving with speed v hits a wall



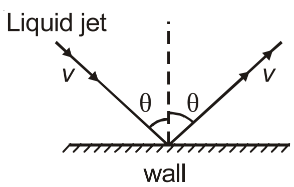
(i) Force required by a pump to move the liquid with this speed is

$$F = v\frac{dm}{dt} = v \times \ Av = \ Av^2$$

(ii) As jet hits a vertical wall and does not rebound, the force exerted by it on the wall is, $F_{wall} = \ Av^2$

(iii) When water rebounds elastically, $F_{wall} = 2\ Av^2$

(iv) For oblique impact as shown, $F_{wall} = 2\ Av^2 \cos\theta$



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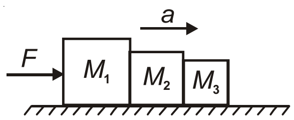
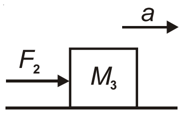
4. The blocks shown are being pushed by a force F. F_1, F_2 are contact forces between M_1 & M_2 and M_2 & M_3 respectively

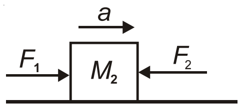


$$a = \frac{F}{M_1 + M_2 + M_3}$$

FBD of M_3


$F_2 = M_3a$

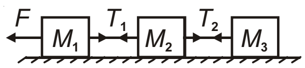
$F_1 - F_2 = M_2a \Rightarrow F_1 = (M_3 + M_2)a$

FBD of M_2


or $F_2 = \dfrac{M_3}{M_1 + M_2 + M_3}F, F_1 = \left(\dfrac{M_2 + M_3}{M_1 + M_2 + M_3}\right)F$

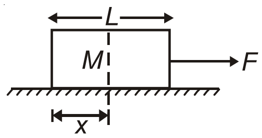
5. The strings are massless. Let T_1 and T_2 be the tensions in the two strings and 'a' be the acceleration.

$$a = \frac{F}{M_1 + M_2 + M_3}, T_1 = \frac{(M_2 + M_3)F}{M_1 + M_2 + M_3}, T_2 = \frac{M_3F}{M_1 + M_2 + M_3}$$

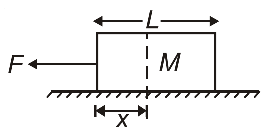


6. Tension in the block at a distance x from left end is given as

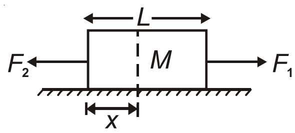
(a) $T_x = \dfrac{Mx}{L} \times \dfrac{F}{M} = \dfrac{Fx}{L}$

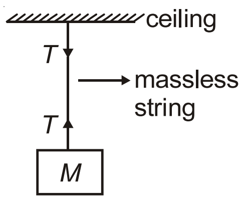


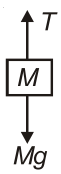
(b) $T_x = \dfrac{F(L - x)}{L}$



(c) $T_x = \dfrac{F_1x}{L} + \dfrac{F_2(L - x)}{L}$



7.




FBD of M w.r.t. ground

(a) When system is stationary

$T - Mg = 0$

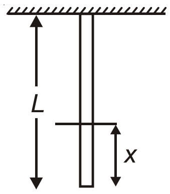
(b) System moves up with acceleration 'a'

$T = M(g + a)$

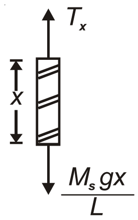
(c) System moves down with acceleration a

$T = M(g - a)$

8. Uniform rope of mass M_s.



FBD of lower portion



The tension in the rope at a distance x from free end is given below for different cases.

(a) Stationary system

$$T_x = \frac{M_s gx}{L}$$

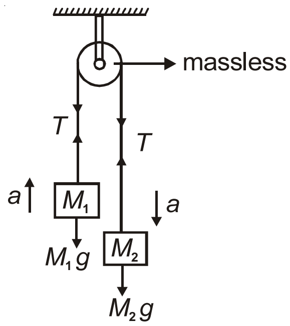
(b) If the rope is accelerating upwards, then $T_x = \frac{M_s x}{L}(g + a)$

(c) If the rope is accelerating downwards, then $T_x = \frac{M_s x}{L}(g - a)\,(g \quad a)$

IMPORTANT PROBLEMS

Pulley mass systems

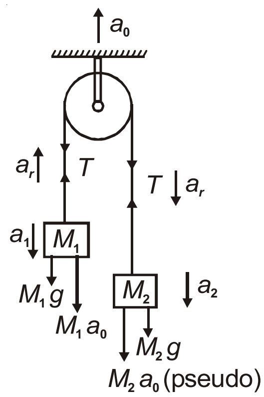
(i) Stationary pulley



$$a = \frac{M_2 - M_1}{M_2 + M_1} g$$

$$T = 2\left(\frac{M_1 M_2}{M_1 + M_2}\right) g$$

(ii) Pulley is moving upward with acceleration a_0



$$T = 2\left(\frac{M_1 M_2}{M_1 + M_2}\right)(g + a_0)$$

The acceleration of each block with respect to pulley is

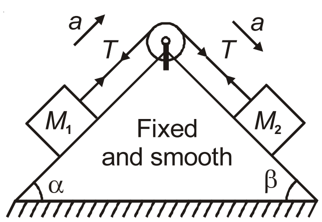
$$a_r = \left(\frac{M_2 - M_1}{M_2 + M_1}\right)(g + a_0)$$

The absolute accelerations of the two blocks are a_1 and a_2

$$a_1 = -(a_0 + a_r) \qquad \left\{ a_0 = \left(\frac{a_1 + a_2}{2}\right) \right\}$$

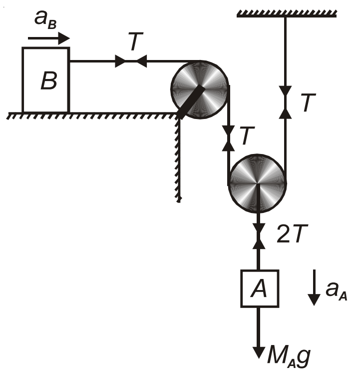
$$a_2 = a_r - a_0$$

(iii) $T = \frac{M_1 M_2 g}{M_1 + M_2}(\sin\alpha + \sin\beta)$



$$a = \left(\frac{M_2 \sin\beta - M_1 \sin\alpha}{M_2 + M_1}\right) g$$

(iv)



$2a_A = a_B$ ...(i)

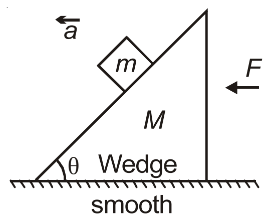
$M_Ag - 2T = M_Aa_A$ ...(ii)

$T = M_Ba_B$ ...(iii)

Two block system :

Case - I :

Let 'm' does not slide down relative to wedge 'M'



The force required is given by

$F = (M + m)g \tan\theta$

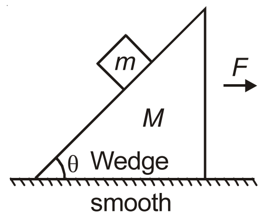
$a = g \tan\theta$ (in horizontal direction w.r.t. ground)

Contact force R between m and M is

$$R = \frac{mg}{\cos\theta}$$

Case - II :

Minimum value of F so that 'm' falls freely is given by



$F = Mg\cot\theta$

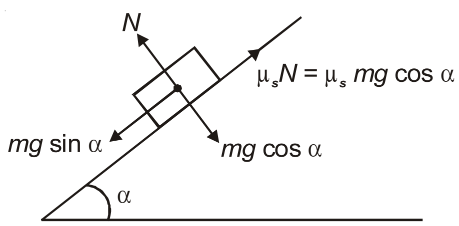
Wedge M moves with acceleration = $g\cot\theta$

The block falls vertically with acceleration 'g'.

Contact force between M and m is zero.

Angle of Repose

Consider a situation in which a block is placed on an inclined plane with co-efficient of friction 'μ'. The maximum value of angle of inclined plane for which the block can remain at rest is defined as angle of repose.

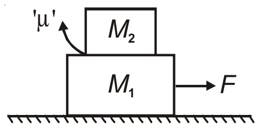


$\alpha = \tan^{-1}(\mu_s)$

Two blocks placed one above the other on smooth ground

Case - I : If a force F is applied on lower block

(a) $F \quad \mu(M_1 + M_2)g$



Smooth

Both blocks move together with same acceleration $a = \dfrac{F}{M_1 + M_2}$

$a_{max} = \mu g$

(b) $F > \mu(M_1 + M_2)g$

M_2 moves with constant acceleration $a_2 = \mu g$

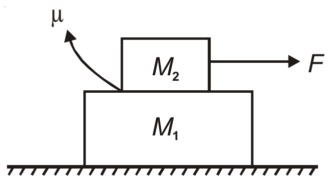
M_1 moves with acceleration $a_1 = \dfrac{F - \mu M_2 g}{M_1}$

M_2 slips backward on M_1.

Case - II : If a force F is applied on upper block

(a) $F \quad \dfrac{\mu(M_1 + M_2)M_2}{M_1}g$, both blocks move together with acceleration $a = \dfrac{F}{M_1 + M_2}$ with $a_{max} = \dfrac{\mu M_2}{M_1}g$.

(b) If $F \quad \dfrac{\mu(M_1 + M_2)M_2 g}{M_1}$,



Smooth

M_1 moves with constant acceleration $a_1 = \dfrac{\mu M_2}{M_1}g$

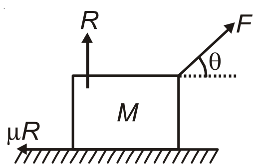
M_2 moves with acceleration $a_2 = \dfrac{F - \mu M_2 g}{M_2}$

M_2 slips forward on M_1.

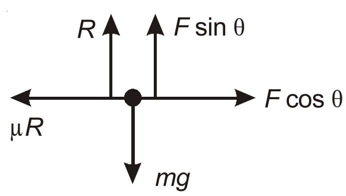
Minimum force required to move a body on a rough horizontal surface

$F \cos\theta > \mu R \quad (R = mg - F\sin\theta)$

$F \quad \dfrac{\mu mg}{\cos\theta + \mu\sin\theta}$

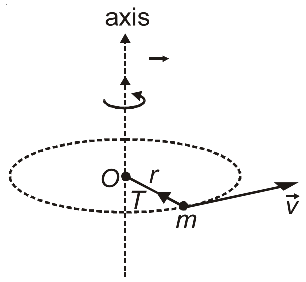


$F_{min} = \dfrac{\mu mg}{\sqrt{1 + \mu^2}}$ at $\theta = \tan^{-1}(\mu)$ $\Rightarrow$



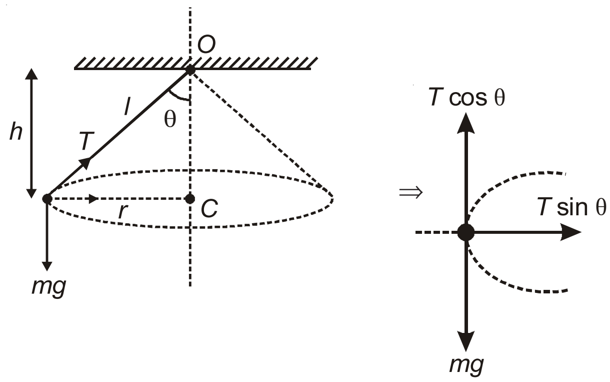
DYNAMICS OF CIRCULAR MOTION

Neglecting Gravity



T = Centripetal force = $\dfrac{mv^2}{r} = m\ {}^2r$

Considering gravity (Conical pendulum)



$T\sin\theta = m\ {}^2r$ ...(1)

$T\cos\theta = mg$

$T = \dfrac{mg}{\cos\theta}$ ...(2)

(a) For θ to be 90º (*i.e.*, string to be horizontal)

$T =$

$\therefore$ It is not possible.

(b) $T\sin\theta = m\ {}^2r = m\ {}^2l\sin\theta$

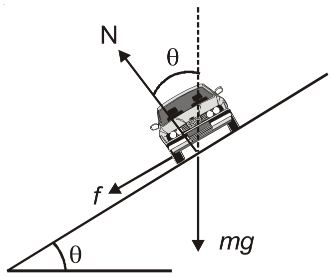
$\Rightarrow\quad T = m\ {}^2l$

(c) Time period = $2\pi\sqrt{\dfrac{h}{g}}$ $(h = l\cos\theta)$

Vehicle negotiating a curve on a banked road

The maximum velocity with which a vehicle can safely negotiate a curve of radius r on a rough inclined road is

$$V_{max} = \sqrt{\frac{rg(\mu + \tan\theta)}{1 - \mu\tan\theta}}$$



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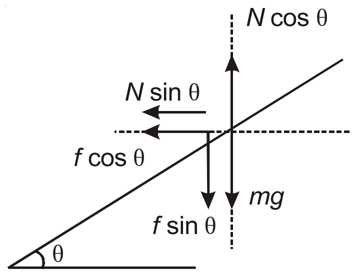

Special Cases :

For a smooth inclined surface $\mu = 0 \Rightarrow v_{max} = \sqrt{rg\tan\theta}$

For a horizontal rough surface, $\theta = 0 \Rightarrow v_{max} = \sqrt{\mu rg}$

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Chapter 4

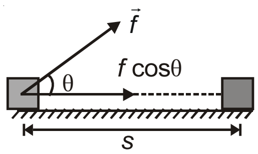
Work, Energy and Power

CONCEPT OF WORK

$$W = \vec{f}.\vec{s} \quad (\vec{s} = \vec{s}_2 - \vec{s}_1)$$

$$= fs\cos\theta$$

$$= s\,(f\cos\theta)$$



KINETIC ENERGY

$$\text{K.E.} = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

WORK-ENERGY THEOREM

$$W_{\text{Total}} = \text{ K.E.}$$

or

$$W_{\text{ext.}} + W_{\text{int.}} = \text{ K.E}$$

Work Done by Spring Force

$$W = -\int_{x_1}^{x_2} kx\,dx = \frac{-1}{2}k(x_2^2 - x_1^2)$$

when $x_2 > x_1$, $W < 0$

when $x_2 < x_1$, $W > 0$

COLLISION

One dimension

$(u_1 > u_2$ and $v_2 > v_1)$

$$\text{KE} = \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2}(u_1 - u_2)^2(1 - e^2)$$

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Following are the important cases

1. **Elastic collision $e = 1$**

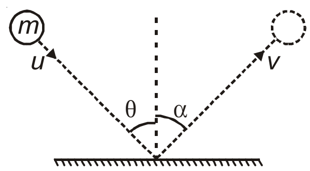
$$v_2 - v_1 = u_1 - u_2$$

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2}u_1 + \frac{2m_2u_2}{m_1 + m_2}$$

$$v_2 = \frac{m_2 - m_1}{m_2 + m_1}u_2 + \frac{2m_1u_1}{m_2 + m_1}$$

KE = 0 $\Rightarrow$ Final KE = Initial KE

2. **Coefficient of restitution = e**



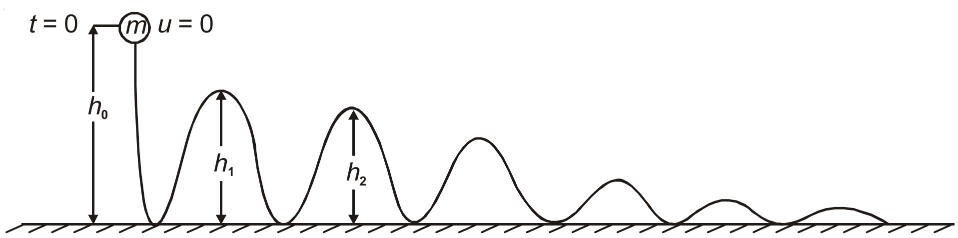
$u \sin\theta = v \sin\alpha$ ...(1)

$eu \cos\theta = v \cos\alpha$ ...(2)

$$\Rightarrow \quad v = u\sqrt{\sin^2\theta + e^2\cos^2\theta}$$

$$\tan\alpha = \frac{\tan\theta}{e} \quad (i.e.\ \alpha > \theta)$$

3. **A ball of mass m is dropped from a height h_0 on an inelastic floor.**



The coefficient of restitution = e

(a) Maximum height after n^{th} bounce is $h_n = e^{2n}h_0$

(b) Speed of rebound after n^{th} bounce = $e^n\sqrt{2gh_0}$

(c) Total distance travelled before the body comes to rest

$$= h_0\left[\frac{1+e^2}{1-e^2}\right]$$

(d) The time after which the body comes to rest

$$= \sqrt{\frac{2h_0}{g}}\cdot\left[\frac{1+e}{1-e}\right]$$

(e) Average force exerted on the ground is mg

(f) Displacement of ball when it stops is h_0

4. Oblique elastic collision

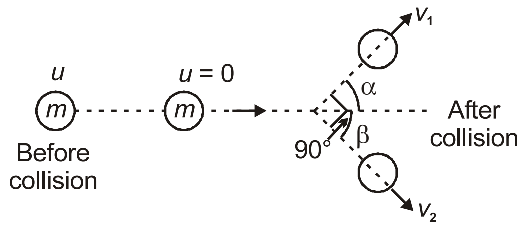
A body of mass m collides with a stationary body of same mass.

(a) $\alpha + \beta = \dfrac{\pi}{2}$ …(1)

(b) $v_1\cos\alpha + v_2\cos\beta = u$ …(2)

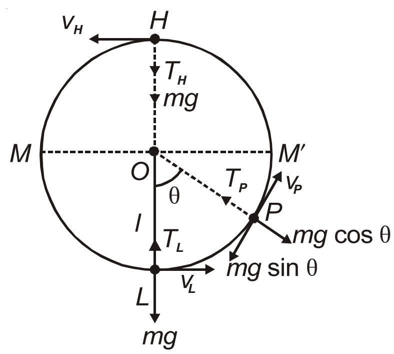
(c) $v_1\sin\alpha = v_2\sin\beta$ …(3)

(d) $u^2 = v_1^2 + v_2^2$ …(4)



MOTION IN A VERTICAL CIRCLE

A particle of mass m is tied to a string of length l whose other end is fixed. The particle can revolve about O in a vertical circle. when it is at position L (lowest point), it is given a speed V_L horizontal. Following results are useful in describing its motion.



1. $a_T = g\sin\theta$ …(1)

2. $a_C = \dfrac{T_p - mg\cos\theta}{m} = \dfrac{v_p^2}{l}$

3. $T_p = \dfrac{mv_p^2}{l} + mg\cos\theta$ …(2)

4. $T_L = mg + \dfrac{mv_L^2}{l}$

5. $v_L \quad \sqrt{2gl}$, it oscillates between M and M'

6. $\sqrt{2gl} \quad v_L \quad \sqrt{5gl}$, it will leave the circular path somewhere between M' and H.

7. When $v_L \quad \sqrt{5gl}$, it completes vertical circle (Also $v_H \quad \sqrt{gl}$)

8. $T_H = -mg + \dfrac{mv_H^2}{l}$

9. $T_L - T_H = 6\,mg$ (always)

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Chapter 5

Rotational Motion

CENTRE OF MASS OF A RIGID BODY (CONTINUOUS MASS DISTRIBUTION)

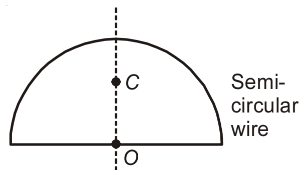
Mathematically position coordinates of the centre of mass of rigid body are given by

$$x_{cm} = \frac{\int x\,dm}{\int dm};\quad y_{cm} = \frac{\int y\,dm}{\int dm};\quad z_{cm} = \frac{\int z\,dm}{\int dm}$$

Centre of mass of some commonly used objects.

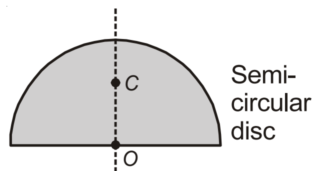
1. Semi circular wire of radius R.

$$OC = \frac{2R}{\pi}$$, where C is centre of mass



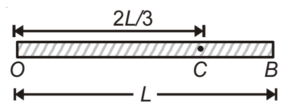
2. Semi circular disc of radius R

$$OC = \frac{4R}{3\pi}$$



3. Non-uniform rod of length L. The linear mass density varies linearly from zero at O to maximum at B.

$$OC = \frac{2L}{3}$$



VELOCITY AND ACCELERATION OF CENTRE OF MASS

Velocity of Centre of Mass

The instantaneous velocity of centre of mass is given by

$$\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2 + ...m_n\vec{v}_n}{\sum_{i=1}^{n} m_i};\qquad \text{or}\qquad \vec{v}_{cm} = \frac{\vec{P}_{system}}{M_{system}}$$

Where $\vec{P}_{system}$ is the total linear momentum of the system of particles.

Acceleration of Centre of Mass

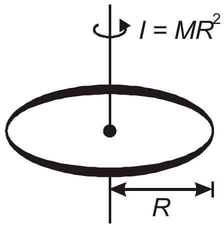
Differentiating $\vec{v}_{cm}$ w.r.t. time we get $\vec{a}_{cm}$ as

$$\vec{a}_{cm} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2 + ...m_n\vec{a}_n}{\sum_{i=1}^{n} m_i}; \qquad \text{or} \qquad \vec{a}_{cm} = \frac{\vec{F}_{ext}}{M_{system}}$$

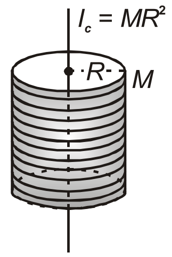
Where $\vec{F}_{ext}$ is the vector sum of forces acting on the particles of system.

MOMENT OF INERTIA OF DIFFERENT OBJECTS

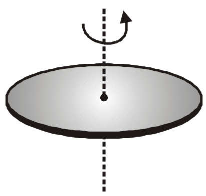
For an axis perpendicular to the plane of the ring



A hollow cylinder

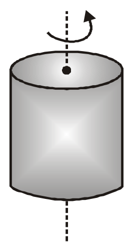


The axis perpendicular to the plane of the disc.



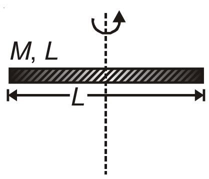
$$I_{cm} = \frac{MR^2}{2}$$

A solid cylinder



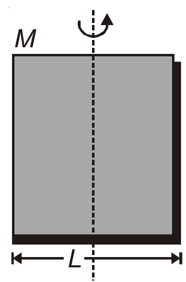
$$I_{cm} = \frac{MR^2}{2}$$

A thin rod



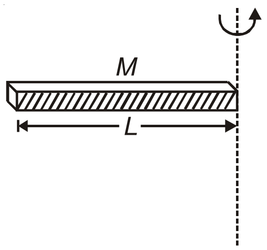
$$I_c = \frac{ML^2}{12}$$

A plate



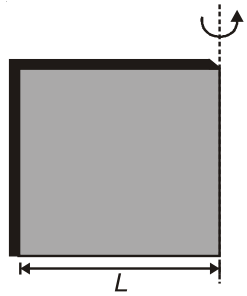
$$I_c = \frac{ML^2}{12}$$

A thin rod about a perpendicular axis through its end



$$I = \frac{ML^2}{3}$$

A plate about one edge



$$I = \frac{ML^2}{3}$$

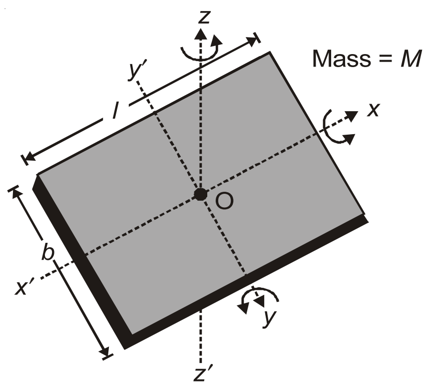
A Rectangular Plate

(a) $I_{xx'} = \dfrac{Mb^2}{12}$

(b) $I_{yy'} = \dfrac{Ml^2}{12}$

(c) $I_{zz'} = I_{xx'} + I_{yy'}$

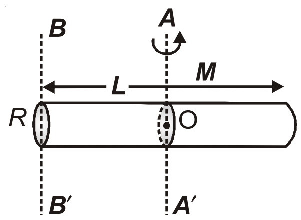
(d) $I_{zz'} = \dfrac{M(l^2 + b^2)}{12}$



A Thick Rod (Solid cylinder)

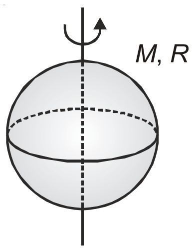
The axis is perpendicular to the rod and passing through the centre of mass

$$I_{AA'} = \frac{ML^2}{12} + \frac{MR^2}{4}; I_{BB'} = \frac{ML^2}{3} + \frac{MR^2}{4}$$



A Solid Sphere

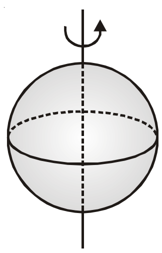
About its diameter



$$I_{cm} = \frac{2}{5}MR^2$$

A Hollow Sphere

About its diameter



$$I_{cm} = \frac{2}{3}MR^2$$

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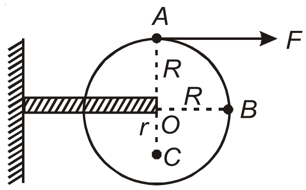
RIGID BODY ROTATION

In this section, rotation of a body about a stationary fixed axis has been discussed

1. Rotating Disc

A tangential force F is applied at the periphery, as a result disc is rotating above an axis passing through its CM, normal to plane of disc



$= F \times R$ [about O]

$I = \frac{1}{2}mR^2$

$\alpha = \frac{}{I} = \frac{2F}{MR}$

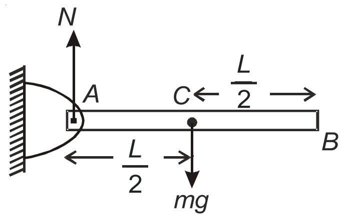
(1) Tangential acceleration of A is $a_A = R\alpha = \frac{2F}{M}$ (along horizontal)

(2) Tangential acceleration of B is $a_B = R\alpha = \frac{2F}{M}$ (vertically downwards)

(3) Tangential acceleration of C is $a_C = r\alpha = \frac{2Fr}{MR}$ (along horizontal, opposite to the direction of tangential acceleration of A)

2. Hinged Rod

The rod is released from rest from horizontal position



$= mg \times \frac{L}{2}$ (about A)

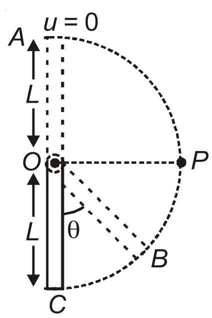
$I = \frac{ML^2}{3}$

$\alpha = \frac{}{I} = \frac{3g}{2L}$

(1) Linear acceleration of COM C is $a_{cm} = \frac{L}{2}\alpha = \frac{3g}{4}$. Also, $N = mg - ma_{cm} = \frac{mg}{4}$

(2) Linear acceleration of point B is $a_B = L\alpha = \frac{3g}{2}$

(3) The rod is released from unstable equilibrium position {from position A}



(a) When at B, $Mg\frac{L}{2}(1+\cos\theta) = \frac{1}{2}\left(\frac{ML^2}{3}\right)^{2}$

$= \sqrt{\frac{6g}{L}} \cos\frac{\theta}{2}$

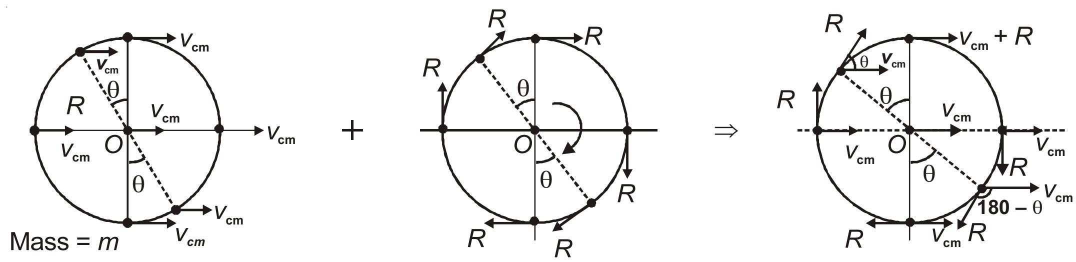
(b) at C, $\theta = 0°$

$= \sqrt{\frac{6g}{l}}$

(c) at P, $\theta = 90°$, $= \sqrt{\frac{3g}{l}}$

ROLLING OF A BODY

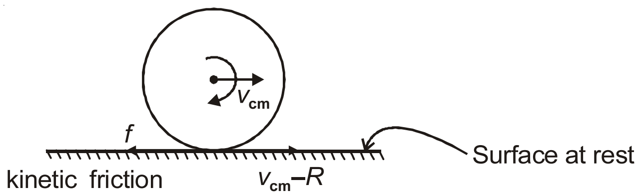
Rolling is combination of Pure Translation and Pure Rotation Motion about Centre of Mass.



Pure translation **Pure rotation (about cm)** **Rolling**

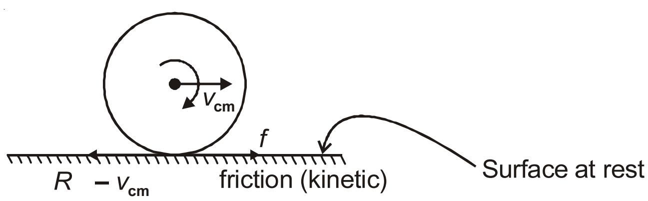
Case - I : Forward slipping

$v_{cm} > R$



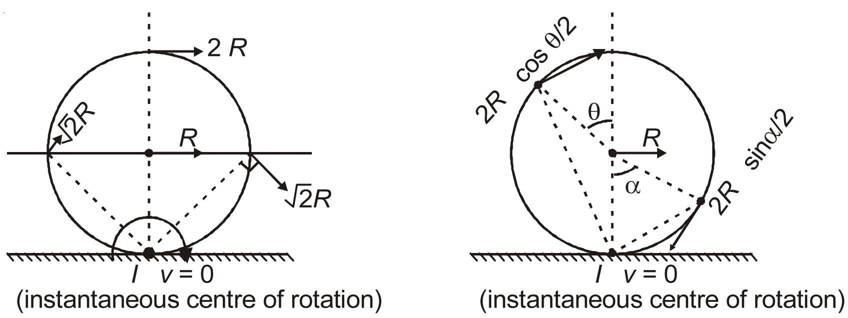
Case - II : Backward slipping

$v_{cm} < R$



***Case - III :* Pure Rolling**

$v_{cm} = R$



Note : *In this case, friction is static. It may or may not be zero.*

Kinetic energy of the body during pure rolling (E)

E = Translational KE + Rotational KE

$= E_T + E_R$

$= \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm} \quad ^2$

$= \frac{1}{2}\left(m + \frac{I_{cm}}{R^2}\right)v_{cm}^2 = \frac{1}{2}mv_{cm}^2\left(1 + \frac{K^2}{R^2}\right)$ (where k is radius of gyration)

$$E = E_T\left(1 + \frac{K^2}{R^2}\right)$$

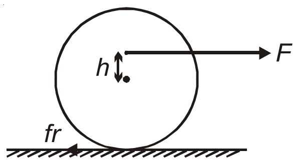
Similarly, $E = E_R\left(1 + \frac{R^2}{K^2}\right)$

Type of body	K	Fraction of total energy translational $X = \left(1 + \frac{K^2}{R^2}\right)^{-1}$	Fraction of total energy rotational $Y = \left(1 + \frac{R^2}{K^2}\right)^{-1}$	$\frac{Y}{X}$
1. Ring or hollow cylinder	R	$\frac{1}{2} = 0.5 = 50\%$	$\frac{1}{2} = 0.5 = 50\%$	1 : 1
2. Spherical Shell	$\sqrt{\frac{2}{3}}R$	$\frac{3}{5} = 0.6 = 60\%$	$\frac{2}{5} = 0.4 = 40\%$	2 : 3
3. Disc or solid cylinder	$\frac{R}{\sqrt{2}}$	$\frac{2}{3} = 0.666 = 66.67\%$	$\frac{1}{3} = 0.333 = 33.33\%$	1 : 2
4. Solid sphere	$\sqrt{\frac{2}{5}}R$	$\frac{5}{7} = 0.714 = 71.4\%$	$\frac{2}{7} = 0.286 = 28.6\%$	2 : 5

Note : *Above values X and Y are independent of mass and radius of the body. They only depend on the type of body.*

Applications

1. A force is applied at the distance h from centre of mass as shown in figure



$$a_{C.M} = \frac{F\left(1 + \frac{h}{R}\right)}{M\left(1 + \frac{K^2}{R^2}\right)}$$

$f_r = \frac{F(K^2 - hR)}{K^2 + R^2} \quad \mu N$ (must be less than μmg for Rolling)

If $\quad h \quad \frac{K^2}{R}$ friction is backward

$h \quad \frac{K^2}{R}$ friction become forward

2. If force is applied at centre of mass then (h = 0)

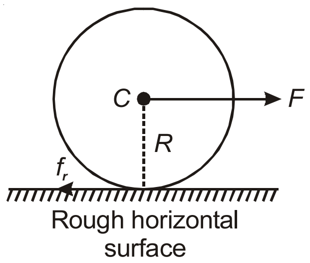
So, $a = \frac{F}{M\left(1 + \frac{K^2}{R^2}\right)}$ and $f_r = \frac{FK^2}{K^2 + R^2} = \frac{F}{\left(1 + \frac{R^2}{K^2}\right)}$

3. If force is applied at highest point ($h = R$)

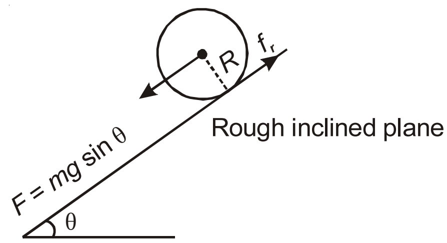
$$a_{\text{C.M.}} = \frac{2F}{M\left(1+\dfrac{K^2}{R^2}\right)}$$

$$f_r = \left(\frac{R^2 - K^2}{K^2 + R^2}\right)F \text{, forward direction}$$

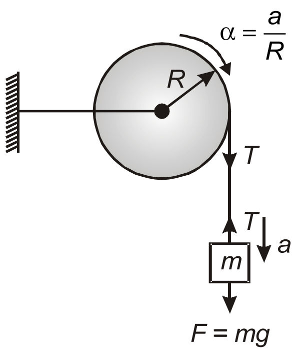
4. (i)



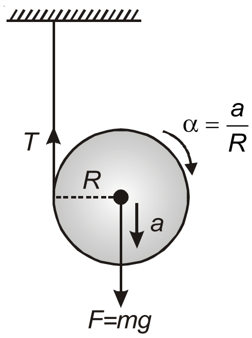
(ii)



(iii)



(iv)



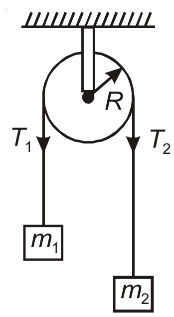
For all the four situations shown above,

$$a = \frac{F}{m + \dfrac{I_{\text{c.m.}}}{R^2}}$$

$$f_r \text{ or } T = \frac{I_{\text{c.m.}}a}{R^2}$$

In the situations described above, the linear acceleration of the moving object can be calculated by same formula, the value of F, and moment of inertia will depend on the kind of problem.

Also consider the following situation.



$$a = \frac{F}{(m_1 + m_2) + \dfrac{I}{R^2}}$$

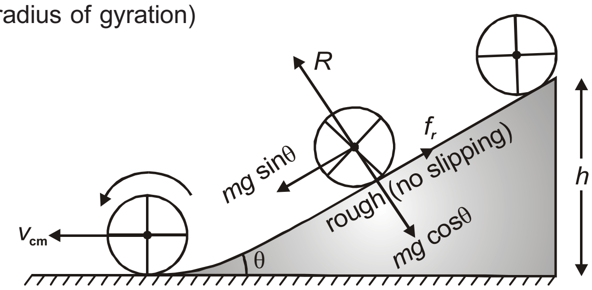
$$T_2 - T_1 = \frac{Ia}{R^2}$$

Here, $F = (m_2 - m_1)\,g$ = Net pulling force

Rolling of a Body on an Inclined Plane

By conservation of energy, $\underset{\text{(Total energy)}}{mgh} = \underset{\text{(Rotatory)}}{\frac{1}{2}I\ ^{2}} + \underset{\text{(Translatory)}}{\frac{1}{2}mv_{cm}^{2}}$

Let $\beta = 1 + \frac{I_{cm}}{MR^2} = 1 + \frac{K^2}{R^2}$ (where k is radius of gyration)



1. $a_{cm} = \dfrac{mg\sin\theta}{m + \dfrac{I_{cm}}{R^2}} = \dfrac{g\sin\theta}{1 + \dfrac{K^2}{R^2}} = \dfrac{g\sin\theta}{\beta}$

2. $v_{cm} = \sqrt{\dfrac{2gh}{\beta}} = \sqrt{2gh\Big/1 + \dfrac{K^2}{R^2}}$

3. Time $= \dfrac{1}{\sin\theta}.\sqrt{\beta\dfrac{2h}{g}}$ $\because$ Force of friction $f_r = \dfrac{mg\sin\theta}{1 + \dfrac{R^2}{K^2}}$

   i.e., $t \propto \sqrt{\beta}$

4. Force of friction $f_r = \dfrac{mg\sin\theta}{1 + \dfrac{R^2}{K^2}}$

5. Instantaneous power $P = (mg\sin\theta)v$

6. Maximum angle of inclination for pure rolling, $\theta_{max} = \tan^{-1}\left(\mu \times \left(1 + \dfrac{R^2}{K^2}\right)\right)$

Ring	:	$\theta_{max} = \tan^{-1}(2\mu)$,
Spherical Shell	:	$\theta_{max} = \tan^{-1}(2.5\ \mu)$
Disc	:	$\theta_{max} = \tan^{-1}(3\mu)$
Solid sphere	:	$\theta_{max} = \tan^{-1}(3.5\ \mu)$.

ANGULAR MOMENTUM

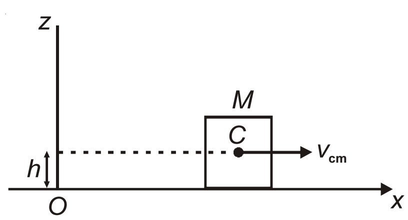
The general formula for angular momentum about any point is

$$\vec{L} = M\vec{v}_{cm}R + I\vec{\ }$$

Case - I : Pure translation

$$|\vec{L}_O| = Mv_{cm}h$$

$$L_C = 0$$



Case - II : Rolling body

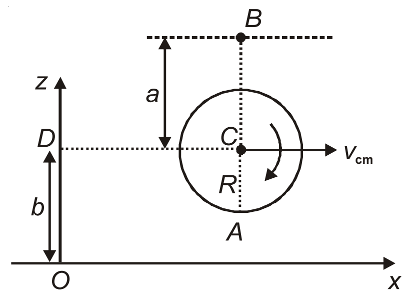
$L_c = I_c$

$L_A = I_C + Mv_{cm}R$

$L_O = I_c + Mv_{cm}b$

$L_B = I_c - Mv_{cm}a$

$L_D = I_c$



Case - III : Centre of mass is fixed

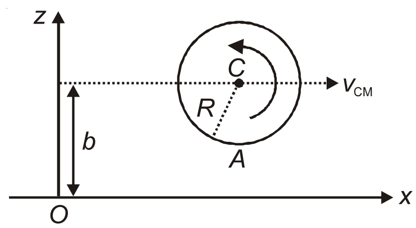
Put $v_{cm} = 0$ in the above results so $L_0 = L_A = L_C = I_C$

Case - IV

$L_c = - I_c$

$L_A = - I_C + Mv_{cm}R$

$L_O = - I_c + Mv_{cm}b$

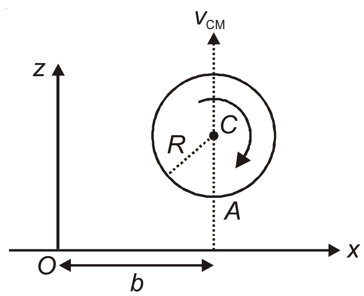


Case - V

$L_c = I_c$

$L_A = I_C$

$L_O = I_c - Mv_{cm}b$



Note : *In all above situations, anticlockwise sense has been assigned a negative sign.*

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Chapter 6

Gravitation

VARIATION IN THE VALUE OF *g*

1. At height *h* (Above earth's surface)

$$g' = \frac{g}{\left[1+\frac{h}{R_e}\right]^2}$$

If $h = R_e$, $g' = \frac{g}{4}$

If $h << R_e$ then $g' = g\left(1-\frac{2h}{R_e}\right)$

2. At depth *"x"* (below earth's surface)

$g' = g\left\{1-\frac{x}{R_e}\right\}$, at the centre of earth $g' = 0$, weight = 0

3. **Due to Rotation of Earth :**

Apparent value of acceleration due to gravity.

$g' = g - R_e\ {}^2\cos^2\lambda$

λ angle of latitude

GRAVITATIONAL FIELD INTENSITY AND POTENTIAL (*V*)

1. Gravitational field intensity

$$\vec{I} = \frac{-GM}{r^2}\hat{r}$$

2. Gravitational potential

$$V = -\frac{W}{m} \Rightarrow V = \frac{-GM}{r} \text{ (units J/kg)}$$

Variation of Intensity and Potential

1. For a spherical shell of mass M and radius R

Case-I : $r < R$ (internal point)

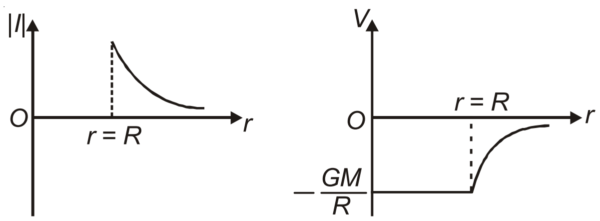
$$I_i = 0,\ V_i = -\frac{GM}{R}$$

Case-II : $r = R$ (on the surface)

$$I_s = \frac{GM}{R^2}, V_s = \frac{-GM}{R}$$

Case-III : $r > R$ (outside the shell)

$$I_o = \frac{GM}{r^2}, V_o = \frac{-GM}{r}$$



2. For Uniform solid Sphere

Case-I : $r < R$ (internal point)

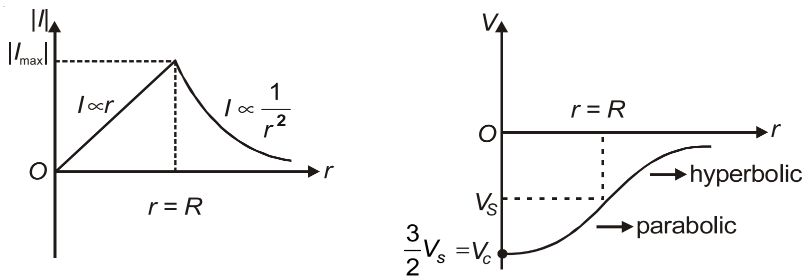
$$I_i = \frac{GMr}{R^3}, V_i = \frac{-GM}{2R^3}(3R^2 - r^2) \text{ At centre } V_c = -\frac{3GM}{2R} = \frac{3}{2}V_{\text{Surface}}$$

Case-II : $r = R$ (on the surface)

$$I_s = \frac{GM}{R^2},\ V_s = \frac{-GM}{R}$$

Case-III : $r > R$ (outside the surface)

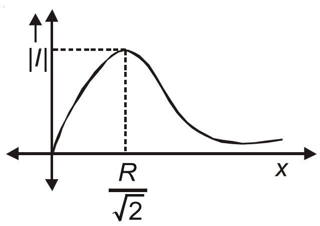
$$I_o = \frac{GM}{r^2},\ V_o = \frac{-GM}{r}$$

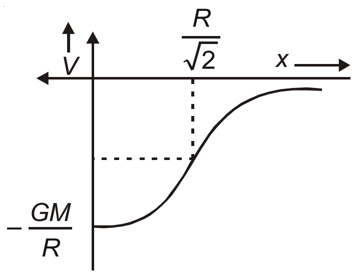


3. Gravitational intensity and potential on the axis of uniform ring of mass M radius R at distance x from centre.

$$I = \frac{GMx}{(R^2 + x^2)^{3/2}}, V = \frac{-GM}{\sqrt{R^2 + x^2}}$$

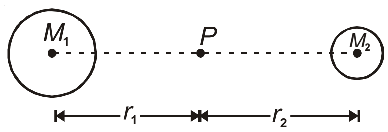
At centre I = 0. I is maximum at $x = \frac{R}{\sqrt{2}}$; $I_{\max} = \frac{2GM}{3\sqrt{3}R^2}$





4. **Neutral point :** The point P at which gravitational field is zero between two massive bodies, is called neutral point.

$$\boxed{\frac{M_1}{r_1^2} = \frac{M_2}{r_2^2}}$$



GRAVITATIONAL POTENTIAL ENERGY

At earth surface $U = -\frac{GM_e m}{R_e}$, At height h, $U = \frac{-GM_e m}{R_e + h}$

Energy required to escape = Escape energy = + $\frac{GM_e m}{R_e}$ = Binding energy.

ESCAPE VELOCITY

$$\Rightarrow \quad v_e = \sqrt{\frac{2GM_e}{R_e}} = \sqrt{\frac{8}{3}\pi G R_e{}^2} = \sqrt{2gR_e}$$

At earth surface, v_e = 11.2 km/s

KEPLER'S LAWS

(1) All planets revolve around the Sun in elliptical orbit having the Sun at one focus.

If e = eccentricity of ellipse then distance of the planet from the Sun at perigee is

$r_p = (1 - e)a$

and distance of the planet from the Sun at apogee is

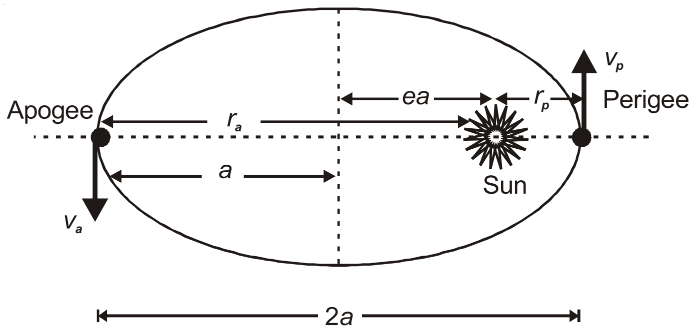
$r_a = (1 + e)a$ (a = semi major axis)

Ratio of orbital speeds at apogee and perigee is

$$\frac{v_a}{v_p} = \frac{r_p}{r_a} = \frac{1-e}{1+e}$$

Ratio of angular velocities at apogee and perigee is

$$\frac{\ _{a}}{_{p}} = \left(\frac{r_p}{r_a}\right)^2 = \left(\frac{1-e}{1+e}\right)^2$$



(2) A planet sweeps out equal area in equal time interval *i.e.,* Areal speed of the planet is constant

$$\frac{dA}{dt} = \frac{1}{2}vr = \frac{L}{2m} = \text{constant } (L \text{ represents angular momentum of planet about the Sun})$$

(3) Square of time period is proportional to cube of semi-major axis of the elliptical orbit of the planet.
i.e., $T^2 \propto a^3$

SATELLITES

Important results regarding satellite motion in circular orbit.

1. **Orbital Velocity (v_0) :** Gravitational attraction of planet gives necessary centripetal force.

$$\Rightarrow \quad v_0 = \sqrt{\frac{GM}{r}}$$

$$v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{gR_e^2}{R_e + h}} \qquad (h = \text{height above the surface of earth})$$

$$\Rightarrow \quad v_e = \sqrt{2}v_0$$

2. **Time Period :** The period of revolution of a satellite is

$$T = \frac{2\pi r}{v_0} = 2\pi r\sqrt{\frac{r}{GM}} = 2\pi\sqrt{\frac{r^3}{GM}}$$

For a satellite orbiting close to the earth's surface ($r \simeq R_e$), the time period is minimum and is given by

$$T_{\min} = 2\pi\sqrt{\frac{R_e^3}{GM}} = 2\pi\sqrt{\frac{R_e}{g}}$$

For earth R_e = 6400 km,

g = 9.8 m/s^2

T_{min} = 84.6 min.= 1.4 h

Thus, for any satellite orbiting around the earth, its time period must be more than $2\pi\sqrt{\frac{R}{g}}$ or 84.6 minutes.

3. Potential Energy (U), Kinetic Energy (K) and Total Energy (E) of satellite

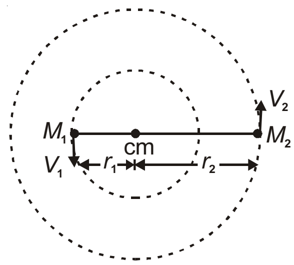
$$U = -\frac{GMm}{r}$$

$$K = \frac{GMm}{2r}$$

$$E = -\frac{GMm}{2r} \quad [K = -E \ \& \ U = 2E]$$

BINARY STAR SYSTEM

Two stars of mass M_1 and M_2 form a stable system when they move in circular orbit about their centre of mass, under their mutual gravitational attraction.



(1) $F = \frac{GM_1M_2}{r^2}$, where r is distance between them (*i.e.,* $r = r_1 + r_2$)

(2) $M_1r_1 = M_2r_2$

(3) $$\frac{GM_1M_2}{r^2} = \frac{M_1V_1^2}{r_1}$$

(4) $$\frac{GM_1M_2}{r^2} = \frac{M_2V_2^2}{r_2^2}$$

(5) $$r_1 = \frac{M_2r}{M_1 + M_2}, \ r_2 = \frac{M_1r}{M_1 + M_2}$$

$$\therefore \quad V_1 = M_2\sqrt{\frac{G}{(M_1 + M_2)r}},$$

$$V_2 = M_1\sqrt{\frac{G}{(M_1 + M_2)r}}$$

when $M_1 = M_2$

$$V_1 = V_2 = \sqrt{\frac{GM}{2r}}$$

$$r_1 = r_2 = \frac{r}{2}$$

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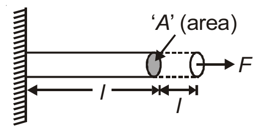
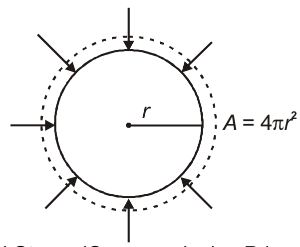
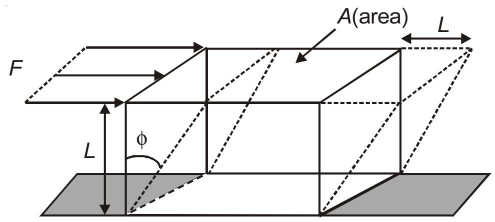
Chapter 7

Properties of Solids and Liquids

Strain	Stress
(1) Longitudinal strain = $\frac{l}{l}$	(1)  Normal Stress (Tensile) = F/A
(2) Volumetric strain = $-\frac{V}{V}$	(2)  Normal Stress (Compressive) = P (pressure)
(3) Shear strain = $\phi = \frac{L}{L}$	(3)  Tangential Stress or Shear Stress = $\frac{F}{A}$

MODULI OF ELASTICITY

(1) Young's modulus of elasticity $Y = \frac{\text{Tensile stress}}{\text{Longitudinal strain}} = \frac{Fl}{A\ l}$

(2) Bulk modulus of elasticity $\beta = \frac{\text{Normal or compressive stress}}{\text{Volumetric strain}} = -V\frac{P}{V}$ or, $\beta = -V\frac{dP}{dV}$

Compressibility = $\frac{1}{\beta}$

(3) Modulus of rigidity or shear modulus or $G = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{F}{A\phi} = \frac{FL}{A\ L}$

Applications :

1. For a wire $Y = \frac{Fl}{A\ \ l} \Rightarrow F = \frac{YA}{l}\ \ l$

   i.e. a wire behaves like a spring of spring constant (k)

   $$k = \frac{YA}{l} \qquad \left(i.e.,\ k \propto \frac{1}{l}\right)$$

2. When this wire is stretched by applying an external force F, and $\ \ l$ is extension produced, then

   (a) Work done by external force = $F\ \ l$

   (b) Work done by restoring force = $\frac{1}{2}F\ \ l$

   (c) Heat produced = $\frac{1}{2}F\ \ l$

   (d) Elastic potential energy stored = $\frac{1}{2}F\ \ l$

   Energy density $U = \frac{1}{2}\frac{F\ \ l}{\text{volume}} = \frac{1}{2}\frac{F\ \ l}{Al}$

   $$= \frac{1}{2}\ \text{stress} \times \text{strain}$$

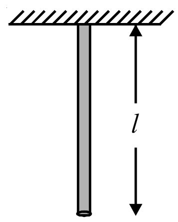
   $$= \frac{1}{2}\frac{(\text{stress})^2}{Y} = \frac{1}{2}Y(\text{strain})^2$$

3. A rod of mass m and length l hangs from a support

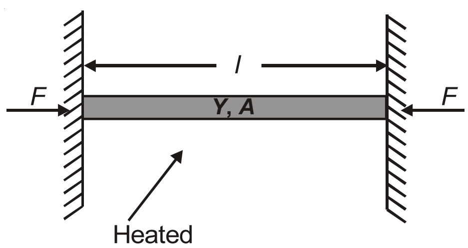
   Area of cross-section = A

   Extension produced due to its own weight,

   $$l = \frac{Mgl}{2AY} = \frac{\ \ gl^2}{2Y} \quad (\ \ = \text{density of wire})$$



Thermal Stress : Rod Fixed between Rigid Support



If $\ \ \theta$= Rise in temperature

Compressive strain = $\frac{\ \ l}{l} = \alpha\ \ \theta$

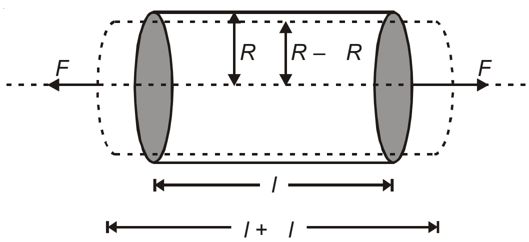
Compressive stress = Y × strain = $Y\alpha\ \ \theta$

$\Rightarrow \quad F = Y\alpha\ \ \theta \times A$

> ***Note :*** *If the rod is placed on horizontal frictionless surface, then stress developed on heating is zero.*

Poisson's Ratio

Consider a uniform bar being stretched by applying two forces at its ends.



Longitudinal strain $= \frac{l}{l}$

Lateral strain $= -\frac{R}{R}$

Poisson's ratio, $= -\frac{R / R}{l / l}$

(a) Theoretically – 1 0.5

(b) Practically 0 0.5

(c) When density of material is constant ⇒ = 0.5

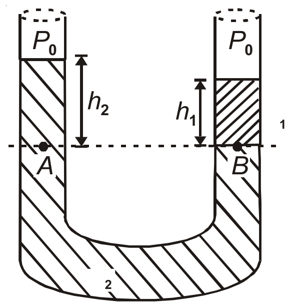
Equilibrium of Different Liquids in a U tube

1. $P_A = P_B$ (as A & B are at same level in same liquid)

 ⇒ $P_0 + h_1 \ {}_1 g = P_0 + h_2 \ {}_2 g$ (where P_0 is atmospheric pressure)

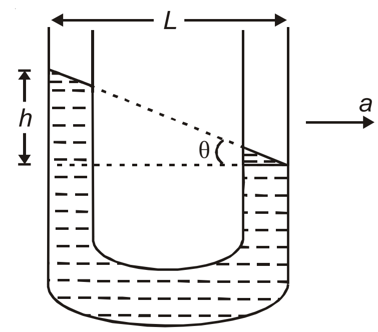
 ⇒ $h_1 \ {}_1 g = h_2 \ {}_2 g$

 ⇒ ${}_1 h_1 = {}_2 h_2$



2. When the U tube accelerates horizontally, difference of levels of liquid satisfies the relation,

$$\tan\theta = \frac{a}{g} = \frac{h}{L}$$



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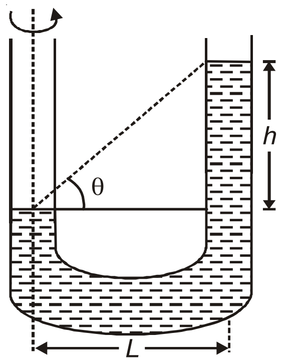
3. When U-tube is rotated about on limb



Here $h = \frac{{}^2L^2}{2g}$ and $\tan\theta = \frac{h}{L} \Rightarrow \tan\theta = \frac{{}^2L}{2g}$.

Excess pressure

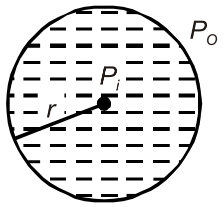
If P_o = Atmospheric pressure

P_i = Inside pressure

then $P_i - P_o$ = Excess pressure

Liquid drop

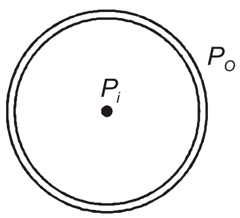
(1)



$$P_i = P_o + \frac{2T}{r}$$

Soap bubble

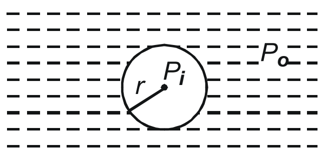
(2)



$$P_i - P_o = \frac{4T}{r}$$

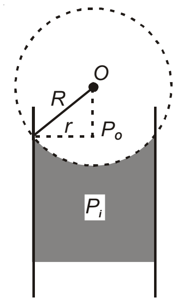
Air bubble

(3)



$$P_i = P_o + \frac{2T}{r}$$

(4) Capillary tube, concave meniscus

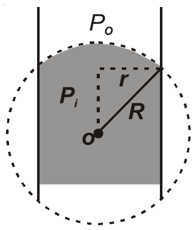


Capillary tube, Concave Meniscus

(a) $P_i = P_o - \frac{2T}{R}$

(b) $F_a \quad \frac{F_c}{\sqrt{2}}$

(5) Capillary tube, convex meniscus.



Convex Meniscus

(a) $P_i = P_o + \frac{2T}{R}$

(b) $F_a \quad \frac{F_c}{\sqrt{2}}$

Combining of Bubbles

1. If the soap bubble coalesce in vacuum, then $P_o = 0$

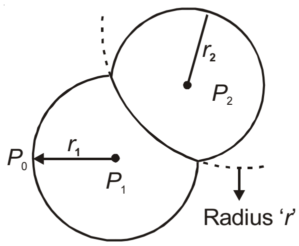
   $\Rightarrow \quad r^2 = r_1^2 + r_2^2$

2. If two soap bubbles come in contact to form a double bubble then

   r = radius of interface, $r_1 > r_2$

   $\frac{1}{r} = \frac{1}{r_2} - \frac{1}{r_1}$

   The interface will be convex towards larger bubble and concave towards smaller bubble because $P_2 > P_1 > P_0$.



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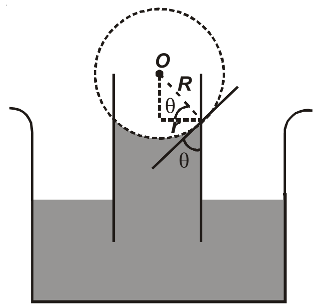
CAPILLARY ACTION

Rise or fall of liquid in a tube of fine diameter.



Ascent formula

$$h = \frac{2T}{R\ \ g} = \frac{2T\cos\theta}{r\ \ g}$$

Energy of a Liquid

Various energies per unit mass :

1. Potential energy/mass = gh
2. Kinetic energy/mass = $\frac{1}{2}v^2$
3. Pressure energy/mass = $\frac{P}{\ }$

Energy Heads

Various energy heads per unit mass :

1. Gravitational head = h
2. Velocity head = $\frac{v^2}{2g}$
3. Pressure head = $\frac{P}{\ \ g}$

BERNOULLI'S THEOREM

It is based on conservation of energy.

For an ideal, non-viscous and incompressible liquid,

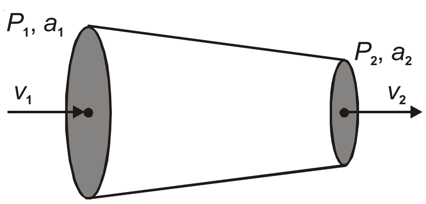
$$\frac{P_1}{\ } + \frac{v_1^2}{2} + gh_1 = \frac{P_2}{\ } + \frac{v_2^2}{2} + gh_2 = \text{constant}$$

Applications of Bernoulli's Theorem

(1) To find rate of flow of liquid $Q = av$. Value of Q in various cases has been given below

Case - (a) :

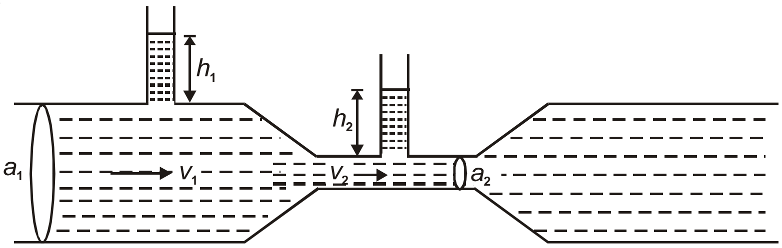
$$Q = a_1a_2\sqrt{\frac{2(P_1 - P_2)}{\ (a_1^2 - a_2^2)}}$$



Case - (b) :

Venturimeter

$$Q = a_1 a_2 \sqrt{\frac{2g(h_1 - h_2)}{a_1^2 - a_2^2}}$$

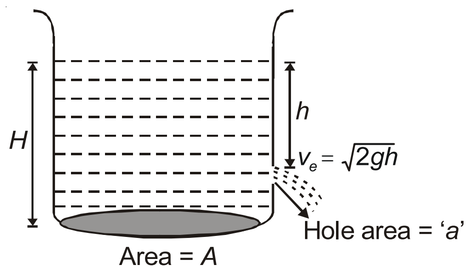


(2) Hole in a tank

(a) Speed of efflux $v_e = \sqrt{2gh}$ (If $a << A$). This is known as Toricelli's theorem.

If a is comparable to A then

$$v_e = \sqrt{2gh}\sqrt{\frac{A^2}{A^2 - a^2}}$$



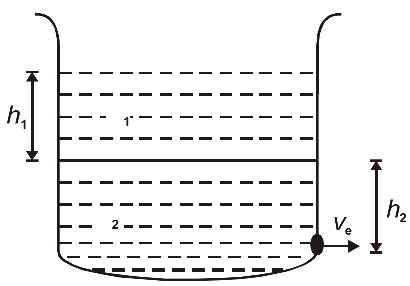
(b) Time taken by water level to fall from h_1 to h_2

$$t = \frac{A}{a}\sqrt{\frac{2}{g}}\ \sqrt{h_1} - \sqrt{h_2}$$

(c) Time taken to completely empty the container by a hole at bottom

$t \propto \sqrt{H}$ [Put $h_1 = H$, $h_2 = 0$]

(d) $v_e = \sqrt{2g\left(h_2 + \frac{{}_1h_1}{2}\right)}$ in the situation shown in figure

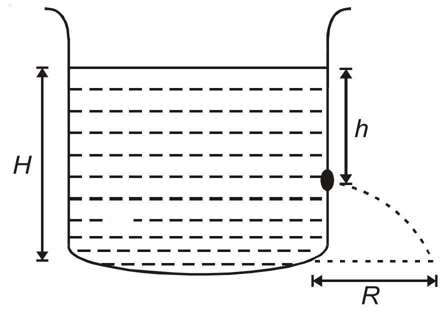


(e) Range of liquid

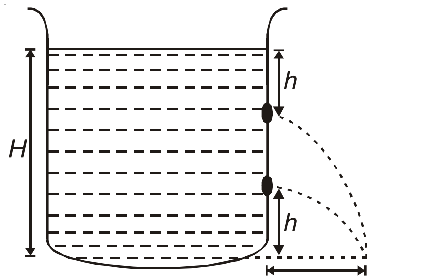
$R = 2\sqrt{h(H-h)}$

For a given value of total height (H)

$R_{max} = H$ when $h = \dfrac{H}{2}$



(f)



$R = 2\sqrt{h(H-h)}$ for both holes

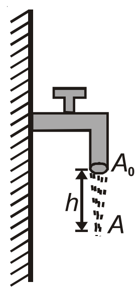
(3) If A_0 = area of cross-section of mouth of tap

A = area of cross-section of water jet at a depth h

$A_0 v_0 = Av = Q$ [rate of flow]

By Bernoulli's theorem $\dfrac{v_0^2}{2} + gh = \dfrac{v_1^2}{2}$

[$\because$ pressure is atmospheric at both points]



Reynold's Number

$$N_R = \frac{vD}{\ } = \frac{\text{Inertial Force}}{\text{Viscous force}}$$

Value of N_R for various cases :

(1) $N_R < 2000$, flow is streamline

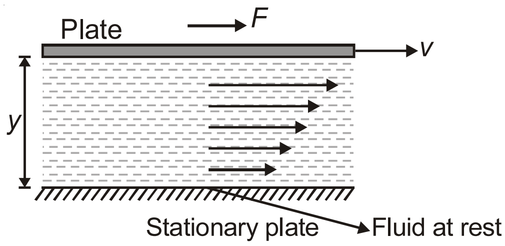
(2) $N_R > 3000$, flow is turbulent

(3) $2000 < N_R < 3000$, flow is unstable

(4) When $N_R = 2000$, flow is critical

$$\frac{vD}{\ } = 2000 \Rightarrow v_c = 2000\frac{\ }{D} \text{ (Critical velocity)}$$

Viscosity and Viscous Force



Viscous force is given in this case by,

$$F = - \quad A\frac{dv}{dy}$$

Units of : SI 1 Pa.s = 10 poise = 1 decapoise

C.G.S 1 dyne/cm^2-s = 1 poise

Poiseuille's Equation

Volume flow rate across a tube with pressure difference between its ends is,

$$Q = \frac{dV}{dt} = \frac{\pi}{8}\frac{Pr^4}{l}$$

Series combination of two tubes

Two tubes of radius r_1, length l_1 and radius r_2, length l_2 are connected in series across a pressure difference of P. Length of a single tube that can replace the two tubes is found using,

$$\frac{l}{r^4} = \frac{l_1}{r_1^4} + \frac{l_2}{r_2^4}$$

STOKES LAW

When a small spherical body of radius r is moving with velocity v through a perfectly homogeneous medium having coefficient of viscosity , it experiences a retarding force given by

$F = 6\pi\ rv$.

Important case :

(1) A body of radius r released from rest in a fluid

If = density of body

 = density of liquid or fluid

Terminal velocity is given by,

$$v_T = \frac{2}{9}\frac{r^2 g}{}(\quad - \quad)$$

Thermal Expansion

When the temperature of a body increases, its all dimensions (length, area, volume) increase.

(1) Coefficient of linear expansion is given by

$$\alpha = \frac{L}{L\ \theta}$$

$L_\theta = L_0 (1 + \alpha\ \theta)$, θ is the change in temperature in °C or K & L_0 is initial length

(2) Coefficient of superficial expansion is given by

$$\beta = \frac{A}{A\ \theta}$$

$$A_\theta = A_0\,(1 + \beta\ \ \theta)$$

(3) Coefficient of cubical expansion is given by

$$= \frac{V}{V\ \theta} \qquad \text{or} \qquad V_\theta = V_0\,(1 + \quad \theta)$$

$$\frac{m}{\ _\theta} = \frac{m}{\ _0}(1 + \quad \theta) \Rightarrow \quad _\theta = \ _0(1 - \quad)\theta$$

An Isotropic body expands equally in all directions and we can obtain the following relations

$$= 3\alpha,\ \beta = 2\alpha \ \text{ or } \ \boxed{\frac{\alpha}{1} = \frac{\beta}{2} = \frac{\ }{3}},\quad = \frac{3}{2}\beta$$

CALORIMETRY

(1) Specific heat capacity or Gram specific heat (c) : If Q heat is given to a substance of mass 'm', and rise in temperature is θ then

$$c = \frac{Q}{m\ \theta}\ \text{(cal/g°C)}$$

(2) Molar heat capacity (C) :

C = Molar mass (M) × specific heat capacity

$$\Rightarrow\ C = Mc = \frac{M.Q}{m\ \theta} = \frac{Q}{n\ \theta}$$

$$\boxed{C = \frac{Q}{n\ \theta}} : n = \frac{m}{M} \text{ is number of moles}$$

(3) Heat capacity of an object is defined as product of mass and specific heat.

(4) In general if Q heat is given to a substance of mass 'm' which increases its temperature by θ then

$\boxed{Q = mc\ \theta}$ c is specific heat capacity

or $\boxed{Q = nC\ \theta}$ C is molar heat capacity, n is number of moles of the substance.

Specific Latent heat

(1) Latent heat of fusion $L_f = \dfrac{Q}{m}$

(2) Latent heat of vaporisation $L_v = \dfrac{Q}{m}$

During phase change (liquid $\rightleftarrows$ solid or liquid $\rightleftarrows$ vapours) temperature remains constant, but internal energy changes.

Water

Specific heat C = 1 cal/gm/°C = 4.2 J/gm/°C = 4200 J/kg/°C

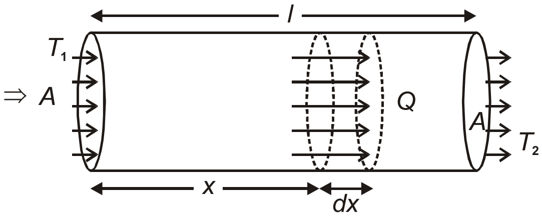
L_f = 80 cal/gm = 336 J/gm

L_v = 540 cal/gm = 2268 J/gm

For ice : C_{ice} = 0.5 cal/g °C = 2100 J/kg °C

Law of Conduction

Consider a rod of length l, cross sectional area A, with its ends maintained at temperatures T_1 & T_2 ($T_1 > T_2$). In steady state



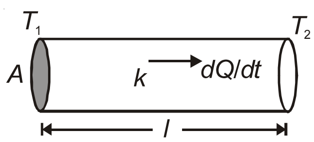
Rate of heat flow across any section is given by

$$H = \frac{dQ}{dt} = -kA\frac{dT}{dx}$$

Here k = Thermal conductivity and $\frac{dT}{dx}$ is known as temperature gradient i.e. rate of change of temperature with distance. k depends only on the nature of the material.

S.I. units of thermal conductivity is $Wm^{-1}K^{-1}$

THERMAL RESISTANCE OF A ROD



In steady state $\frac{dQ}{dt} = kA\frac{(T_1 - T_2)}{l}$

Thermal resistance, $R = \frac{l}{kA}$ $\left[\text{as in current electricity } R = \frac{l}{A} = \frac{l}{A}\right]$

Weidmann – Franz law

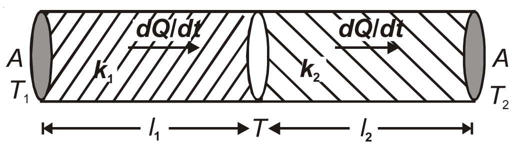
$\frac{k}{T}$ = constant ⇒ a substance which is good conductor of heat (silver) is also a good conductor of electricity (mica & human body is exception to above law)

where is electrical conductivity

Composite Rod :

(1) Series

In steady state



$$R = \frac{l_1}{k_1 A} + \frac{l_2}{k_2 A} = \frac{l_1 + l_2}{kA}$$

Where k = effective thermal conductivity given by

For same area of cross section,

$$k = \frac{l_1 + l_2 + ... + l_n}{\frac{l_1}{k_1} + \frac{l_2}{k_2} + ... + \frac{l_n}{k_n}}$$

For two slabs of equal length

$$k = \frac{2k_1k_2}{k_1 + k_2}$$

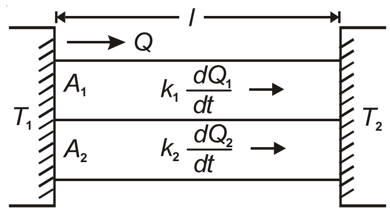
Temperature of junction

$$T = \frac{\frac{k_1}{l_1}T_1 + \frac{k_2}{l_2}T_2}{\frac{k_1}{l_1} + \frac{k_2}{l_2}}.$$

For same geometrical dimensions,

$$T = \frac{k_1T_1 + k_2T_2}{k_1 + k_2}$$

(2) In parallel



$$\frac{dQ}{dt} = \frac{dQ_1}{dt} + \frac{dQ_2}{dt}$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \quad \Rightarrow \frac{k(A_1 + A_2)}{l} = \frac{k_1A_1}{l} + \frac{k_2A_2}{l}$$

where k = effective coefficient of thermal conductivity given by

$$k = \frac{k_1A_1 + k_2A_2}{A_1 + A_2}$$

Example, for two slabs of equal area $k = \frac{k_1 + k_2}{2}$

STEFAN'S LAW

The radiant energy emitted by a perfectly black body per second per unit area (emissive power) is directly proportional to the fourth power of the absolute temperature of the body.

$R \propto T^4$ $\qquad$ $R = \ \ T^4$

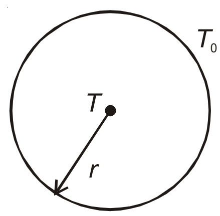
$R = \frac{\text{Power}}{\text{Area}} \Rightarrow$ P = $A \ \ T^4$ $\qquad$ ($\ \ = 5.67 \times 10^{-8}\ \text{Wm}^{-2}\ \text{K}^{-4}$)

For other bodies $P = \varepsilon A \ \ T^4$, ε is emissivity of the body.

Rate of heat loss

For a sphere of radius r at a temperature T placed in a surrounding of temperature T_0, the rate of heat loss is $\frac{dQ}{dt} = 4\pi r^2 \varepsilon \ \ (T^4 - T_0^4)$, where ε is emissivity.

Rate of cooling

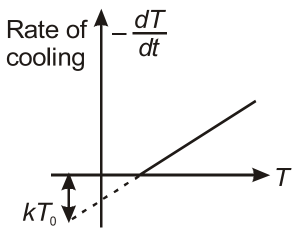


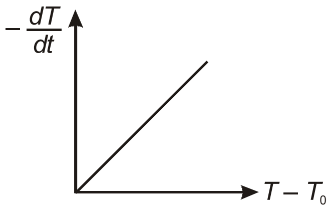
For a sphere of radius r, density and specific heat capacity s.

The rate of fall in temperature is given by $\frac{dT}{dt} = -\frac{3\varepsilon \ \ (T^4 - T_0^4)}{sr}$

Newton's Law of Cooling

If the temperature T of a body is not much different from surrounding temperature T_0, then rate of cooling of a liquid is directly proportional to the difference in the temperature of liquid T and temperature of surroundings (T_0) *i.e.*





$$\text{Rate of cooling} = \left(-\frac{dT}{dt}\right) \propto (T - T_0)$$

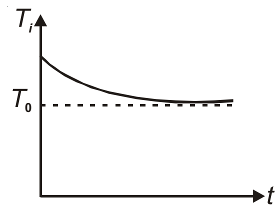
$$-\frac{dT}{dt} = \alpha(T - T_0)$$

Results

(1) $T_f = T_0 + (T_i - T_0)e^{-\alpha t}$, where T_i is initial temperature, T_f is temperature after time t.

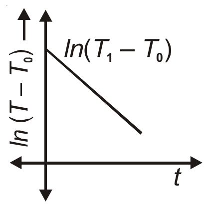
(2) Another form $\alpha t = \log\left|\frac{T_i - T_0}{T_f - T_0}\right|$

(3) $\frac{-dT}{dt} = \frac{4\varepsilon A\ T_0^3}{mc}(T - T_0)$, $\begin{bmatrix} m = \text{mass of body} \\ c = \text{specific heat} \\ A = \text{surface area} \\ \varepsilon = \text{emissivity} \end{bmatrix}$



(4) Another approximate formula is

$$\frac{T_1 - T_2}{t} = \alpha\left(\frac{T_1 + T_2}{2} - T_0\right)$$



Above formula gives time 't' taken by the body to cool down from T_1 to T_2. T_0 is temperature of surrounding.

(5) If temperature of a body changes from θ_1 to θ_2 in time 't' and changes from θ_2 to θ_3 in next time then

$$\frac{\theta_2 - \theta_0}{\theta_1 - \theta_0} = \frac{\theta_3 - \theta_0}{\theta_2 - \theta_0} \quad (\theta_0 = \text{temperature of environment})$$

(6) If equal masses of two liquids having same surface area and finish, cool from same initial temperature to same final temperature with same surrounding, then

$$\frac{t_1}{t_2} = \frac{k_2}{k_1} = \frac{c_1}{c_2}\ ; \qquad c_1\ \&\ c_2 \text{ are specific heats}$$

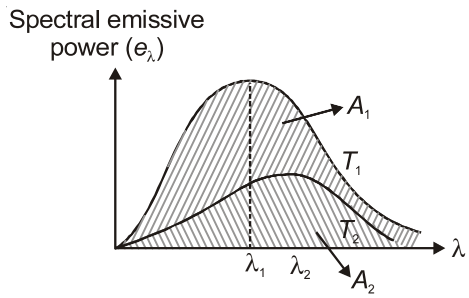
WIEN'S DISPLACEMENT LAW

This law states that the wavelength corresponding to maximum intensity for a black body is inversely proportional to the absolute temperature of the body

$$\lambda_m = \frac{b}{T}$$

where b is a constant known as Wien's constant

Results



$\boxed{\dfrac{\lambda_1}{\lambda_2} = \dfrac{T_2}{T_1}}$ $\boxed{\dfrac{A_1}{A_2} = \dfrac{T_1^4}{T_2^4}}$

(1) $\lambda_{max}\, T = b$

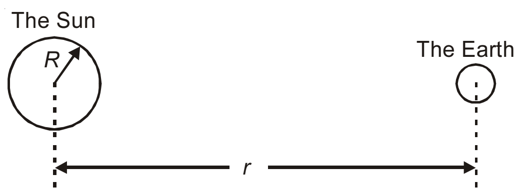
(2) $b = 2.898 \times 10^{-3}$ m-K

(3) Area under $e_\lambda - \lambda$ graph = T^4 (Total emissive power)

(4) If the temperature of the black body is made two fold, λ_{max} becomes half, while area becomes 16 times.

(5) Temperature of the Sun,

If T = temperature of sun, then total energy radiated by sun per second = $T^4\,(4\pi R^2)$



Intensity at distance r from the sun (*i.e.*, on earth)

$$I = \text{(Power radiated)(Area)} = \frac{T^4 R^2}{r^2} = S, \text{ where S is called solar constant } [S = 1.4 \text{ kW / m}^2]$$

$$\text{So, } T = \left[\left(\frac{r^2}{R^2}\right)\frac{S}{}\right]^{1/4} = \left[\left(\frac{1.5\times10^8}{7\times10^5}\right)^2 \times \frac{1.4\times10^3}{5.67\times10^{-8}}\right]^{\frac{1}{4}} \quad 5800 \text{ K}$$

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Chapter 8

Kinetic Theory of Gases and Thermodynamics

Pressure Exerted by the Gas

The pressure of the gas is due to continuous bombardment of the gas molecules against the walls of the container. According to kinetic theory, the pressure exerted by an ideal gas is given by

$$\boxed{P = \frac{1}{3}\frac{M}{V}\bar{v}^2}$$

M = Mass of the enclosed gas

V = Volume of the container

$\bar{v}^2$ = Mean square speed of molecules

or $P = \frac{1}{3}\frac{M}{V}v_{rms}^2$ $\quad$ V_{rms} = Root mean square velocity

or $P = \frac{1}{3}\frac{Nm}{V}\bar{v}^2$ $\quad$ N = Number of molecules

$$\boxed{P = \frac{}{3}v_{rms}^2}$$

= density of gas

m = Mass of the molecule

$\bar{v}^2$ = Mean square speed of molecules

Speeds of gas molecules :

(i) Root mean square speed,

$$v_{rms} = \sqrt{\frac{3RT}{M_w}} = \sqrt{\frac{3P}{}} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + \ldots\ldots v_n^2}{n}}$$

Here, M_W is molecular weight in kg.

$$v_{rms} = \sqrt{\frac{3\,kT}{m}}$$, k = Boltzmann's constant, m = mass of one molecule in kg

(ii) Average speed $v_{avg} = \sqrt{\frac{8RT}{\pi M_w}} = \sqrt{\frac{8P}{\pi}} = \frac{v_1 + v_2 + v_3 + \ldots\ldots\ldots + v_n}{n}$

(iii) v_{mp} = Most probable speed is defined as the speed corresponding to which there are maximum number of molecules.

$$v_{mp} = \sqrt{\frac{2RT}{M_w}} = \sqrt{\frac{2P}{\ }} = \sqrt{\frac{2\,kT}{m}}$$

= Density of gas
M_w = Molecular weight
R = Gas constant
P = Pressure of gas
m = Mass of one molecule

Order of magnitude : $v_{rms} > v_{avg} > v_{mp}$

$$v_{rms} : v_{av} : v_{mp} = \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2} \simeq \sqrt{3} : \sqrt{2.5} : \sqrt{2}$$

Relation between C_P & C_V :

(i) $C_P - C_V = R \qquad (C_P > C_V)$

(ii) $\dfrac{C_P}{C_V} =$

(iii) $C_v = \dfrac{f}{2}R$

Gas	Degrees of freedom (f)	$U = \frac{f}{2}nR\ \ T$	$C_V = \frac{U}{n\ \ T}$	C_P=C_V + R	$= \frac{C_P}{C_V}$
Monoatomic	3 (Translational)	$\frac{3}{2}nR\ \ T$	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3}$
Diatomic	3(Trans) + 2(Rot)	$\frac{5}{2}nR\ \ T$	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5}$
Non-Linear Poly atomic	3 (Trans) + 3 (Rot)	$3nR\ \ T$	$3\,R$	$4\,R$	$\frac{4}{3}$

For a mixture of two gases A and B containing n_A and n_B number of moles.

(i) $f_{\text{mix}} = \dfrac{(n_A f_A + n_B f_B)}{n_A + n_B}$

(ii) $_{\text{mix}} = 1 + \dfrac{2}{f_{\text{mix}}}$

$C_{v_{\text{mix}}} = \dfrac{f_{\text{mix}}R}{2}$

$C_{p_{\text{mix}}} = C_{v_{\text{mix}}} + R$

Thermodynamic Process

(1) Melting process : (Change of state, solid to liquid)

$Q = \ \ U + W$

$mL_f = \ \ U + 0$ [W = 0 as volume remains nearly constant]

(2) Boiling process : (Change of state, liquid to vapours)

$mL_v = \ \ U + P[V_2 - V_1]$

V_2 = volume of vapours

V_1 = volume of liquid

When 1 g of water vapourises isobarically at atmospheric pressure. U = 2091 J, $P = 1.01 \times 10^5$ Pa, V_1 = 1 cm^3, V_2 = 1671 m^3.

(3) Isochoric process : Volume is constant

$dV = 0 \Rightarrow W = 0$ [dV = change in volume]

$Q = nC_V\ T = \ U$

$$\Rightarrow \quad C_V = \frac{U}{n\ T}$$

(4) Isobaric process : Pressure is constant

P = constant, $dW = PdV$

$\Rightarrow \quad W = P\ V = nR\ T$

$Q = nC_P\ T = \ U + W$

$nC_P\ T = nC_V\ T + nR\ T$

$\Rightarrow \quad C_P = C_V + R$

$$\frac{U}{f} = \frac{W}{2} = \frac{Q}{f+2} \quad \text{or} \quad \frac{U}{1} = \frac{W}{-1} = \frac{Q}{}$$

Fraction of total heat converted to internal energy = $\dfrac{U}{Q} = \dfrac{1}{}$

Fraction of total heat converted to work is, $\dfrac{W}{Q} = \dfrac{-1}{}$

(5) Isothermal process : Temperature is constant

$PV = K \Rightarrow dT = 0 \Rightarrow dU = 0,\ C =$

as $PV = nRT$ (Constant)

So $P = \dfrac{nRT}{V}$

Work done in isothermal process

$$W = Q = nRT \log_e\left(\frac{V_2}{V_1}\right) = 2.303\ nRT \log_{10}\left(\frac{V_2}{V_1}\right) = 2.303\ nRT \log_{10}\left(\frac{P_1}{P_2}\right)$$

(6) Adiabatic process : Heat exchanged (Q) is zero

$PV \ = K$ [Equation of adiabatic process]

As $Q = 0$, $nC\ T = 0$ or $C = 0$

Also, $0 = nC_V\ \ T + W$ [by first law of thermodynamics]

Now, $\boxed{W = -\ U}$ i.e. $W = -\ nC_V\ \ T = -\dfrac{nR}{-1}\ (T_2 - T_1)$

(7) Polytropic Process

PV^x = Constant

$$W = \frac{nR\ T}{1-x}$$

Molar heat capacity $\boxed{C = C_V + \dfrac{R}{1-x}}$

(8) Cyclic process : System returns to its initial state (P, V & T)

For the overall process $T = 0 \Rightarrow U = 0$

$\Rightarrow \quad Q = W$

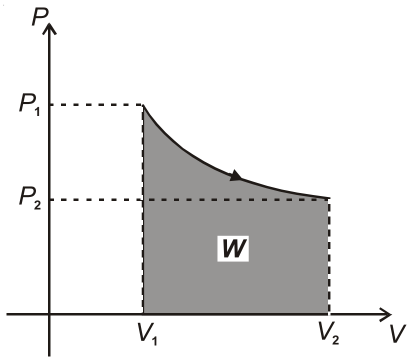
Indicator Diagram :

P-V graph of a process is called indicator diagram. Area under P-V graph represents the work done in a process.

Small work done, $dW = PdV$

Total work done, $W = \int dW = \int PdV$

= Area of curve ($P - V$) bounded with volume axis



CARNOT ENGINE

Heat supplied = Q_1

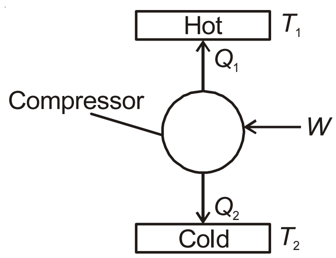
Heat rejected = Q_2 $\quad\Rightarrow\quad Q_1 - Q_2 = W$

% efficiency, $= \dfrac{W}{Q_{\text{supplied}}} \times 100 = \dfrac{Q_1 - Q_2}{Q_1} \times 100\%$

Coefficient of performance, $\beta = \dfrac{\text{Heat Supplied}}{W_{\text{total}}} = \dfrac{Q_1}{W} = \dfrac{Q_1}{Q_1 - Q_2}$

$\beta = \dfrac{T_1}{T_1 - T_2}$ (for ideal pump)

Refrigerator : In a refrigerator, W work is done on the working substance, Q_2 heat is absorbed from lower temperature T_2 and Q_1 heat is rejected to higher temperature T_1. ($T_1 > T_2$).



Coefficient of performance $\beta = \dfrac{\text{Heat exchange from sink}}{W_{\text{total}}} = \dfrac{Q_2}{Q_1 - Q_2}$

$\Rightarrow \beta = \dfrac{T_2}{T_1 - T_2}$

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Chapter 9

Oscillations and Waves

Periodic Function

If $f(t + T) = f(t)$ then function 'f' is periodic with period T.

Harmonic Motion

When oscillatory motion of a particle can be expressed in terms of sine or cosine functions, it is said to be a harmonic motion.

SIMPLE HARMONIC MOTION

When a motion can be expressed in terms of a single sine or cosine (sinusoidal) function, the motion is said to be Simple Harmonic Motion (SHM). For SHM, force ∝-(displacement)

$\Rightarrow$ $F \propto -x$

$\Rightarrow$ $F = -kx$ [Restoring Force]

$\Rightarrow$ $a = -\frac{k}{m}x$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \quad \text{or} \quad \boxed{\frac{d^2x}{dt^2} + {}^2x = 0}$$ (This equation represents the differential equation of S.H.M.)

Velocity and acceleration of a particle executing S.H.M.

If $x = A \sin \; t$, A is amplitude (maximum displacement from mean position)

$\Rightarrow$ $v = \frac{dx}{dt} = A \; \cos \; t$

or, $v = A \; \sin\left(\; t + \frac{\pi}{2}\right)$, maximum speed = A

i.e., **velocity leads displacement by** $\frac{\pi}{2}$. (This is always true in SHM)

(a) Dependence of velocity v with displacement from mean position (x)

$$v = \; \sqrt{A^2 - x^2}$$

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(b) Acceleration

$$a = \frac{dv}{dt} = -A\ ^2 \sin\ t$$, maximum acceleration = $A\ ^2$

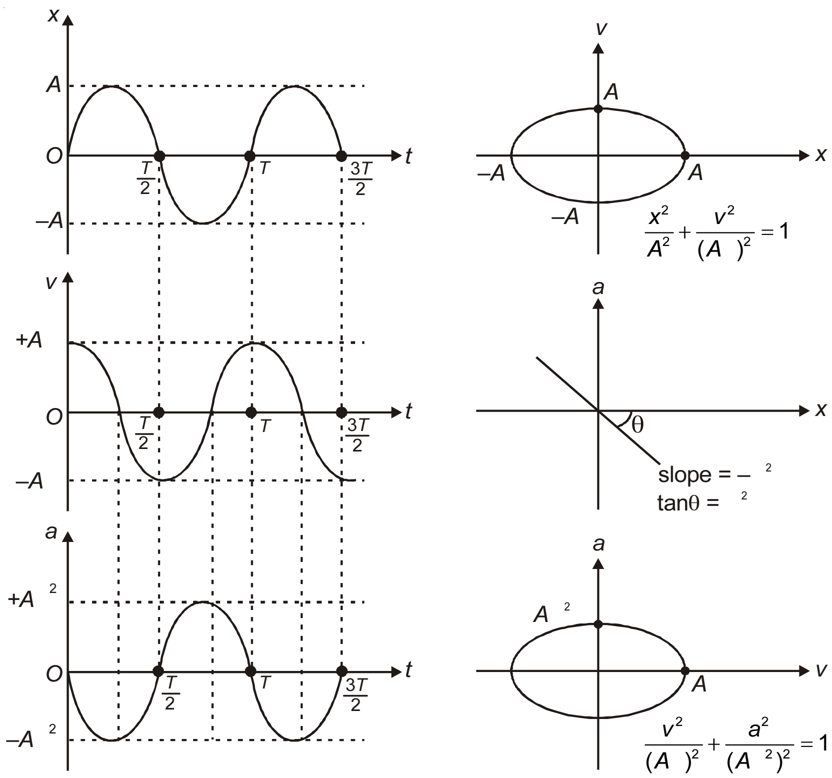
$\Rightarrow$ $a = A\ ^2 \sin(\ t + \pi)$

i.e., **acceleration leads velocity by $\frac{\pi}{2}$. Acceleration and displacement are in opposite phase**.

(c) Dependence of acceleration with position, is $a = -\ ^2x$

Graphical representation of variation of position, velocity and acceleration

(For $x = A \sin\ t$)



Energy in SHM

Salient points regarding energy in SHM :

Oscillating quantity	Time period	Frequency
Displacement	T	f
KE	$T/2$	$2f$
PE	$T/2$	$2f$
|KE ~ PE|	$T/4$	$4f$
Total Energy		0

1. $KE_{avg} = \frac{1}{4}m\ {}^2A^2$.

2. $KE_{max} = \frac{1}{2}m\ {}^2A^2$ at mean position.

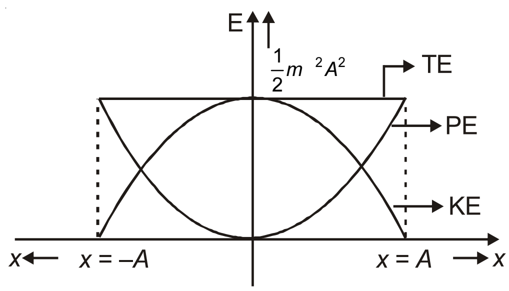
3. KE_{min} = zero at extreme position.

4. $PE_{avg} = \frac{1}{4}m\ {}^2A^2$.

5. $PE_{max} = \frac{1}{2}m\ {}^2A^2$ at extreme position.

6. Total energy, $E = \frac{1}{2}\ m\ {}^2A^2$ which is constant *i.e.* doesn't depend on *x*

7. Both kinetic and potential energy vary parabolically with *x*.



TE = Constant

TE = (1/2)$m\ {}^2A^2$

8. PE = KE at $x = \frac{A}{\sqrt{2}}$ and $t = \frac{T}{8}$. (Starting from mean position towards +*x*).

SIMPLE PENDULUM

Time period of oscillation of simple pendulum of length *l* for small angular amplitude is given by $T = 2\pi\sqrt{\frac{l}{g}}$

where *g* is the effective acceleration due to gravity, directed along the length of pendulum when it is at mean position.

SOME IMPORTANT POINTS :

On changing various factors, T changes as :

1. If length '*l*' is changed, $\frac{T}{T} = \frac{1}{2}.\frac{l}{l}$

2. If gravity '*g*' is changed, $\frac{T}{T} = -\frac{1}{2}.\frac{g}{g}$

Simple Pendulum in Lift

Effective $g' = |g + a|$, where $\vec{a}$ is pseudo acceleration. $T = 2\pi\sqrt{\frac{l}{g'}}$

Simple Pendulum of Length Comparable to the Radius of Earth

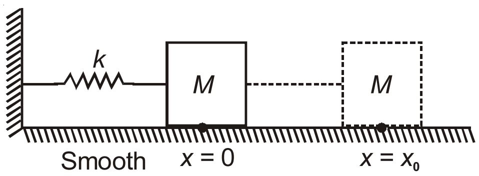
Time period of such a pendulum is given by,

$$T = 2\pi\sqrt{\frac{1}{g\left(\frac{1}{l}+\frac{1}{R_e}\right)}}$$

OSCILLATION OF SPRING

Horizontal Oscillations

The spring is pulled/pushed from $x = 0$ to $x = x_0$ and released.



The block executes SHM

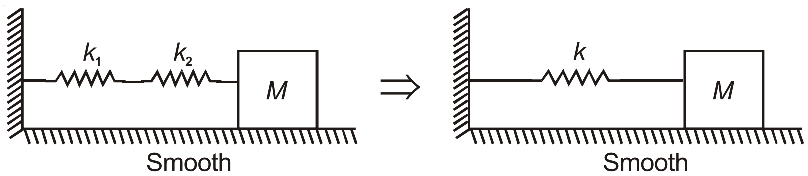
(1) Amplitude of oscillation = x_0

(2) Time period $T = 2\pi\sqrt{\frac{M}{k}}$

COMBINATIONS OF SPRINGS

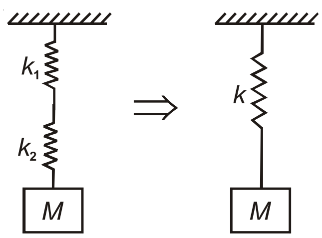
Series Combination :

(1)



$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$ or Effective spring constant, $k = \frac{k_1k_2}{k_1 + k_2}$, $T = 2\pi\sqrt{\frac{M}{k}}$

(2) $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$

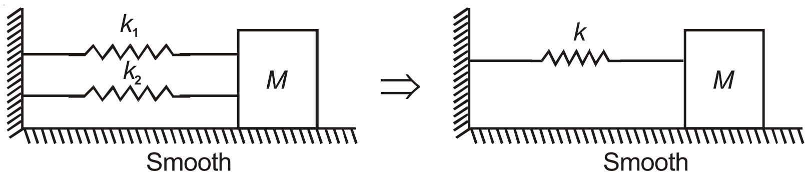


Effective spring constant, $k = \frac{k_1k_2}{k_1 + k_2}$

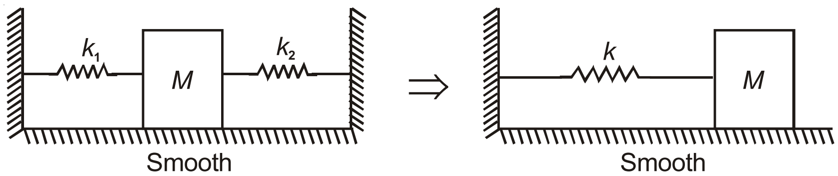
$T = 2\pi\sqrt{\frac{M}{k}}$

Parallel Combination

(1) Effective spring constant, $k = k_1 + k_2$, $T = 2\pi\sqrt{\dfrac{M}{k}}$

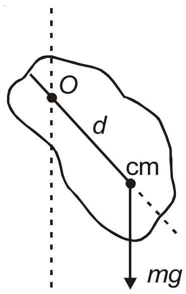


(2) Effective spring constant, $k = k_1 + k_2$ $T = 2\pi\sqrt{\dfrac{M}{k}}$



Physical Pendulum

Figure shows an extended body (called physical pendulum) pivoted about point O, which is at a distance d from its centre of mass.



Time period of oscillation, $T = 2\pi\sqrt{\dfrac{I}{mgd}}$ I = moment of inertia of the body about pivoted point.

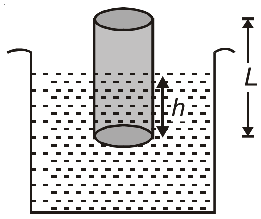
d is the distance of centre of mass from suspension point

Oscillation of a Floating Cylinder

If = density of cylinder material

= density of fluid (<)

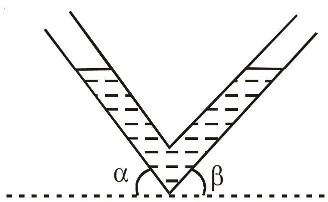
then, $T = 2\pi\sqrt{\dfrac{L}{g}} = 2\pi\sqrt{\dfrac{h}{g}}$



Oscillations of a Liquid in a Tube

$$T = 2\pi\sqrt{\frac{l}{g(\sin\alpha + \sin\beta)}}$$

l = total length of liquid column



Superposition of SHMs

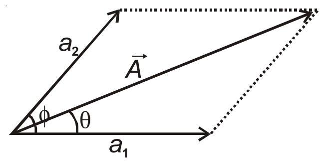
Consider two SHMs along the same line

If $y_1 = a_1 \sin \quad t$

$y_2 = a_2 \sin (\quad t + \phi)$

then, equation of resultant SHM is given by,

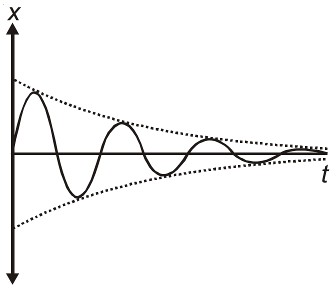
$y = y_1 + y_2 = A \sin (\quad t + \theta)$



where, $A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos\phi}$ & $\theta = \tan^{-1}\left(\frac{a_2 \sin\phi}{a_1 + a_2 \cos\phi}\right)$

Damped Oscillations

If there is any dissipative force like viscous force in SHM, then the amplitude of the particle decreases with time such type of oscillations are known as damped oscillations.



(i) Differential equation for damped oscillation $m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$,

where b = Coefficient of damping.

(ii) Displacement-time equation, $x = A(t) \sin(\quad t + \phi)$

(iii) Amplitude of damped oscillation, $A(t) = A_0 e^{-\frac{b}{2m}t}$, where A_0 = Initial amplitude.

(iv) Angular frequency of damped oscillation, $' = \sqrt{{}_0^2 - \left(\frac{b}{2m}\right)^2}$, where ${}_0 = \sqrt{\frac{k}{m}}$ =natural frequency.

Speed of mechanical waves

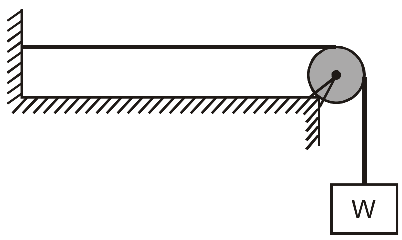
(a) Transverse wave in a stretched string

T = tension in the string

μ = mass per unit length

D = diameter of string

= density



$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{\text{stress}}{\text{density}}} = \frac{2}{D}\sqrt{\frac{T}{\pi}}$

$\boxed{\mu = A \quad = \frac{\pi D^2}{4}}$

$v = \sqrt{\frac{T}{A}}$

(b) Transverse wave in a long bar is given by

$v = \sqrt{\frac{Y}{\ }}$ where Y = Young's modulus,   = Density of material

(c) Longitudinal Waves

(i) In liquid $v = \sqrt{\frac{\beta}{\ }}$ β = bulk modulus of elasticity

  = density

(ii) In gases $v = \sqrt{\frac{\beta}{\ }}$. β is bulk modulus of the gas. This value is not a fixed value for gases

Case - I : Suggested by Newton

Taking isothermal process $\beta = P \Rightarrow v = \sqrt{\frac{P}{\ }}$

Put P = 1 atm,   = 1.23 kg/m^3 $\Rightarrow$ v = 280 m/s (more than 15% error)

Case - II : Corrected by Laplace

For Adiabatic $\beta = \ P$

$\Rightarrow$ $v = \sqrt{\frac{\ P}{\ }}$. Taking   = 1.4, we get, $\boxed{\text{For air } v = 20\sqrt{T}}$

v = 330 m/s

> **Note :** *Propagation of sound in air is adiabatic.*

Factors affecting speed of sound

$$v = \sqrt{\frac{\ RT}{M}} = \sqrt{\frac{\ P}{\ }}$$

(1) v is independent of pressure (If temperature is kept constant)

(2) $v \propto \frac{1}{\sqrt{\ }}$ or $v \propto \frac{1}{\sqrt{M}}$ (If temperature is kept constant)

(3) Velocity of a wave depends on medium, not on the frequency of source

(4) $v \propto \sqrt{T}$

(5) Velocity of sound in humid air is more because its density is less than that of dry air.

(6) Velocity of sound in humid hydrogen is less than in dry hydrogen due to similar reason.

SOUND WAVES

These are mechanical and longitudinal waves. They propagate in form of compressions and rarefactions. Particle displacements can be represented by wave function

$S = A \sin(\ t - kx)$

As particles oscillate, pressure variation takes place according to the wave function.

$P = \ P_0 \cos(\ t - kx)$, P_0 = maximum pressure variation

Characteristic of Sound

Loudness : Sensation of sound produced in human ear is due to amplitude. It depends upon intensity, density of medium, presence of surrounding bodies,

(a) Intensity of Wave

$$I = 2\pi^2 f^2 A^2 v$$

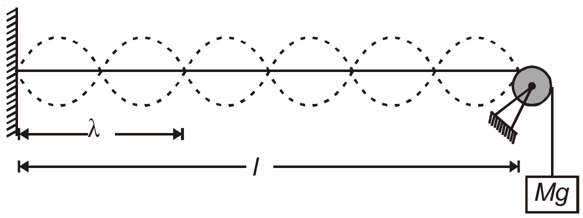
$$= \frac{1}{2} \;{}^2 A^2 \; v$$

$I \propto f^2$ and $I \propto A^2$

(b) Intensity Level or (Sound Level) (β)

$$\beta = 10\log_{10}\left(\frac{I}{I_0}\right) \qquad \begin{bmatrix} I_0 = \text{minimum intensity of audible sound} = 10^{-12}\,\text{W/m}^2 \\ I = \text{measured intensity} \end{bmatrix}$$

Sonometer : In this case, transverse stationary waves are formed.



The wire vibrates in n loops, then

$$l = \frac{n\lambda}{2} \text{ or } \lambda = \frac{2l}{n}$$

Velocity $v = \sqrt{\frac{T}{\mu}}$ where 'μ' is mass per unit length of wire.

$$\Rightarrow \quad \nu_n = \frac{v}{\lambda} = \frac{nv}{2l} = \frac{n}{2l}\sqrt{\frac{T}{\mu}}$$

Pipe length l	Fundamental Mode	Ist Overtone	$(n-1)^{th}$ overtone	Ratio of Successive frequency
Open	$\nu = \frac{V}{2l}$ Ist Harmonic	$\nu = \frac{V}{l}$ 2nd Harmonic	$\nu = n\frac{V}{2l}$ n^{th} Harmonic	1 : 2 : 3 : 4
Closed	$\nu = \frac{V}{4l}$ Ist Harmonic	$\nu = \frac{3V}{4l}$ 3rd Harmonic	$\nu = (2n-1)\frac{V}{4l}$ $(2n-1)^{th}$ Harmonic	1 : 3 : 5 : 7

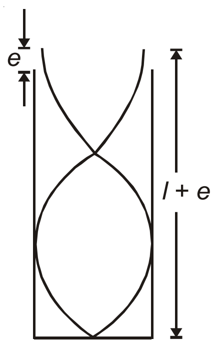
> ***Note :*** *Even numbered (i.e., 2nd, 4th .....) harmonics do not exist in close organ pipe.*

End correction (e) :

The antinodes are formed slightly out side the open end. The distance of antinode from open end of the pipe is called end correction. It depends on radius of pipe. (e = 0.6 r)

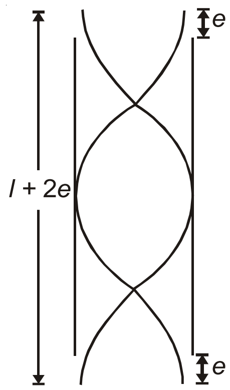
Thus, we have,

For closed organ pipe



$$\nu = \frac{(2n-1)V}{4(l+e)}$$

For open organ pipe



$$\nu = \frac{nV}{2(l+2e)}$$

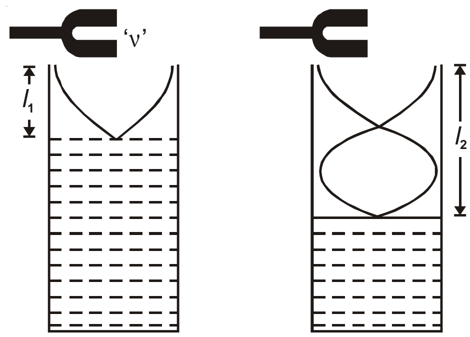
Resonance Tube:

If resonance is obtained first at length l_1.

then at length l_2, then

$\boldsymbol{\lambda = 2(l_2 - l_1)}$

$\Rightarrow$ distance between two successive lengths is $\frac{\lambda}{2}$



Interference

Consider two waves of same frequency and wavelength,

$y_1 = a_1 \sin(\ t - kx)$, $I_1 = Ca_1^2$

$y_2 = a_2 \sin(\ t - kx + \phi)$, $I_2 = Ca_2^2$

Equation of resultant wave is,

$$y = y_1 + y_2 = A\sin(\ t - kx + \theta), \text{ where } A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\phi} \text{ and } \theta = \tan^{-1}\left(\frac{a_2\sin\phi}{a_1 + a_2\cos\phi}\right)$$

Resultant Intensity is given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

DOPPLER'S EFFECT

If a wave source and a observer are moving relative to each other, the frequency observed by the receiver (f) is different from the actual source frequency (f_0) given by,

$$f = f_0\left(\frac{v \quad v_0}{v \mp v_s}\right)$$

where v = speed of sound, v_0 = speed of observer, v_s = speed of source

❑ ❑ ❑

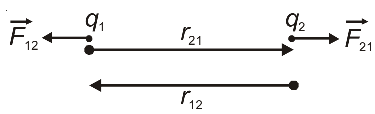
Chapter 10

Electrostatics

Coulomb's Law in Vector Form

$$\overrightarrow{F_{12}} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{|\overrightarrow{r_{12}}|^2}\hat{r}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{|\overrightarrow{r_{12}}|^3}\overrightarrow{r_{12}}$$

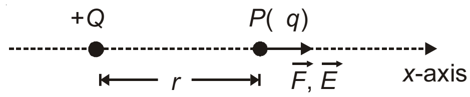
$$\overrightarrow{F_{21}} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{|\overrightarrow{r_{21}}|^2}\hat{r}_{21} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{|\overrightarrow{r_{21}}|^3}\vec{r}_{21}$$



ELECTRIC FIELD

Electric Field due to a Point Charge (*Q*) :

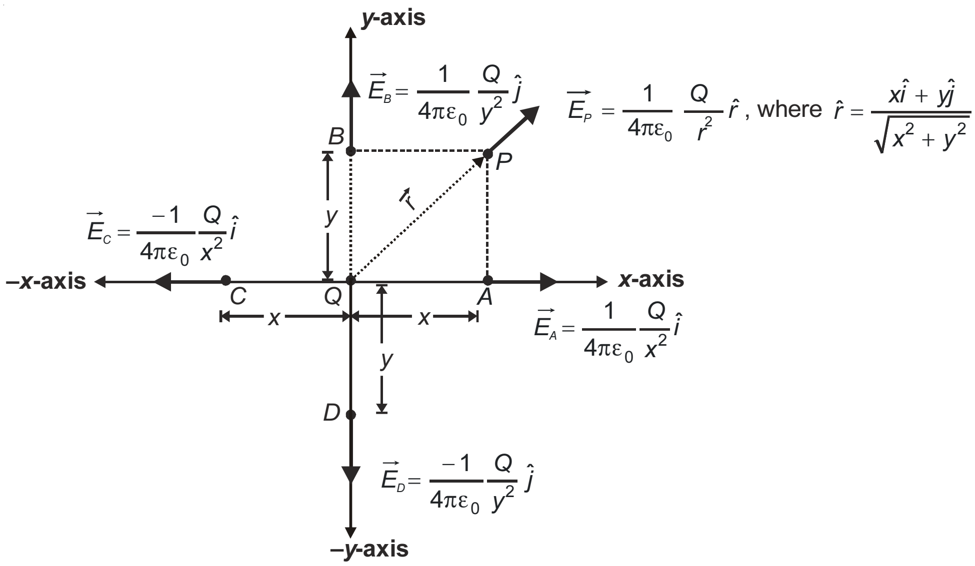
$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{Q\ q}{r^2}\hat{i}\ ,\quad \vec{E} = \frac{\vec{F}}{q} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{i}$$



$$\Rightarrow \quad \vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{i}$$

Application

(i) Direction of Electric Field at Various Points (when charge *Q* is placed at origin) :

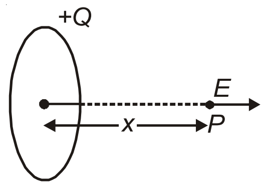


(ii) **Electric field due to a uniformly charged ring on its axis.**

$$E_{\text{axis}} = \frac{Qx}{4\pi\varepsilon_0 (R^2 + x^2)^{3/2}}$$

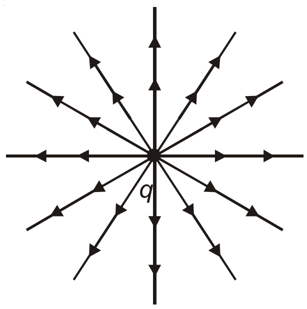
$E_{\text{centre}} = 0$

At $x = \frac{R}{\sqrt{2}}$, E is maximum.

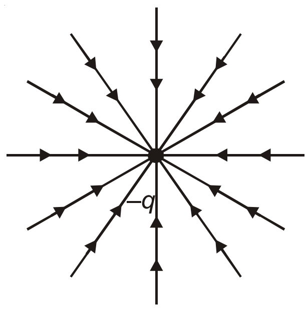


Electric Lines of Force due to Various Configurations

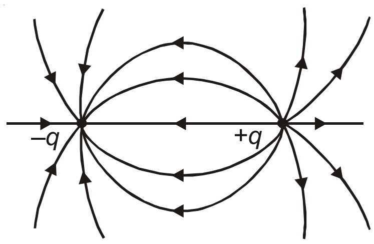
(1) Isolated point charge (+)



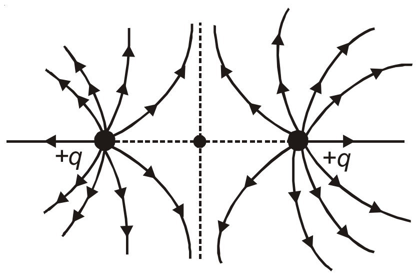
(2) Isolated point Charge (–)



(3) Electric dipole

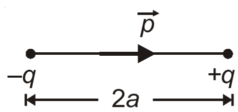


(4) Two identical charges



ELECTRIC DIPOLE

An arrangement of two equal and opposite charges separated by some distance.



Electric Field due to an Electric Dipole

1. For a point P on axial line

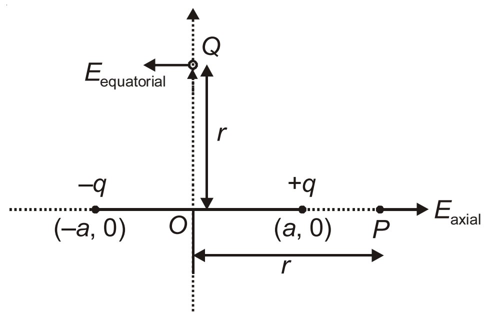
$$E_{\text{axial}} = \frac{2pr}{4\pi\varepsilon_0 (r^2 - a^2)^2}$$

For an ideal dipole ($r >> a \Rightarrow r^2 - a^2 \quad r^2$)

$$\Rightarrow E_{\text{axial}} = \frac{2p}{4\pi\varepsilon_0 r^3}$$

2. For a point Q on equatorial line

$$E_{\text{equatorial}} = \frac{-p}{4\pi\varepsilon_0 (r^2 + a^2)^{3/2}}$$

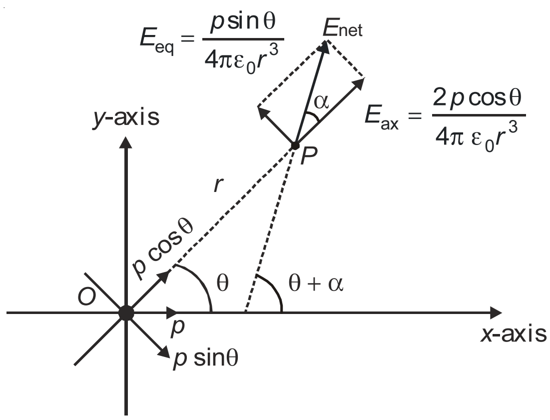


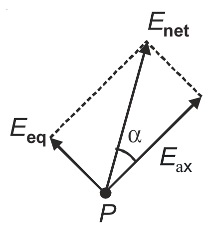
For an ideal dipole ($r >> a \Rightarrow r^2 + a^2 \quad r^2$)

$$\Rightarrow E_{\text{equatorial}} = \frac{-p}{4\pi\varepsilon_0 r^3}$$

3. For an ideal dipole $E_{\text{equatorial}} = \frac{-E_{\text{axial}}}{2}$ (For same distance from centre of dipole).

4. Electric field at any point in the plane of a **short dipole**
P is a point in x-y plane at a distance r from the centre of dipole, such that OP makes an angle θ with dipole moment.





(a) $E_{\text{net}} = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}\sqrt{1+3\cos^2\theta}$

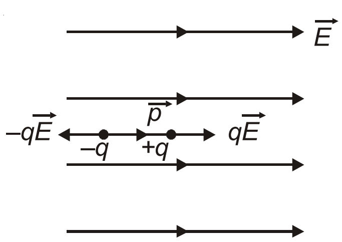
(b) $\tan\alpha = \frac{E_{eq}}{E_{ax}} = \frac{1}{2}\tan\theta \quad \Rightarrow \quad \tan\alpha = \frac{1}{2}\tan\theta$

(c) The net electric field makes an angle $\theta + \alpha$ with dipole moment.

(d) When $\vec{E} \quad \vec{p} \Rightarrow \theta + \alpha = 90° \quad \Rightarrow \quad \theta = \tan^{-1}\sqrt{2}$

Electric Dipole Placed in a Uniform Electric Field (Torque on dipole in uniform electric field)

Case-1 : $\vec{p} \parallel \vec{E}$

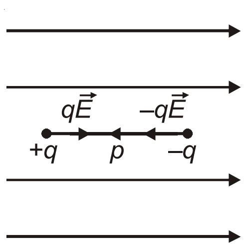


(a) Net force = $q\vec{E} - q\vec{E} = 0$

(b) Net torque = Zero

(c) Stable equilibrium

Case-2 : $\vec{p} \parallel (-\vec{E})$

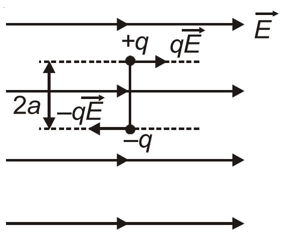


(a) Net force = $q\vec{E} - q\vec{E} = 0$

(b) Net torque = Zero

(c) Unstable equilibrium

Case-3 : $\vec{p}$ $\vec{E}$



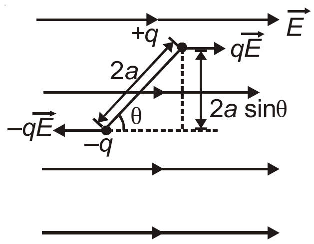
(a) Net force = Zero

(b) $= qE \times 2a = pE$ (This is the maximum value)

In vector form $= \vec{p} \times \vec{E}$

(c) Translational equilibrium but not in rotational equilibrium.

Case-4 : $\vec{p}$ makes an angle θ with $\vec{E}$



(a) Net force = Zero

(b) $= \vec{p} \times \vec{E}$ or $= p\ E \sin\theta$

(c) Translational equilibrium but not in rotational equilibrium.

Potential Energy of Dipole

1. The external work required to change the orientation from θ_1 to θ_2 is

   $W_{ext} = -\ pE[\cos\theta_2 - \cos\theta_1]$

2. Change in potential energy of dipole is

   $U_2 - U_1 = -pE[\cos\theta_2 - \cos\theta_1]$

3. Potential energy of dipole is

   $U = -pE\cos\theta = -\vec{p}.\vec{E}$

ELECTRIC FLUX

The number of field lines that pass through a surface is directly proportional to flux of electric field through that area.

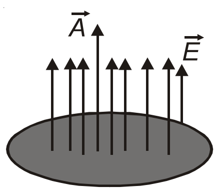
Mathematically, $\phi = \vec{E}\,.\,\vec{A}$ (If E is uniform and the surface in planar.)

In general, $\phi = \int \vec{E}\,.\,d\vec{A}$

Units : $\frac{\text{N m}^2}{\text{C}}$ or V-m

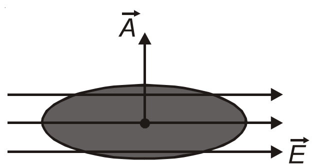
Important cases :

(1) $\vec{E}\,||\,\vec{A}$



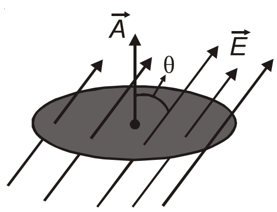
$\phi = EA$

(2) $\vec{E}$ $\vec{A}$



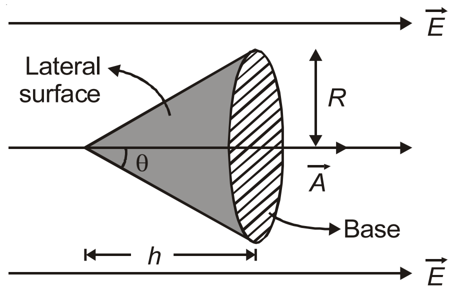
$\phi = 0$

(3) $\vec{E}$ and $\vec{A}$ make an angle θ



$\phi = EA\cos\theta$

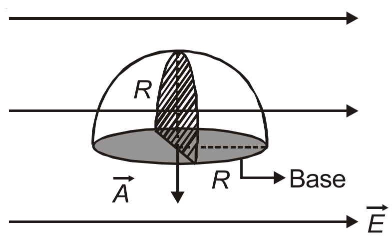
(4)



$\phi_{base} = \vec{E}\,.\,\vec{A} = E \times \pi R^2$

$\phi_{lateral} = -E \times \pi R^2$ ($\because$ Field lines enter through curved surface)

(5)



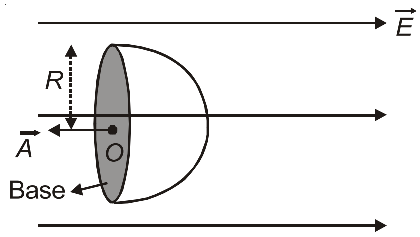
$\phi_{base} = 0$

$\phi_{curved} = 0$ (Total flux that enters = Total flux that leave)

$\varphi_{entered} = -E \times \left(\frac{1}{2}\pi R^2\right)$

$\phi_{leaving} = E \times \frac{\pi R^2}{2}$

(6)

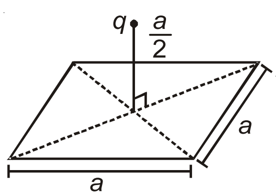


$\phi_{base} = -E \times \pi R^2$

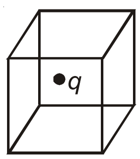
$\phi_{curved} = +E \times \pi R^2$

Electric Flux

Some frequently asked cases :

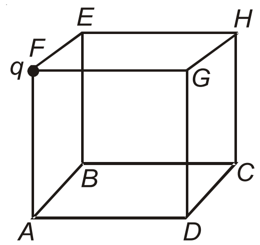
1. 

$\phi_{square} = \frac{q}{6\varepsilon_0}$

2. 

$\phi_{cube} = \frac{q}{\varepsilon_0}$

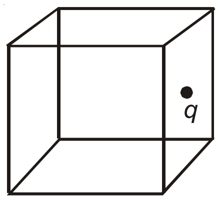
$\phi_{each\ face} = \frac{q}{6\varepsilon_0}$

3. 

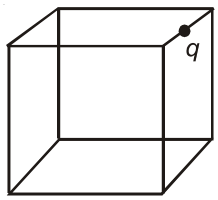
$\phi_{ABCD} = \frac{q}{24\varepsilon_0} = \phi_{DCHG} = \phi_{BCHE}$

$\phi_{ABEF} = 0 = \phi_{ADGP} = \phi_{FGHE}$

$\phi_{cube} = \frac{q}{8\varepsilon_0}$

4. 

$\phi_{\text{cube}} = \frac{q}{2\varepsilon_0}$

5. 

$\phi_{\text{cube}} = \frac{q}{4\varepsilon_0}$

Important results for fields due to different bodies (derived by Gauss Law)

1. Point charge $Q : \frac{kQ}{r^2}$.
2. Shell of charge with charge Q and radius $R : \frac{kQ}{r^2}$ (outside) and zero (inside).
3. Sphere of charge with charge Q and radius $R : \frac{kQr}{R^3}$ (inside) and $\frac{kQ}{r^2}$ (outside).
4. Infinite line of charge with linear charge density $\lambda : \frac{2k\lambda}{r}$ (perpendicular to line charge).
5. Infinite plane surface of charge with charge density $: \frac{}{2\varepsilon_0}$.
6. Infinite conducting sheet of charge with charge density $: \frac{}{\varepsilon_0}$.

Electric Potential Difference (V)

1. It is the work done against electric field in moving a unit positive charge from one point to other. That is

   $V_2 - V_1 = -\int_1^2 \vec{E}.d\vec{r}$.

2. V for two points at a distance r_1 and r_2 from a point charge Q

   $V_2 - V_1 = \quad V = KQ\left[\frac{1}{r_2} - \frac{1}{r_1}\right]$

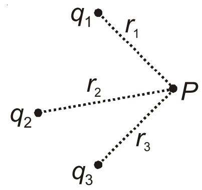
3. Change in potential energy of a charge q when moved across V is $U = q.\ V$.
4. V between two points in electric field does not depend on path.

ELECTRIC POTENTIAL (V)

1. V at a point is work done against electric field in moving a unit positive test charge from infinity to that point is $V = -\int^{r} \vec{E}.dr$.

2. Potential due to a point charge Q at a distance r is $V = \frac{KQ}{r}$.

3. Potential due to a dipole at distance r at angle θ is $V = \frac{Kp\cos\theta}{r^2}$



4. Potential due to system of point charges is $V_P = \left(\frac{Kq_1}{r_1} + \frac{Kq_2}{r_2} + \frac{Kq_3}{r_3}\right)$.

   If V and E are functions of x, then $V_2 - V_1 = -\int_{x_1}^{x_2} E\,dx$.

   In general, $V_2 - V_1 = \int_{x_1}^{x_2} E_x dx - \int_{y_1}^{y_2} E_y dy - \int_{z_1}^{z_2} E_z dz$

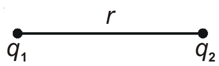
Relation between Electric Field and Potential

1. $E_x = -\frac{V}{x}$, $E_y = -\frac{V}{y}$, $E_z = -\frac{V}{z}$.

2. If V is a function of single variable r, $E = -\frac{dV}{dr}$.

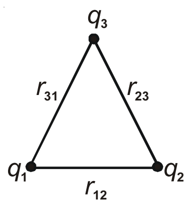
Electric Potential Energy

1. For a two point charge system



$$U = \frac{Kq_1q_2}{r}$$

2. For a three point charge system



$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1q_2}{r_{12}} + \frac{q_2q_3}{r_{23}} + \frac{q_3q_1}{r_{31}}\right]$$

CAPACITOR

It is a device used to store electric energy in the form of electric field.

CAPACITANCE

Capacitance of a conductor is measure of ability of conductor to store electric charge and hence electric energy on it.

When charge is given to a conductor its potential increases. It is found that

$V \propto Q$

or, $Q \propto V$

$Q = CV$

where C is the capacitance and its unit is farad (F).

***C* depends on**

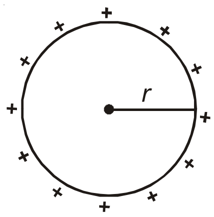
Shape and size of conductors and their relative placement w.r.t. each other.

Medium surrounding the conductor.

Capacitance of Isolated Spherical Conductor

$C = 4\pi\varepsilon_0 r$

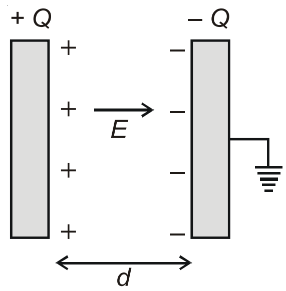
Capacitance of Earth $C_e = 4\pi\varepsilon_0 R_e \;\; = 711\ \mu F$



Capacitance of a Parallel Plate Capacitor

1. Electric field in between plates

$$E = \frac{Q}{A\varepsilon_0} = \frac{}{\varepsilon_0}$$



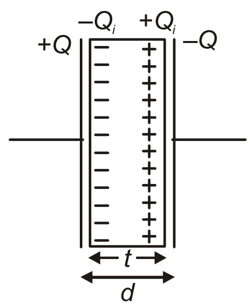
2. Potential difference between the plates = $\frac{Qd}{A\varepsilon_0} = \frac{d}{\varepsilon_0}$

3. Capacitance = $\frac{\varepsilon_0 A}{d}$

4. Force of attraction between the plates = $\frac{Q}{2A\varepsilon_0} = \frac{{}^2A}{2\varepsilon_0} = \frac{QE}{2}$

Parallel Plate Capacitor with Dielectric Slab

(a) Induced charge $Q_i = -Q\left(1 - \frac{1}{K}\right)$, K is dielectric constant.



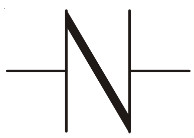
(b) Capacitance, $C = \frac{\varepsilon_0 A}{(d-t) + \frac{t}{K}}$.

(c) For conducting slab, K =

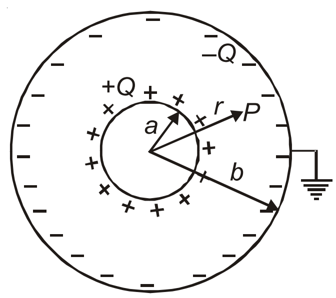
$\Rightarrow Q_i = -Q$ and $C = \frac{\varepsilon_0 A}{d - t}$

(d) The capacitance of a parallel plate capacitor is C. If its plates are connected by an inclined conducting rod, the new capacitance is infinity.

$\boxed{C = }$



Spherical Capacitor



1. Potential difference between plates

$$V = KQ\left[\frac{b-a}{ba}\right]$$

2. Electric field at any point P between plates

$$E = \frac{KQ}{r^2}$$

3. Potential at any point P between plates

$$V = \frac{KQ}{r} - \frac{KQ}{b}$$

4. Capacitance $C = \dfrac{4\pi\varepsilon_0 ab}{b-a}$

5. If the inner surface is grounded, capacitance $C = \dfrac{4\pi\varepsilon_0 b^2}{b-a}$

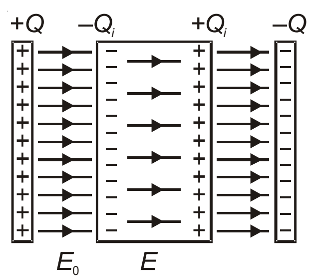
Cylindrical Capacitance of a Long Capacitor

Potential difference between plates

$$V = \frac{2KQ}{l} ln\left(\frac{b}{a}\right)$$

Dielectric Polarisation

When a dielectre slab is placed between the plates of capacitor its polarisation take place. Thus a charge $-Q_i$, appear on its left face and $+Q_i$ appears on its right face.

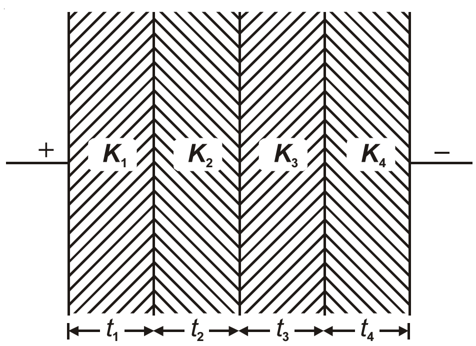


$$Q_i = Q\left(1 - \frac{1}{k}\right)$$

$$E_0 = \frac{Q}{A\varepsilon_0};\ E = \frac{Q}{A\varepsilon_0 k} = \frac{E_0}{k}$$

Effective Capacitance in Some Important Cases

1. $C = \dfrac{\varepsilon_0 A}{\dfrac{t_1}{K_1} + \dfrac{t_2}{K_2} + \dfrac{t_3}{K_3} + \dfrac{t_4}{K_4}}$

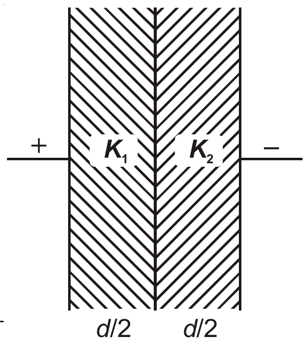


For two dielectrics,

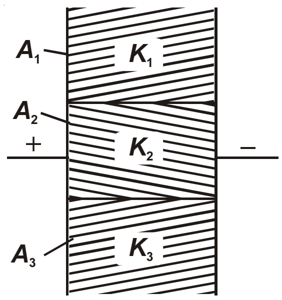
If $t_1 = t_2 = \frac{d}{2}$

$$C = \frac{\varepsilon_0 A}{\frac{d}{2K_1} + \frac{d}{2K_2}} = \frac{2\varepsilon_0 A}{d\left(\frac{1}{K_1} + \frac{1}{K_2}\right)}$$

$$\Rightarrow \quad C = \left(\frac{2K_1K_2}{K_1 + K_2}\right)\frac{\varepsilon_0 A}{d} \quad K_{eq} = \frac{2K_1K_2}{K_1 + K_2}$$



2. $C = \frac{\varepsilon_0[K_1A_1 + K_2A_2 + K_3A_3]}{d}$

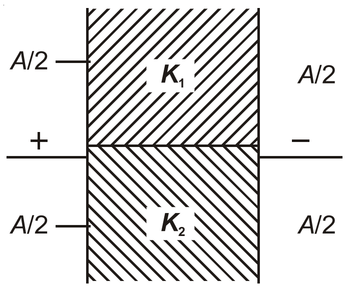


For two dielectrics,

If $A_1 = A_2 = \frac{A}{2}$

$$\Rightarrow C = \frac{\varepsilon_0\left(K_1\frac{A}{2} + K_2\frac{A}{2}\right)}{d}$$

$$\Rightarrow C = \left(\frac{K_1 + K_2}{2}\right)\frac{\varepsilon_0 A}{d} \quad K_{eq} = \frac{K_1 + K_2}{2}$$



COMBINATION OF CAPACITORS

1. **Capacitors in Series** (three capacitors)

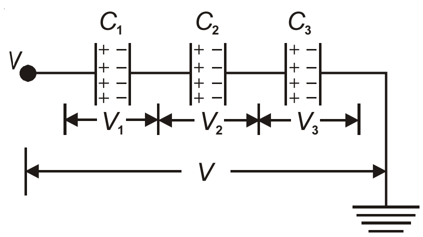
$V_1 = \frac{Q}{C_1}$, $V_2 = \frac{Q}{C_2}$ and $V_3 = \frac{Q}{C_3}$

$V = V_1 + V_2 + V_3$

$$V = Q\left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right]$$

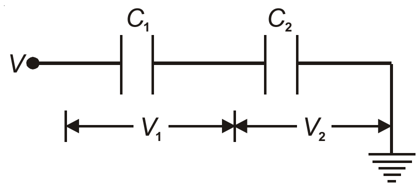
$$V = \frac{Q}{C_{eq}}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$



2. Two Capacitors in Series

$$V_1 = \frac{Q}{C_1} \qquad V_2 = \frac{Q}{C_2}$$



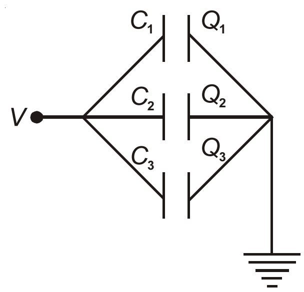
$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$C_{eq} = \frac{C_1C_2}{C_1 + C_2} \qquad Q = C_{eq}V$$

Potential Dividing Rule

$$V_1 = \frac{C_2}{C_1 + C_2}V \quad V_2 = \frac{C_1}{C_1 + C_2}V$$

3. Capacitors in Parallel



$Q_1 = C_1V$, $Q_2 = C_2V$, $Q_3 = C_3V$

$\Rightarrow Q = C_1V + C_2V + C_3V$

$\Rightarrow Q = (C_1 + C_2 + C_3)V$ and $Q = C_{eq}V$

$C_{eq} = C_1 + C_2 + C_3$

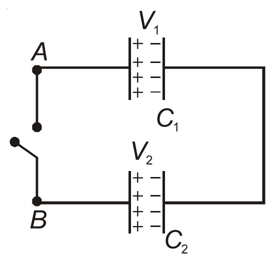
Energy Stored in a Capacitor

Energy stored in a capacitor of capacitance C, charge Q and potential difference V across it is given by

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$$

Sharing of Charge

Case-1 : Two capacitors charged to potentials V_1 and V_2 are connected end to end as shown

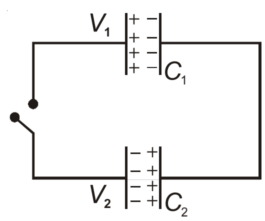


(a) Final common potential $V = \dfrac{C_1V_1 + C_2V_2}{C_1 + C_2}$

(b) Charge flown through key $= \dfrac{C_1C_2}{C_1 + C_2}(V_1 - V_2)$ in the direction A to B.

(c) Loss of energy $= \dfrac{C_1C_2}{2(C_1 + C_2)}(V_1 - V_2)^2$

Case-2 : If positive terminal is connected to negative terminal

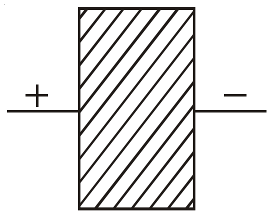


(a) Final common potential $V = \dfrac{C_1V_1 - C_2V_2}{C_1 + C_2}$

(b) Loss of energy $= \dfrac{C_1C_2}{2(C_1 + C_2)}(V_1 + V_2)^2$

Inserting a Dielectric Slab

1 When battery is disconnected (isolated) (charge is constant)



Q_0 = initial charge

C_0 = initial capacitance

V_0 = initial potential

E_0 = initial energy

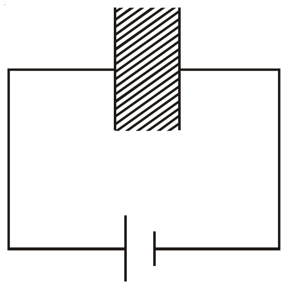
(a) New capacitance = KC_0

(b) New potential difference = $\dfrac{Q_0}{KC_0} = \dfrac{V_0}{K}$

(c) New energy stored = $\dfrac{1}{2}(KC_0)\left(\dfrac{V_0}{K}\right)^2 = \dfrac{E_0}{K}$

(d) Note that charge on each plate remains same.

2. When battery is connected (voltage is constant)



(a) $C = KC_0$

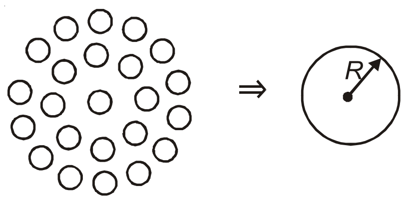
(b) $V = V_0$

(c) $Q = KQ_0$

(d) $E = \dfrac{1}{2}(KC_0)(V_0)^2 = KE_0$

Combining Charged Drops

When n droplets of radius r_0 having equal charge Q_0 combined to form a bigger drop of radius R.



(a) $n\dfrac{4}{3}\pi r_0^{\,3} = \dfrac{4}{3}\pi R^3$

$\Rightarrow R = n^{1/3}r_0$

(b) $C = n^{1/3}C_0$

(c) Total charge = nQ_0

(d) $V = \dfrac{nQ_0}{C} = \dfrac{nQ_0}{n^{1/3}C_0} = n^{2/3}V_0$

(e) Total energy = $\dfrac{1}{2}\dfrac{Q^2}{C} = \dfrac{(nQ_0)^2}{2n^{1/3}C_0} = n^{5/3}U_0$

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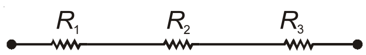
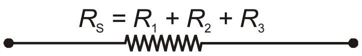
Chapter 11

Current Electricity

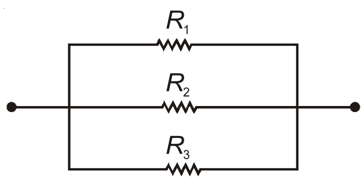
GROUPING OF RESISTORS (SERIES AND PARALLEL COMBINATION)

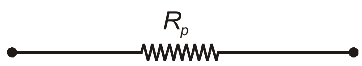
1. **Series Grouping**





2. **Parallel Grouping**

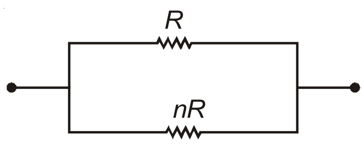


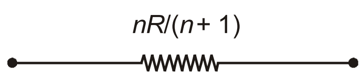


$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

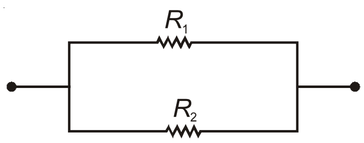
Illustrations :

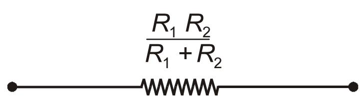
1.



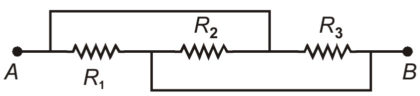


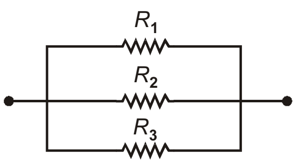
2.

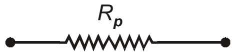




3.







$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Cell Terminology

1. **EMF (*E*)**

   The potential difference across the terminals of a cell when no current is being drawn from it.

2. **Internal Resistance (*r*)**

   The opposition to flow of current inside the cell.

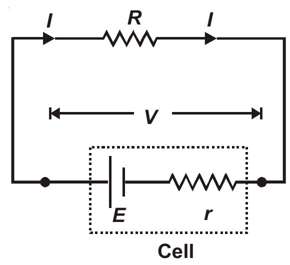
3. **Terminal Potential Difference**

It is the potential difference across the terminals of a cell when current is supplied by it.

$E = IR + Ir, \quad V = IR$

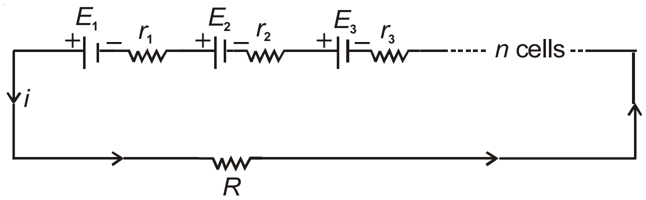
$E - V = Ir$ .... (i)

$r = \left(\frac{E - V}{V}\right)R$



Grouping of Cells (Series and Parallel Combination)

1. **Series Grouping :**



(a) $E_{\text{equivalent}} = E_1 + E_2 + E_3 + \ldots\ldots\ E_n$

(b) $r_{\text{equivalent}} = r_1 + r_2 + r_3 + \ldots\ldots\ r_n$

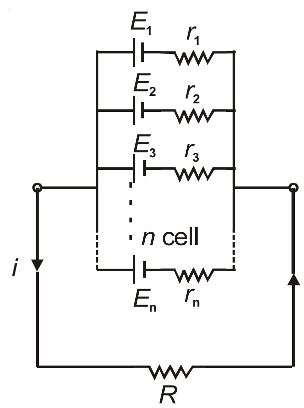
(c) Current $i = \dfrac{\sum E_i}{\sum r_i + R}$

> ***Note :*** *If polarity of m cells are made reverse in the series combination of n identical cells then equivalent emf $E_{equivalent}$ = (n – 2m)E and internal resistance $r_{equivalent}$ = nr*

2. **Parallel Grouping :**

(a) $E_{\text{equivalent}} = \dfrac{\dfrac{E_1}{r_1} + \dfrac{E_2}{r_2} + \dfrac{E_3}{r_3} + \ldots\ldots}{\dfrac{1}{r_1} + \dfrac{1}{r_2} + \dfrac{1}{r_3} + \ldots\ldots..}$

(b) $r_{\text{equivalent}} = \dfrac{1}{\dfrac{1}{r_1} + \dfrac{1}{r_2} + \dfrac{1}{r_3} + \ldots\ldots}$



(c) If all cells have equal emf. E and internal resistance r then

$E_{\text{equivalent}} = E$

$r_{\text{equivalent}} = \dfrac{r}{n}$

$\Rightarrow$ Current $i = \dfrac{E}{\dfrac{r}{n} + R}$

3. **Mixed Grouping of Cell :**

   Let n cells be connected in series in one row and m rows of cells in parallel. If cells are identical each of emf E.

   Total number of cells = mn

   Total emf = nE

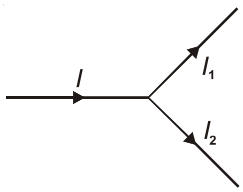
   Let i to be current through external load R.

   $$i = \frac{mnE}{mR + nr}$$

   i will be maximum if $mR = nr$

KIRCHHOFF'S LAWS

1. **Junction Rule :** It is based on conservation of charge.



$$I = I_1 + I_2$$

2. **Loop Rule :** It is based on conservation of energy.

   (a) For any closed loop, total rise in potential + total fall in potential = 0.

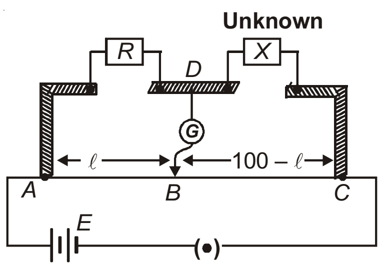
   (b) For any open part from a point A to point B, if V_A is potential at A and V_B is potential at B, then as we move from A to B.

   V_A + total rise in potential + total fall in potential = V_B.

> ***Note :*** *By convention rise in potential is taken as positive and fall in potential is taken as negative.*

METER BRIDGE

It is based on Wheatstone bridge principle. It is used to find an unknown resistance.



When there is no deflection in galvanometer then bridge is called balanced and for balanced bridge $\frac{P}{Q} = \frac{R}{S}$,

$$\frac{R}{\ell} = \frac{X}{100-\ell} \quad \Rightarrow \quad \text{Unknown } X = R\left(\frac{100-\ell}{\ell}\right)$$

> ***Note :*** *Location of null point is independent of resistivity or area of cross-section of wire AB.*

INSTRUMENTS

1. **Ammeter**

   (a) Shunt resistance is added in parallel to the galvanometer coil to make it into an ammeter.

   Shunt resistance $S = \left(\dfrac{i_g}{i - i_g}\right)G$.

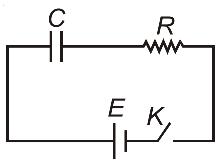
2. **Voltmeter**

   (a) A large resistance is added in series to the galvanometer coil to make it into a voltmeter.

   $$R = \frac{V}{i_g} - G$$

R-C CIRCUIT

1. **Charging**



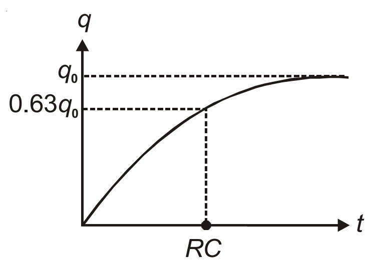
Key K is closed at t = 0.

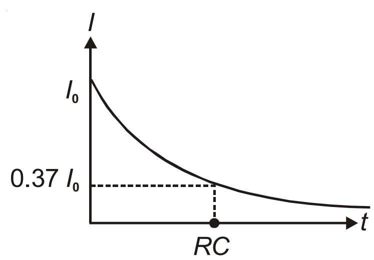
Current starts flowing and charge of capacitor starts increasing.

At any instant t, q is charge on capacitor. I is current in the circuit.

(a) $q = q_0\,[1 - e^{-t/RC}]$ where $q_0 = EC$ is maximum charge

(b) $I = \dfrac{E}{R}e^{-t/RC} = I_0 e^{\frac{-t}{RC}}$



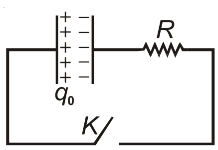


$\Rightarrow$ At $t = 0$, $I = E/R$

(c) RC = time constant. During charging, in $t = RC$, $q = 0.63q_0$.

(d) In RC circuit an uncharged capacitor behaves like closed switch at t = 0 and open switch at t = .

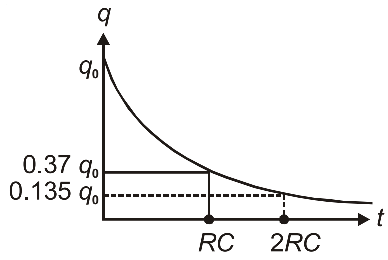
2. **Discharging RC Circuit**



Key K is closed at t = 0

(a) $q = q_0 e^{-t/RC}$

(b) $I = -\dfrac{dq}{dt} \Rightarrow I = +I_0 e^{-t/RC}$ where, $I_0 = \dfrac{q_0}{RC}$



(c) RC = time constant ( )

at $t = 0$, $q = q_0$

at $t = RC$, $q = 0.37\ q_0$

at $t = 2RC$, $q = 0.135\ q_0$

(d) The charge and potential difference both decay exponentially like radioactive decay with half-life = 0.693 RC

HEATING EFFECT OF CURRENT

Joule's Law

When a constant current I is passed through a device having resistance R, then the amount of heat produced in time t

$H = I^2Rt$ in joules

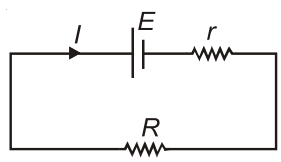
$\Rightarrow \quad H = \dfrac{I^2Rt}{J}$ in calories, where J = mechanical equivalent of heat = 4.186 or 4.2 J/cal

In general of a variable current I passes through a resistor, heat produced across R in time t is $H = \int_0^t I^2 R\,dt$

Maximum Power Transfer Theorem

In an electrical circuit, the maximum power can be drawn from the battery when external resistance is same as the internal resistance of the battery. Power drawn in the external resistor is

$$P = I^2R = \frac{E^2R}{(R+r)^2} \Rightarrow \frac{dP}{dR} = E^2\,\frac{(R+r)^2 - 2R(R+r)}{(R+r)^4} = 0$$



$\Rightarrow \quad R = r$

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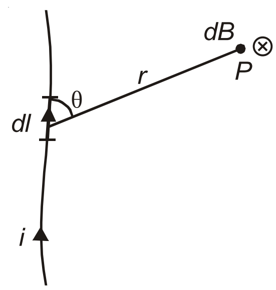
Chapter 12

Magnetic Effects of Current and Magnetism

MAGNETIC EFFECTS OF CURRENT

BIOT-SAVART LAW

Magnetic field due to current carrying element is given by



$$dB = \frac{\mu_0}{4\pi}\frac{idl\sin\theta}{r^2}$$

$$\overrightarrow{dB} = \frac{\mu_0}{4\pi}\frac{i(\overrightarrow{dl}\times\vec{r})}{r^3}$$

The direction of magnetic field due to small element dl is in the direction of $\overrightarrow{dl}\times\vec{r}$

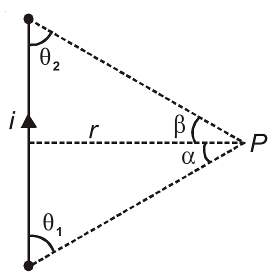
Units : 1. S.I. unit of magnetic field is tesla (T)

2. CGS unit of magnetic field is gauss.

3. 1 gauss = 10^{-4} tesla.

Magnetic Field Due to Straight Current Carrying Wire

Magnetic field at P



$$B = \frac{\mu_0 i}{4\pi r}(\cos\theta_1 + \cos\theta_2)$$

or $$B = \frac{\mu_0 i}{4\pi r}(\sin\alpha + \sin\beta)$$

1. For an infinite long wire

$$\theta_1 = \theta_2 = 0 \text{ or } \alpha = \beta = \frac{\pi}{2},\ B = \frac{\mu_0 i}{2\pi r}$$

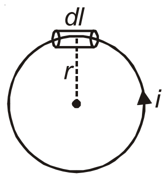
2. For semi-infinite long wire

$$\theta_1 = 0°,\ \theta_2 = 90°, \text{ or } \alpha = \frac{\pi}{2}, \beta = 0,\ B = \frac{\mu_0 i}{4\pi r}$$

CIRCULAR LOOP

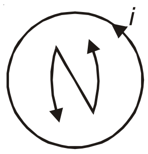
1. At the centre of a current loop

$$dB = \frac{\mu_0 i dl \sin 90}{4\pi r^2}$$

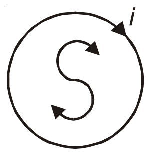


$$B = \frac{\mu_0 i}{4\pi r^2}\int dl = \frac{\mu_0 i}{4\pi r^2}\times 2\pi r$$

$$B = \frac{\mu_0 i}{2r} \text{ (Outward).}$$

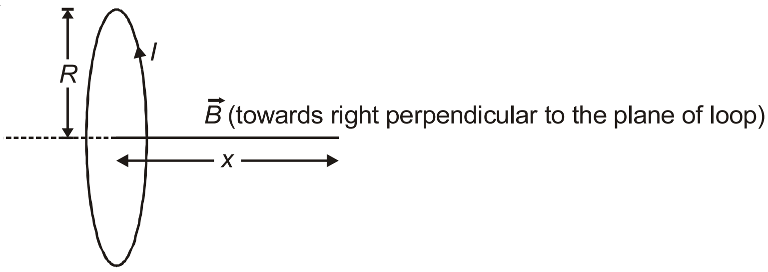


(Outward field perpendicular to the plane of loop)



(Inward field perpendicular to the plane of loop)

2. On the axis of a loop



$$B = \frac{\mu_0}{4\pi}\frac{2\times I\times \pi R^2}{(R^2+x^2)^{3/2}} = \frac{\mu_0}{4\pi}\frac{2M}{(R^2+x^2)^{3/2}}$$

For $x >> R$, $B = \frac{\mu_0}{4\pi}\frac{2M}{x^3}$ [Current carrying loop acts as an magnetic dipole]

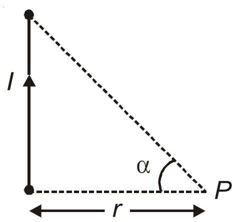
where $M = I \times \pi R^2$ is called magnetic dipole moment

$\vec{M} = I\vec{A}$

S.I. Unit : A-m^2

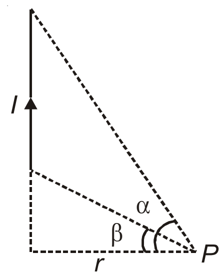
Various Cases of Magnetic Field (Straight Wire and Circular Loop)

1.



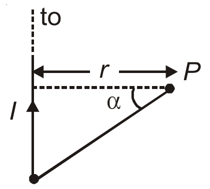
$$B_P = \frac{\mu_0 I}{4\pi r}[\sin\alpha] \otimes$$

2.



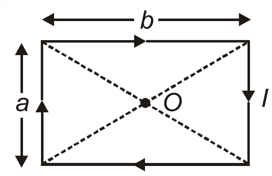
$$B_P = \frac{\mu_0 I}{4\pi r}[\sin\alpha - \sin\beta] \otimes$$

3.



$$B_P = \frac{\mu_0 I}{4\pi r}[\sin\alpha + 1] \otimes$$

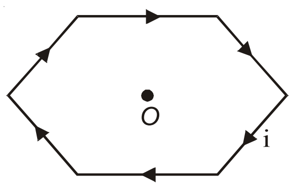
4.



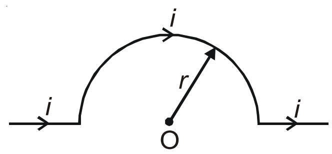
$$B_0 = \frac{8\mu_0 I}{4\pi}\ \frac{\sqrt{a^2+b^2}}{ab}$$

when $a = b$ $B_0 = \dfrac{8\sqrt{2}\mu_0 I}{4\pi a}$

5.



6.

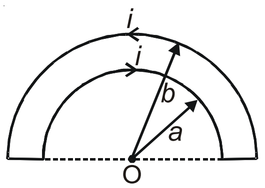


$$B_0 = \frac{\mu_0 i}{4r} \otimes$$

Magnetic field at O

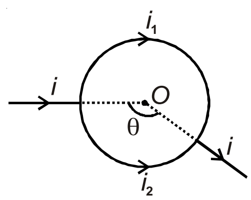
$B_0 = \dfrac{\sqrt{3}\mu_0 i}{\pi a}$ where 'a' is length of each side of regular hexagon

7.



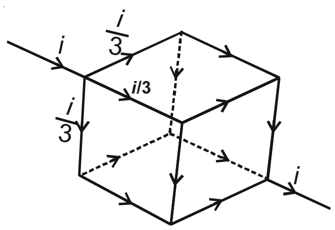
$$B_0 = \frac{\mu_0 i}{4}\left(\frac{1}{a} - \frac{1}{b}\right) \otimes$$

8.



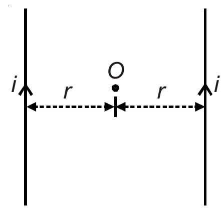
$B_0 = 0$ (for any value of θ)

9.



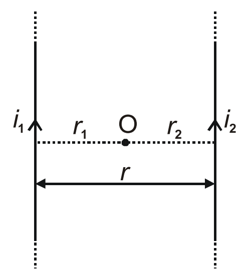
At the centre of cube, $B = 0$

10.



At O, $B = 0$

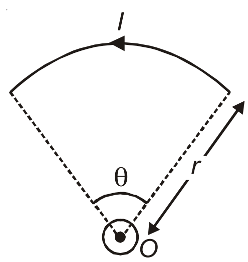
11.



At O, $B = 0$, such that

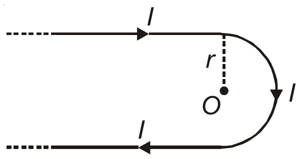
$$r_1 = \frac{i_1}{i_1 + i_2} r; r_2 = \frac{i_2 r}{i_1 + i_2}$$

12.

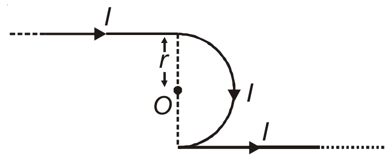


$$B_0 = \frac{\mu_0 I}{2r} \times \frac{\theta}{2\pi} \odot$$

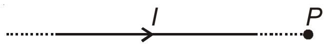
(θ is radian)

13. 

$B_0 = \frac{\mu_0 I}{2\pi r} + \frac{\mu_0 I}{4r} \otimes$

14. 

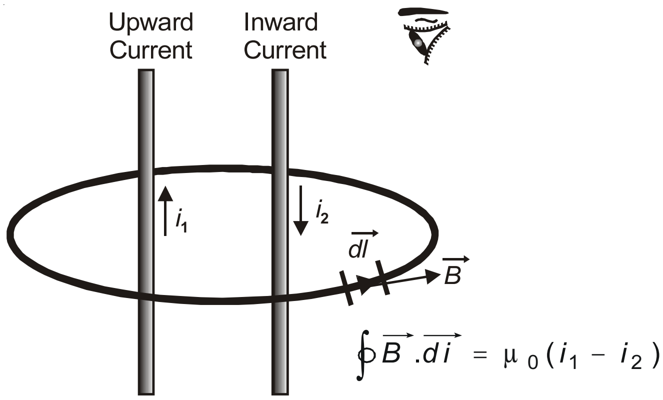
$B_0 = \frac{\mu_0 I}{4r} \otimes$

15. 

$B_P = 0$ (at point P)

AMPERE CIRCUITAL LAW

It states that the line integral of resultant magnetic field over a closed path is equal to μ_0 times the algebraic sum of the current threading the closed path in free space. Ampere's circuital law has the form

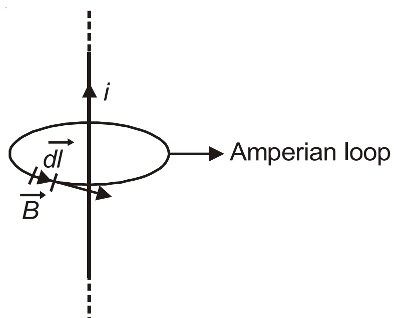


$$\oint \vec{B}.\vec{dl} = \mu_0 i_{enc}$$

Here $\oint \vec{B}.\vec{dl}$ implies the integration of scalar product $\vec{B}.\vec{dl}$ around a closed loop called an **Amperian loop**. The current i_{enc} is the net current encircled by the loop.

Applications of Ampere's Circuital Law

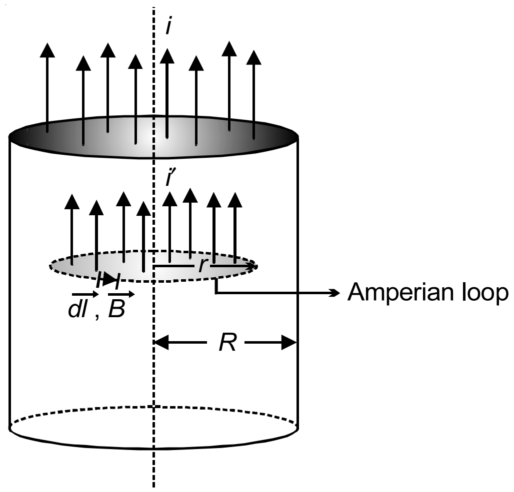
(1) Magnetic field due to a long thin current carrying wire

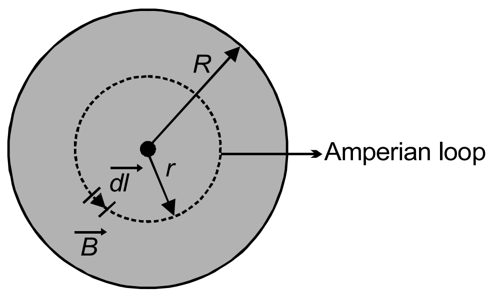


$$\oint \vec{B}\cdot\vec{dl} = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

(2) Magnetic field inside a long straight current carrying conductor



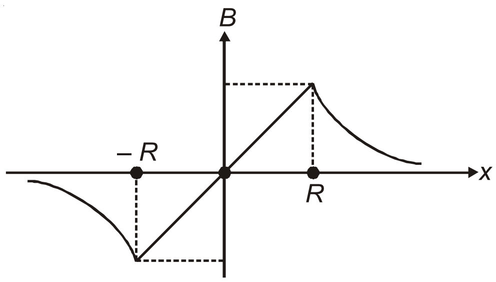


$$I' = \frac{I}{nR^2} \times \pi r^2$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(\frac{I}{nR^2} \pi r^2 \right)$$

$$B = \frac{\mu_0}{2\pi} \frac{r}{R^2} \qquad \text{i.e. } B_{in} \propto r$$

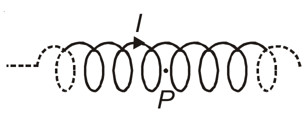
Graphical variation of magnetic field



(3) Inside a hollow tube of current, magnetic field is zero.

SOLENOID

A long solenoid having number of turns/length '*n*' carries a current *I*.

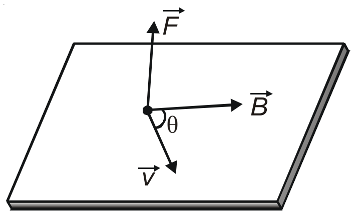


The magnetic field *B* is given by,

$B_P = \mu_0 nI$ (in between)

$B_{end} = \frac{\mu_0 nI}{2}$ (near one end)

Force on a Moving Charge in Magnetic Field



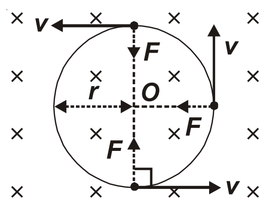
$\vec{F} = q(\vec{v} \times \vec{B})$

$F = qvB \sin\theta$

$\vec{F}$ is perpendicular to both $\vec{v}$ & $\vec{B}$ and is in the direction of $\vec{v} \times \vec{B}$ when $q > 0$ and opposite to $\vec{v} \times \vec{B}$ when $q < 0$.

MOTION OF CHARGED PARTICLE IN A UNIFORM MAGNETIC FIELD $\vec{B}$.

Case-1 : $\vec{v}$ $\vec{B}$

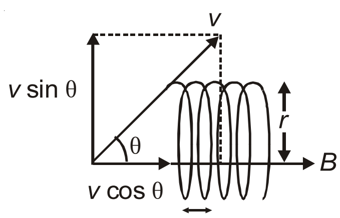


$F = qvB = \dfrac{mv^2}{r}$ $\Rightarrow r = \dfrac{mv}{qB} = \dfrac{p}{qB} = \dfrac{\sqrt{2km}}{qB}$ (k is kinetic energy). As $k = \dfrac{p^2}{2m}$.

Results

1. During revolution its speed is constant
2. During revolution its kinetic energy is constant
3. Work done by the magnetic force is zero
4. Velocity and momentum change continuously in direction, not in magnitude.
5. $T = \dfrac{2\pi r}{v} = \dfrac{2\pi m}{qB}$ (Independent of speed and radius)
6. Frequency $f = \dfrac{qB}{2\pi m}$ (Cyclotron frequency)

Case-2 : If $\vec{v}$ is not perpendicular to $\vec{B}$. Let θ be the angle between $\vec{v}$ and $\vec{B}$.



The particle moves in a helical path such that

$$r = \frac{mv\sin\theta}{qB}, T = \frac{2\pi m}{qB}, \text{ Pitch} = v\cos\theta \times T$$

Case-3 : If charged particle is moving parallel or anti-parallel to field $\vec{B}$ then force is zero and it moves in a straight line

Cyclotron

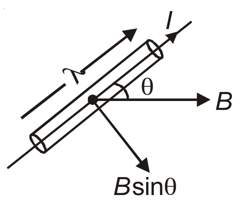
A cyclotron is a machine used to accelerate charge particle (like proton, deutron, α-particle). It uses both electric and magnetic field. Speed of particle is increased by electric field.

$$\text{Cyclotron frequency} = \frac{qB}{2\pi m}$$

$$K_{\max} = \frac{q^2B^2R_{\max}^2}{2m}$$

To accelerate electrons Betatron and Synchrotron are used. A synchrotron accounts for the variation in mass with speed. A Betatron uses the induced electric field produced by a time varying magnetic field to accelerate charged particles.

Force on a current carrying conductor in uniform field



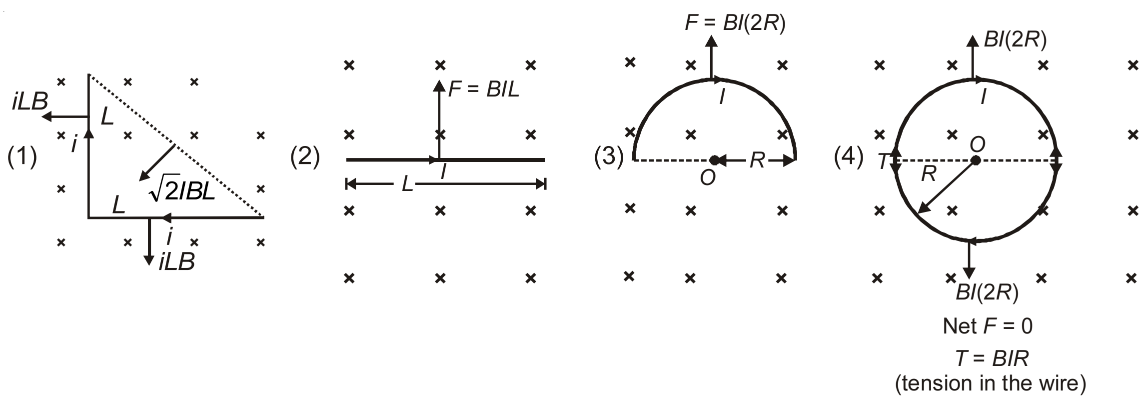
I = Current through the conductor, ℓ = Length of the conductor

$$F = IB\ell\sin\theta$$

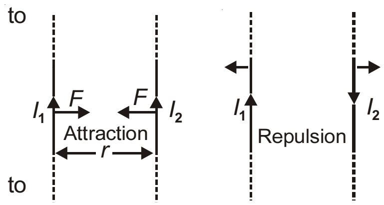
$$\vec{F} = I(\vec{\ell}\times\vec{B})$$

The direction of $\vec{\ell}$ is always in the direction of current.

Some Important Cases



FORCE BETWEEN TWO CURRENT CARRYING WIRES



F (force/unit length) is given by $F = \dfrac{\mu_0 I_1 I_2}{2\pi r}$.

The force on a segment of length 'l' is $= \dfrac{\mu_0 I_1 I_2}{2\pi r} \times l$

Force on a small current carrying segment placed near long and perpendicular current carrying wire.

$$F_{PQ} = \frac{\mu_0 i_1 i_2}{2\pi} \log\left(1 + \frac{l}{d}\right)$$

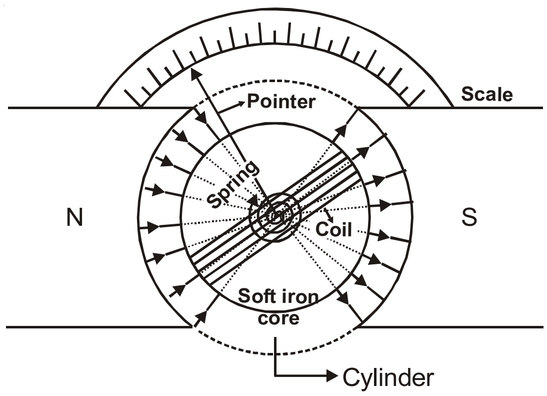
$_{PQ}$ 0

MOVING COIL GALVANOMETER

It is a device used to measure small current through the circuit.

Principle

When a current carrying coil is suitably placed in a magnetic field, torque acts on it. In moving coil galvanometer radial field is used which is obtained from magnet having concave shape poles. In this type of field plane of the coil is always parallel to the magnetic field so maximum torque acts on it.



$= NIAB$ [ϕ = 90° due to radial field] (ϕ – Angle between $\vec{M}$ and $\vec{B}$)

$= C\theta$ (θ = Angle of twist)

where θ is angle turned by the pointer and C is restoring torque/twist in the suspension wire.

$$I = \frac{C\theta}{NBA}$$

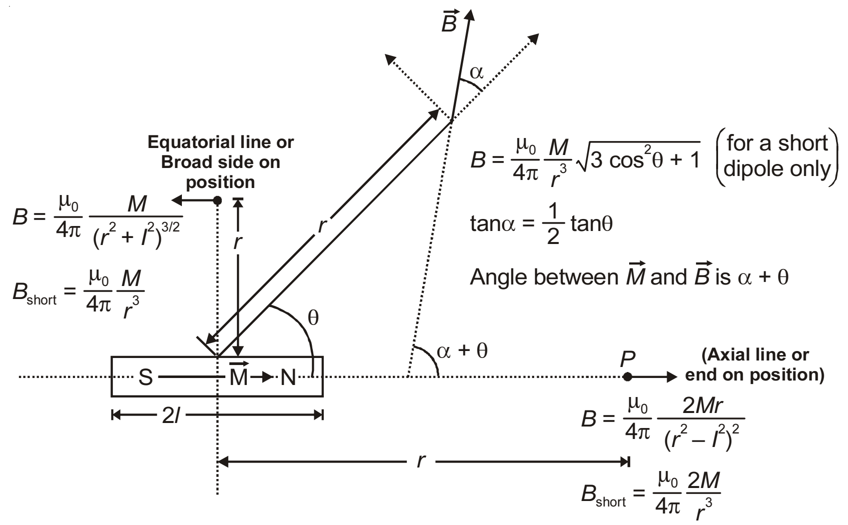
Current sensitivity = $\dfrac{\theta}{I} = \dfrac{NBA}{C}$

Voltage sensitivity = $\dfrac{\theta}{V} = \dfrac{\theta}{IR} = \dfrac{NBA}{CR}$

MAGNETISM

BAR MAGNET

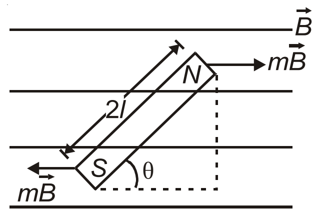
Magnetic field due to a Bar Magnet



1. $\vec{B}_{axial}$ is parallel to $\vec{M}$.
2. $\vec{B}_{equatorial}$ is antiparallel to $\vec{M}$.
3. $\vec{B}$ $\vec{M}$ when θ + α = 90° *i.e.*, θ = 90° – α

   as $\tan\alpha = \frac{1}{2}\tan\theta \Rightarrow \cot\theta = \frac{1}{2}\tan\theta$ or $\tan\theta = \sqrt{2}$.

TORQUE ON A BAR MAGNET IN MAGNETIC FIELD



= $mB \times 2l \sin\theta$

= $m \times 2l \times B \sin\theta$

⇒ = $MB \sin\theta$

Results :

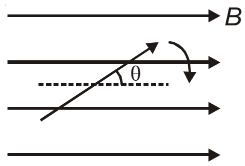
1. $\vec{M}\times\vec{B}$, $_{max} = MB$ [when θ 90°],

   $_{min} = 0$ [when θ = 0 or 180°]
2. Net force on dipole is zero.
3. $U = -\vec{M}.\vec{B}$ $U_{min} = -MB$ at θ = 0°

   $U_{max} = MB$ at θ = 180°
4. Work done by external agent in rotating bar magnet from angular position θ_1 to θ_2 is $W = MB\,[\cos\theta_1 - \cos\theta_2]$
5. A bar magnet kept in a non-uniform magnetic field experience a net force and may experience a torque.

Oscillations of a Bar Magnet in Magnetic Field

For small displacements from equilibrium position, bar magnet oscillates simple harmonically such that



Angular frequency $=\sqrt{\dfrac{MB}{I}}$;

Time period $T = 2\pi\sqrt{\dfrac{I}{MB}}$

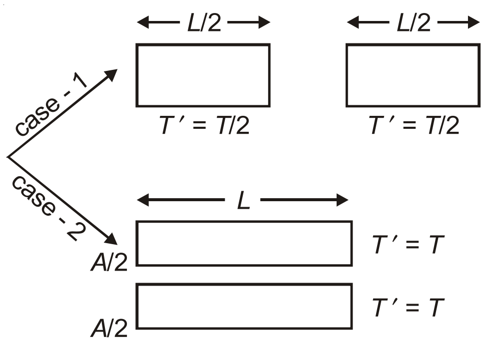
Some important cases related to time period are given below

A bar magnet of length L is,

Pole strength $\propto$ Area of cross-section of magnetic dipole ($m \propto A$)

$\vec{M} = m \times 2\vec{l}$

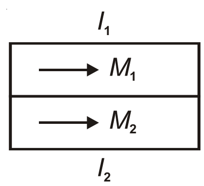
1. Cut into two halves



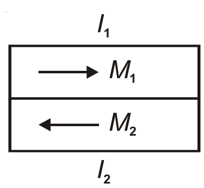
where $T = 2\pi\sqrt{\dfrac{I}{MB}}$

2. Two bar magnets having magnetic moments M_1, M_2 and moment of inertias I_1, I_2 are joined as shown.

(a) $T_1 = 2\pi\sqrt{\dfrac{I_1 + I_2}{(M_1 + M_2)B}}$



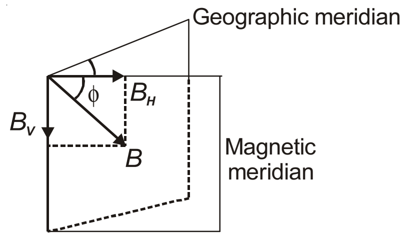
(b) $T_2 = 2\pi\sqrt{\dfrac{I_1 + I_2}{(M_1 - M_2)B}}$



$\Rightarrow \dfrac{T_2^2 + T_1^2}{T_2^2 - T_1^2} = \dfrac{M_1}{M_2}$

EARTH'S MAGNETIC FIELD

The basic components of earth's magnetic field at a place are shown

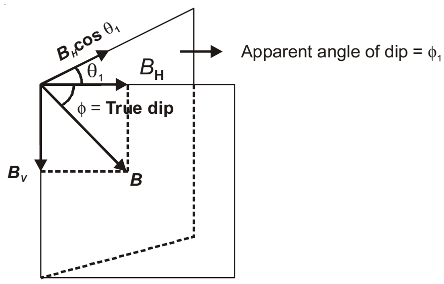


1. = Angle of declination
2. ϕ = Angle of dip/inclination
3. $B_H = B \cos \phi$
4. $B_V = B \sin \phi$
5. $B_H^2 + B_V^2 = B^2$

Note : *The needle of a vertical compass in magnetic meridian points toward B.*

6. $\tan\phi = \dfrac{B_V}{B_H}$

7. When the dip circle is not in magnetic meridian and dip circle is at an angle of θ_1 to magnetic meridian.



[ϕ = True dip angle, ϕ_1 and ϕ_2 = apparent dip angle in two arbitrary perpendicular)

$\tan\phi_1 = \dfrac{B_V}{B_H \cos\theta_1}$ (apparent dip) [Vertical component remains same]

$\Rightarrow \quad \tan\phi_1 = \dfrac{\tan\phi}{\cos\theta_1}$

8. $\cot^2\phi = \cot^2\phi_1 + \cot^2\phi_2$, plane

PARA, DIA AND FERROMAGNETIC SUBSTANCES

All the elements of the nature are studied under the action of magnetic field and classified into three parts according to following properties.

1. **Magnetic Intensity (Magnetising Force) :** $H = \dfrac{B_0}{\mu_0}$

   B_0 is magnetic field in vacuum

   SI unit = A/m

2. **Intensity of Magnetisation :** Magnetic moment developed/volume

   $I = \dfrac{M}{V}$ (Unit A/m)

$\Rightarrow \quad I = \dfrac{\text{Pole strength}}{\text{area}} \qquad \begin{bmatrix} \because M = m \times l \\ \because V = A \times l \end{bmatrix}$

3. **Magnetic Induction or Magnetic Flux Density (*B*) :** Number of magnetic field lines crossing per unit area normally through a magnetic substance.

$B = B_0 + \mu_0 I$

$B = \mu_0 H + \mu_0 I$

$B = \mu_0 (H + I)$

$\begin{bmatrix} B_0 = \text{applied magnetic field} \\ \mu_0 I = \text{magnetic field due to magnetisation} \end{bmatrix}$

4. **Magnetic Susceptibility :** $_m = \frac{I}{H}$ (no unit)

5. **Magnetic Permeability :** $\mu = \frac{B}{H} \Rightarrow B = \mu H$

From above $B = \mu_0(H + I)$

$\Rightarrow \quad \mu H = \mu_0 (H + I)$

$\frac{\mu}{\mu_0} = 1 + \frac{I}{H}$

$\mu_r = 1 + {}_m$ where μ_r = relative permeability.

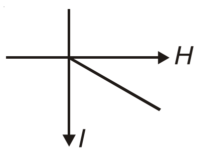
Curie Law

Magnetic susceptibility of paramagnetic material is inversely proportional to its absolute temperature.

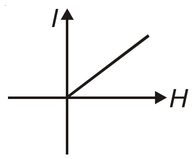
$_m \propto \frac{1}{T}$

Variation of *I* with *H*

1. Diamagnetic



2. Paramagnetic

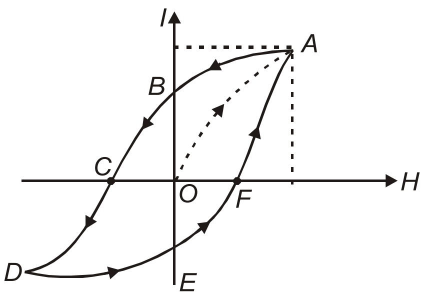


3. Ferromagnetic (Hysteresis)

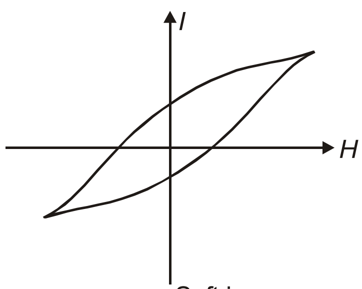
OB = Retentivity (residual magnetism even after magnetising field is reduced to zero)

OC = Coercivity (reverse magnetic field required to reduce residual magnetism to zero)

Area *ABCDEFA* = Energy loss/cycle during magnetisation and demagnetisation.



Steel
(High coercivity, high retentivity)



Soft iron
(Low coercivity, low retentivity)

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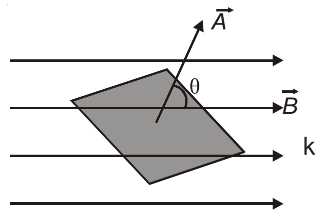
Chapter 13

Electromagnetic Induction

MAGNETIC FLUX

$\phi = \vec{B}.\vec{A} = BA\cos\theta$ (θ = Angle between $\vec{B}$ and $\vec{A}$)



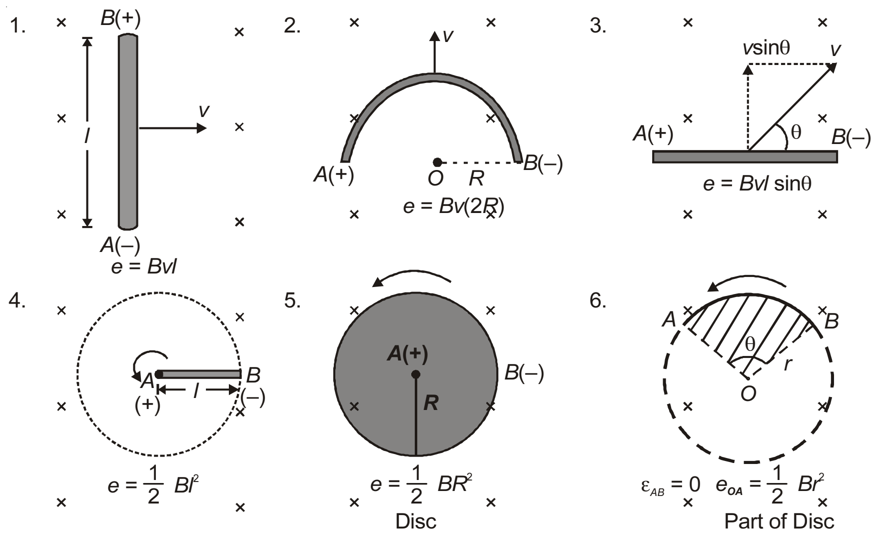
Faraday's Laws of Electromagnetic Induction

$e = -\dfrac{d\phi_B}{dt}$ (for a loop), for a plane coil having N turns

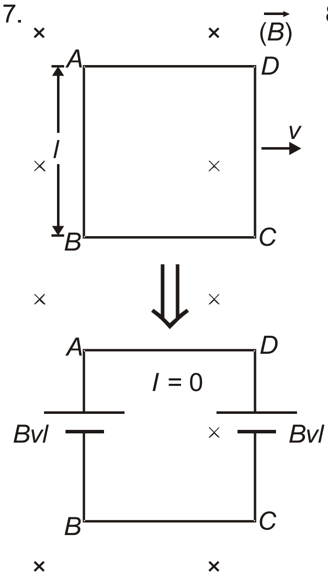
$$e = -\frac{Nd\phi_B}{dt} = -\frac{d(N\phi_B)}{dt}$$

> ***Note :*** *Negative sign indicates opposition (explained by Lenz's law).*

Induced EMF in different cases



7.



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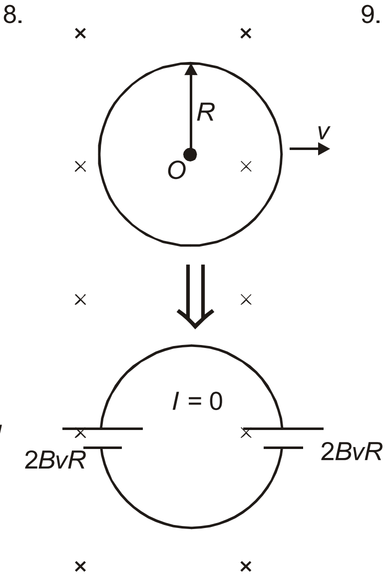

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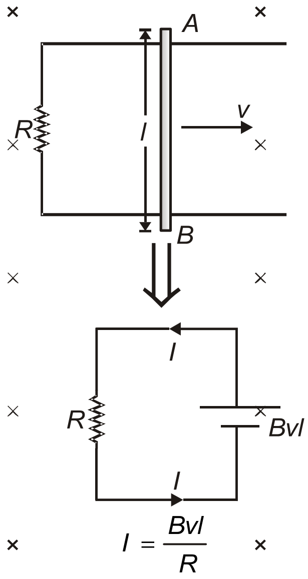

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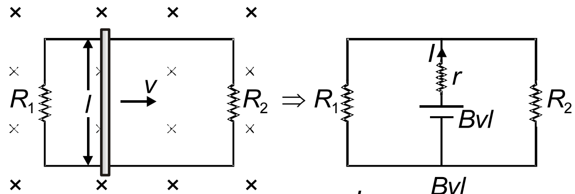
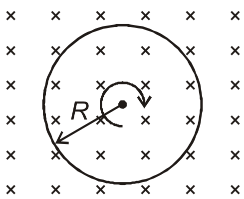


Sliding rod also has a resistance '*r*'

$$I = \frac{Bvl}{r + \dfrac{R_1 R_2}{R_1 + R_2}}$$

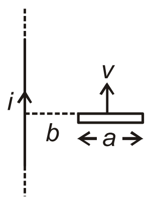
11.



$E_{\text{induced}} = 0$

Ring

12.



$$\varepsilon = \frac{\mu_0 i v}{2\pi} \ln\left(1 + \frac{a}{b}\right)$$

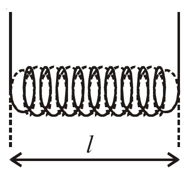
MUTUAL AND SELF INDUCTANCE

Mutual Induction

Important cases :

1. Mutual inductance of two long solenoids :

$$M = \frac{\mu_0 N_1 N_2 A}{l}$$



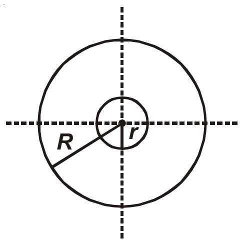
N_1 = Number of turns in one solenoid

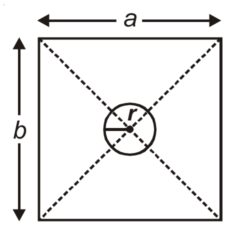
N_2 = Number of turns in other solenoid

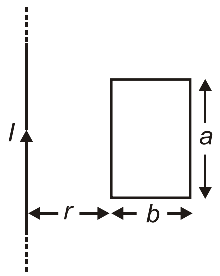
A = Area of cross-section of narrower solenoid

l = Length of solenoid

2. Two loops : $R >> r$







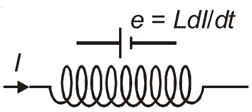
(a) $M = \dfrac{\mu_0 \pi r^2}{2R}$

(b) $M = \dfrac{\mu}{4\pi} \dfrac{8\sqrt{a^2+b^2}}{ab} \pi r^2$
(For r << a, r << b)

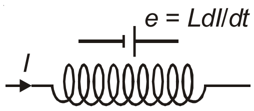
(c) $M = \dfrac{\mu_0}{2\pi} a \log\left(\dfrac{r+b}{r}\right)$

Direction of Induced emf

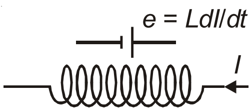
(a) I is increasing



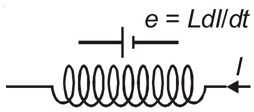
(b) I is decreasing



(c) I is increasing

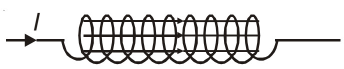


(d) I is decreasing



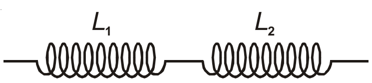
Energy in Inductor

Energy $U_B = \dfrac{1}{2} LI^2$, Energy density $= \dfrac{1}{2}\dfrac{B^2}{\mu_0}$



Combination of Inductors

1. **Inductor in series**



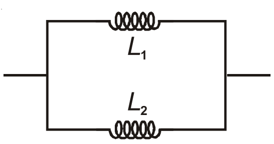
When mutual inductance is neglected $L = L_1 + L_2$

When mutual inductance is considered $L = L_1 + L_2 \pm 2M$

'+' sign is taken when flux of one coil supports the flux in other coil. '–' sign is taken when flux of one coil opposes the flux of other.

Here, $M = K\sqrt{L_1 L_2}$ where K = coefficient of coupling.

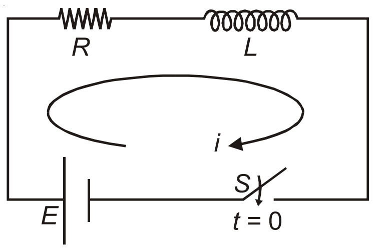
2. **Inductor in parallel**



$\dfrac{1}{L} = \dfrac{1}{L_1} + \dfrac{1}{L_2}$ (Neglecting mutual induction)

L-R- CIRCUIT WITH D.C. SOURCE

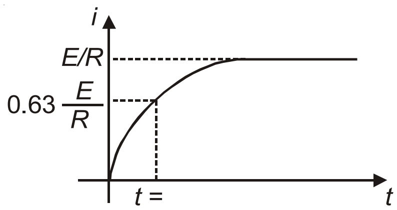
Growth of Current



At $t = 0$ the switch S is closed

$$i = \frac{E}{R}\left(i - e^{-t/}\right). \quad \text{Here} \quad = \frac{L}{R}$$

Graph for variation of current (i) with time (t) is shown here.

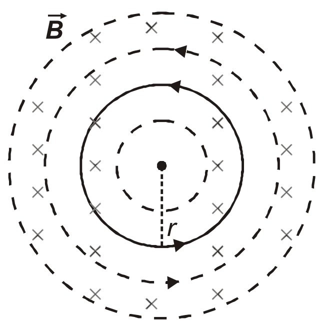


INDUCED ELECTRIC FIELD

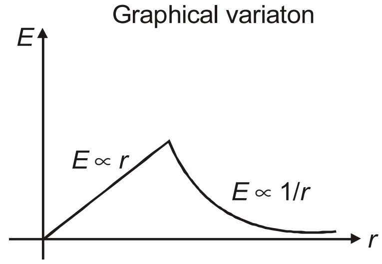
A changing magnetic field produces an electric field. This field is non-conservative and always forms closed loop. Consider a region of magnetic field. The magnetic field strength is increasing at a rate $\frac{d\vec{B}}{dt}$. This creates anticlockwise electric field lines. Electric field strength at distance r from point O is given by

$E = \frac{dB}{dt} \times \frac{r}{2}$ For inside point

$E = \frac{R^2}{2r}\frac{dB}{dt}$ For outside point

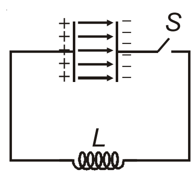


Graphical variaton



LC OSCILLATIONS

A charged capacitor is connected to an inductor and switch is closed at $t = 0$.

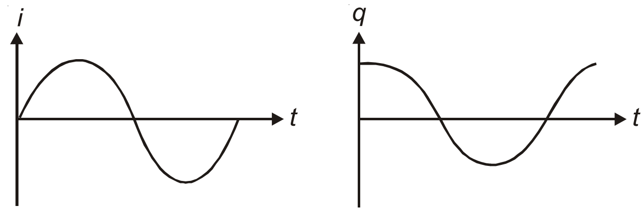


The charge and current vary sinusoidally as,

$q = q_0 \cos t$ [$\because$ at $t = 0$, $q = q_0$]

$i = i_0 \sin t$ [$\because$ at $t = 0$, $i = 0$]

Graphical representation of this variation is as shown.



$i_0 = \frac{q_0}{\sqrt{LC}}, \quad = \frac{1}{\sqrt{LC}}, \nu = \frac{1}{2\pi\sqrt{LC}}$ is frequency of *LC* oscillations

The maximum charge and maximum current in *LC* oscillation are related as $\frac{1}{2}LI_{max}^2 = \frac{q_{max}^2}{2C}$

$q_{max} = I_{max}\sqrt{LC}$

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Chapter 14

Alternating Current and Electromagnetic Waves

AVERAGE AND RMS VALUE OF AC

1. Mean or Average value for time interval '*t*'

$$E_{\text{mean}} = \frac{1}{t}\int_0^t E\,dt,\ \ I_{\text{mean}} = \frac{1}{t}\int_0^t I\,dt$$

The total charge flown in time '*t*' in a wire is $Q = I_{\text{mean}} \times t$

2. Root Mean Square (RMS) Value

RMS value of ac is equal to that value of dc, which when passed through a resistance for a given time interval will produce the same amount of heat as produced by the ac when passed through the same resistance for same time. RMS values are also known as **virtual or effective value.**

$$E_{\text{rms}}^2 = \frac{1}{t}\int_0^t E^2\,dt,\ \ I_{\text{rms}}^2 = \frac{1}{t}\int_0^t I^2\,dt$$

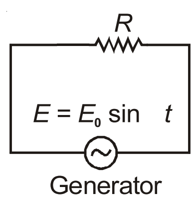
PHASOR AND AC CIRCUITS

Different ac Circuits

1. Resistive Circuit

$I = I_0 \sin \quad t$

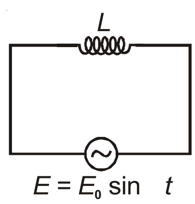
$$I_0 = \frac{E_0}{R}$$



2. Inductive Circuit

$I = I_0 \sin(\quad t - \pi/2)$

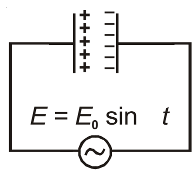
$$I_0 = \frac{E_0}{X_L}, \text{ where } X_L = \quad L = 2\pi f L$$



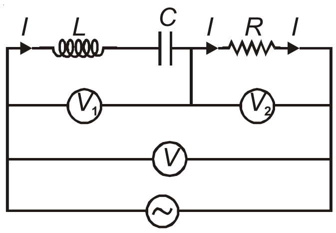
Capacitive Circuit

1. $I = I_0 \sin(\quad t + \pi/2)$

2. $I_0 = \dfrac{E_0}{X_C}$, where $X_C = \dfrac{1}{\quad C} = \dfrac{1}{2\pi f C}$



SERIES LCR CIRCUIT



$E = E_0 \sin \; t$

$V = \dfrac{E_0}{\sqrt{2}}$ = rms value of applied voltage

V_1 = rms voltage across L-C = $V_L - V_C$

V_2 = rms voltage across R = V_R

Phase Relationship

⇒ I and V_R are in same phase in case of resistance.

⇒ V_L leads I by 90° in case of inductor.

⇒ V_C lags behind I by 90° in case of capacitor.

⇒ In general, $V = \sqrt{V_R^2 + (V_L - V_C)^2}$

$$= I\sqrt{R^2 + (X_L - X_C)^2}$$

(a) Impedance = $Z = \sqrt{R^2 + (X_L - X_C)^2}$

(b) Power factor = $\cos\phi = \dfrac{R}{Z} = \dfrac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$

POWER CONSUMED IN AN A.C. CIRCUIT

$$P_{av} = \frac{1}{T}\int_0^T EIdt$$

If $E = E_0 \sin \; t$ and $I = I_0 \sin (\; t + \phi)$

$$P_{av} = \frac{E_0 I_0}{2}\cos\phi$$

$$= \frac{E_0}{\sqrt{2}} \cdot \frac{I_0}{\sqrt{2}}\cos\phi$$

$= E_v I_v \cos\phi$ [E_v = Virtual or rms voltage, I_v = Virtual or rms current]

$\cos\phi$ is known as power factor.

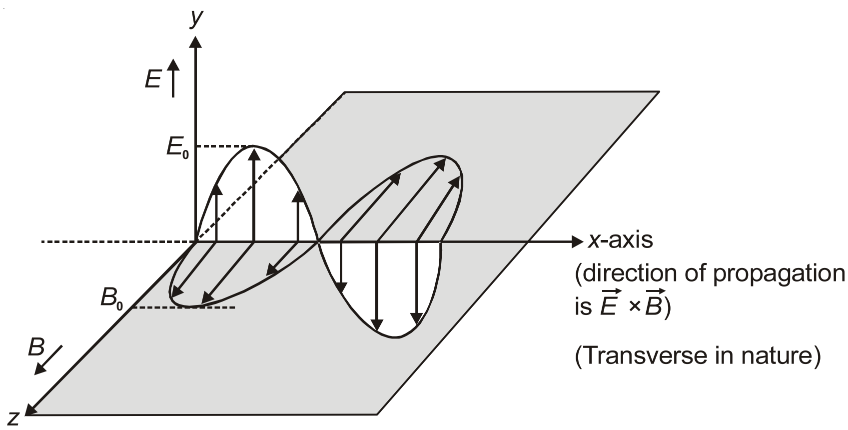
The inductor and capacitor are wattless components of ac circuit as their power consumption is zero.

ELECTROMAGNETIC WAVES

Electromagnetic waves are non-mechanical waves. They transport energy and momentum and carries information. They are transverse in nature and hence they can be polarised.

An electromagnetic wave is the one constituted by oscillating electric and magnetic fields which oscillate in two mutually perpendicular planes; the wave itself propagates in a direction, perpendicular to both the directions of oscillations of electric and magnetic fields.

Important Points Regarding Electromagnetic Waves Characteristics



1. Both electric and magnetic field vary in phase with each other.

2. $\vec{E}_y = E_0 \sin(\ t - kx)\hat{j}$, $\vec{B}_z = B_0 \sin(\ t - kx)\hat{k}$, speed $v = \frac{}{k}$

3. In vacuum, $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$. In a medium $v = \frac{1}{\sqrt{\mu\varepsilon}}$.

   Refractive index of the medium $\mu = \frac{c}{v} = \sqrt{\mu_r \varepsilon_r}$.

4. $\frac{E}{B} = \frac{E_0}{B_0} = \frac{E_{\text{rms}}}{B_{\text{rms}}} = c$ (speed) $= \frac{}{k}$

5. Average energy density $U_{\text{av}} = \frac{1}{2}\varepsilon_0 E_0^2 = \frac{1}{2}\frac{B_0^2}{\mu_0} = \frac{1}{4}\varepsilon_0 E_0^2 + \frac{1}{4}\frac{B_0^2}{\mu_0}$

6. Poynting vector $\vec{S}$ represents instantaneous intensity at a point

   $\vec{S} = \frac{\vec{E}\times\vec{B}}{\mu_0}$ or $S = \frac{EB}{\mu_0}$. Here $\vec{E}\times\vec{B}$ represents the direction of flow of energy.

7. **Intensity :** Average rate of flow of energy per unit time per unit area in the direction of wave propagation.

   $I = S_{\text{av}} = \frac{E_{\text{rms}}B_{\text{rms}}}{\mu_0} = \frac{1}{2}\frac{E_0 B_0}{\mu_0}$

   $I = c \times U_{\text{av}}$ or $U_{\text{av}} = \frac{I}{c}$

https://neetmedicalacademy.blogspot.com

For a point source of power P, Intensity at a distance r from the source is $I = \dfrac{P}{4\pi r^2}$.

Now $U_{av} = \dfrac{I}{c} = \dfrac{P}{4\pi r^2 c}$

$$\Rightarrow \quad \frac{1}{2}\varepsilon_0 E_0^2 = \frac{P}{4\pi r^2 c} \quad \Rightarrow \quad E_0 = \sqrt{\frac{P}{2\pi r^2 c\,\varepsilon_0}}$$

$$B_0 = \frac{E_0}{c} = \sqrt{\frac{P\mu_0}{2\pi r^2 c}}$$

8. Radiation pressure (for normal incidence)
   (a) For perfectly absorbing surface $P = I/c$
   (b) For perfectly reflecting surface $P = 2I/c$
   (c) For all other surfaces $I/c < P < 2I/c$
   (d) For a surface of reflectance r

   $$P = \frac{I}{c}(1 + r)$$

9. The direction of wave propagation is always along $\vec{E} \times \vec{B}$ and E, B, c are always perpendicular.
10. Speed of electromagnetic waves only depends upon the medium. All the electromagnetic waves travels with equal speed (*i.e.* 3×10^8 m/s) in vacuum.
11. When a electromagnetic wave travels from one medium to another then its speed changes but frequency remains same.
12. Speed of electromagnetic waves in any medium = $\dfrac{c}{\mu}$ [where μ is the refractive index of the electromagnetic waves]

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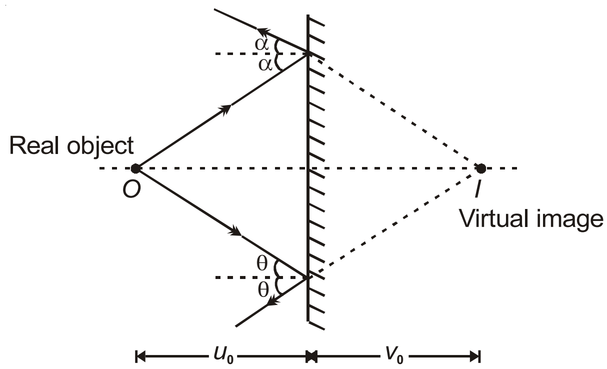
Chapter 15

Optics

PLANE MIRROR

Following are some important points regarding image formation by plane mirror :

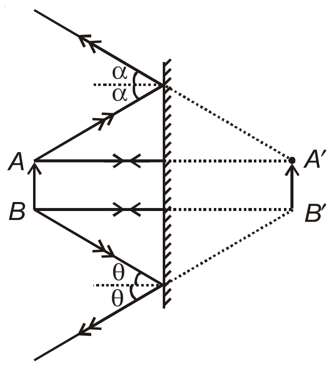


1. Object distance = Image distance, $|u_0| = |v_0|$.

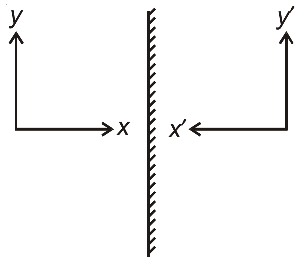
2. $\frac{du}{dt} = -\frac{dv}{dt}$, *i.e.*, speed of object w.r.t. mirror is equal to speed of image w.r.t. mirror. For example, if object is at rest and mirror is moving with velocity x towards object then velocity of image will be $2x$.

3. For extended object, it can be seen that height of object = height of image.

   $\Rightarrow$ Magnification = $\dfrac{\text{Height of image}}{\text{Height of object}} = 1$



4. Image is laterally inverted, i.e., it has front back reversed.



5. Keeping incident ray fixed, if a plane mirror is rotated by an angle θ, reflected ray rotates by an angle 2θ.
6. For two mirrors inclined at an angle 'θ'. Number of images formed by the mirrors for a point object are

(a) $\frac{360}{\theta}-1$, if $\frac{360}{\theta}$ = even number

(b) $\frac{360}{\theta}-1$, when $\frac{360}{\theta}$ = odd and object is placed symmetrically.

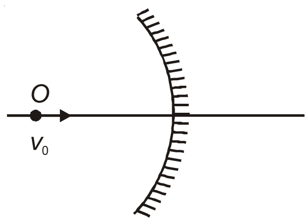
(c) $\frac{360}{\theta}$, when $\frac{360}{\theta}$ = odd and object is placed unsymmetrically.

REFLECTION FROM CURVED SURFACES

1. $f=\frac{R}{2}$
2. $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$
3. $m=\frac{I}{O}=-\frac{v}{u}=\frac{f-v}{f}=\frac{f}{f-u}$

where,
f = Focal length
u = Object distance
v = Image distance
I = Size of image
O = Size of object

4. Magnification is negative for inverted image and positive for erect image.
5. If object O is moving with velocity v_0 as shown in figure then velocity of image $v_{image} = -m^2v_0$



$\because \quad \frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$\Rightarrow \quad \frac{dv}{dt}=-\frac{v^2}{u^2}\left(\frac{du}{dt}\right)$

$\Rightarrow \quad v_{image} = -m^2v_0$

6. If object is moving with velocity v_0 as shown in figure then velocity of real image $v_{image} = mv_0$

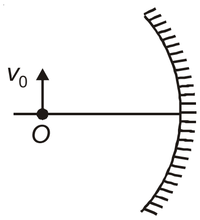
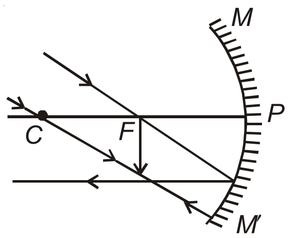
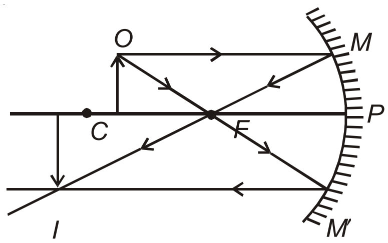
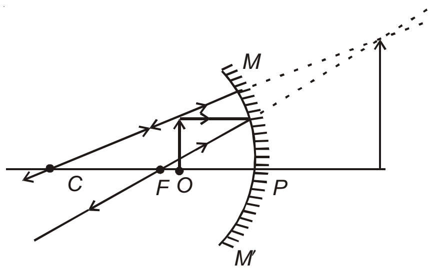


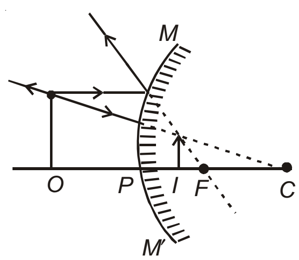
Image formation for Concave Mirror

Table : Position and nature of image for a given position of object

S.No.	Position of object	Ray diagram	Details of image
1.	At infinity		Real inverted, very small [m (0$^-$)]; at focus or in focal plane.
2.	Between F and C		Real, inverted, large ($m < -1$) beyond C
3.	Between F and P		Virtual, erect ($m > +1$) Behind the mirror

For Convex Mirror

Table: Position and nature of image for a given position of object

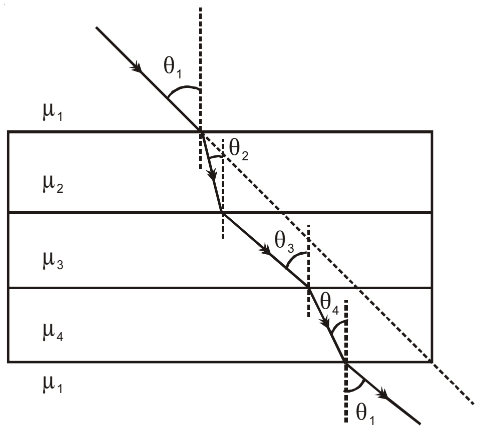
S.No.	Position of object	Ray diagram	Details of image
1.	In front of mirror		Virtual, erect, diminished ($m < +1$) between P and F

Laws of Refraction (Snell's Law)

Consider a light ray crossing a series of media as shown.

$\mu_1\sin\theta_1 = \mu_2\sin\theta_2 = \mu_3\sin\theta_3 = \mu_4\sin\theta_4 = \mu_1\sin\theta_1$

$\Rightarrow$ $\mu\sin\theta$ = Constant



Incident and refracted are parallel
if they are in same medium

REAL DEPTH AND APPARENT DEPTH

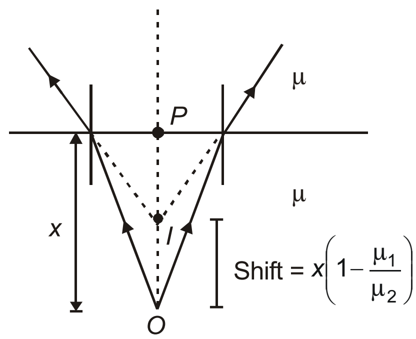
Consider that rays from object *O* are going from medium 2 to 1 ($\mu_2 > \mu_1$) and image is formed at *I*.

$\Rightarrow$ $\dfrac{OP}{\mu_2} = \dfrac{PI}{\mu_1}$ (By Snell's law)

$\Rightarrow$ $PI = \dfrac{\mu_1}{\mu_2}OP = \dfrac{\mu_1}{\mu_2}x$

$\because$ $\mu_1 < \mu_2$, $PI < OP$

$\Rightarrow$ Shift $= x\left(1 - \dfrac{\mu}{\mu}\right)$



$\Rightarrow$ Velocity of image = $\dfrac{\mu}{\mu}$ (Velocity of object)

Glass-slab

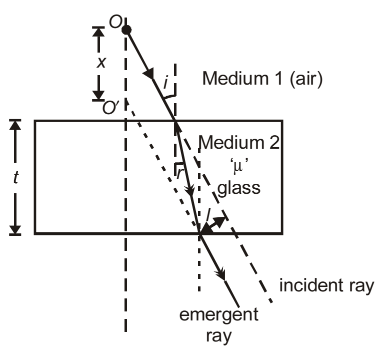
Different results for refraction by glass slab are :

1. Incident and emergent rays are parallel.

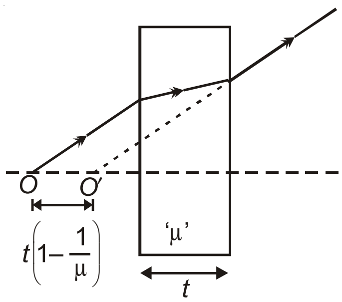
2. l = lateral displacement = $\dfrac{t\sin(i-r)}{\cos r}$

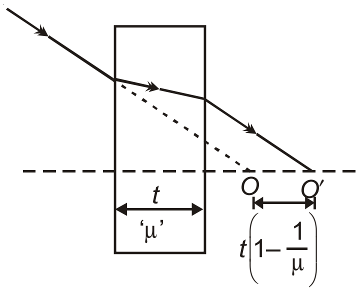
   for small values of i, $\sin(i - r)$ $i - r$ and r 0.

   $\Rightarrow$ $l = t\left[i - \dfrac{i}{\mu}\right] = ti\left[1 - \dfrac{1}{\mu}\right]$

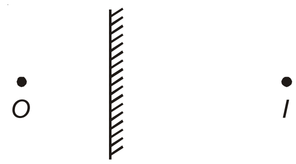


3. I I_{max}, when i 90º

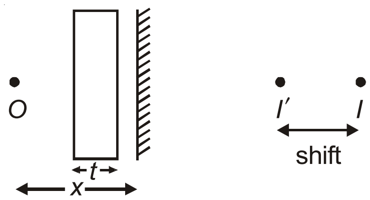
4. 

5. 

If an object is placed at distance x from plane mirror then image will be formed at distance of x from mirror.



Now if a glass slab of thickness t is introduced between object and mirror then image will shift toward object by $2t\left(1-\frac{1}{\mu}\right)$



Critical Angle

If a ray is travelling from optically denser medium to optically rarer medium, then critical angle may be defined as the angle of incidence in denser medium corresponding to which angle of refraction in rarer medium is 90°.

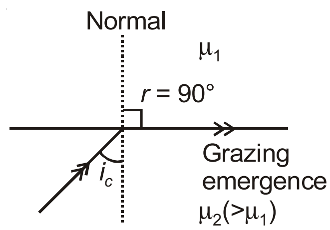
If μ_1 = refractive index of rarer medium

μ_2 = refractive index of denser medium

and i_c = critical angle

then $\mu_2 \sin i_c = \mu_1 \sin 90°$

$\Rightarrow \quad \sin i_c = \frac{\mu_1}{\mu_2}$

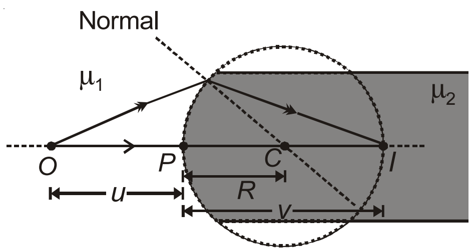


Spherical Refracting Surface

For object O, image formed by refracting surface is at I.

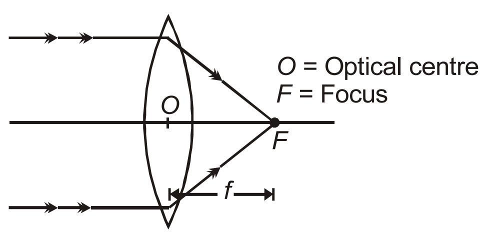
Object distance (u) and image distance (v) are related as

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

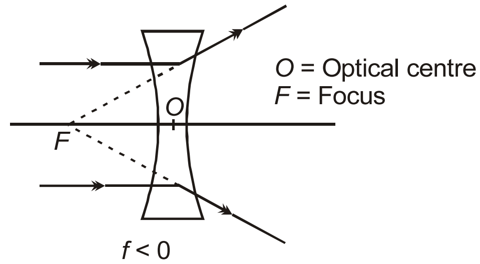


Some Important Relations for Lenses

Converging (Convex) lens **Diverging (Concave) lens**



$f > 0$



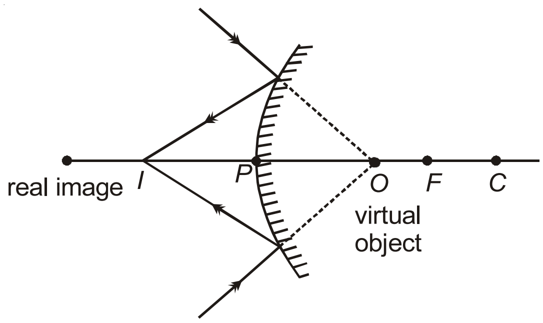
$f < 0$

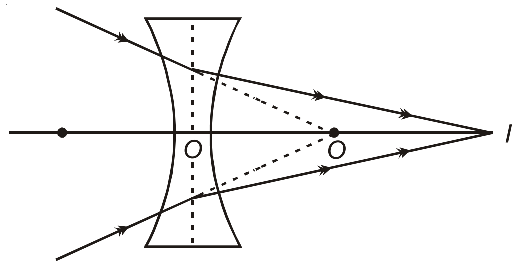
1. Lens formula $\dfrac{1}{v} - \dfrac{1}{u} = \dfrac{1}{f}$

2. Linear magnification $m = \dfrac{I}{O}$ m 0 for inverted image; m 0 for erect image

3. In lenses, $m = \dfrac{v}{u} = \dfrac{f}{f+u} = \dfrac{f-v}{f}$

4. A convex mirror or a concave lens can form a real image if object is virtual as shown.





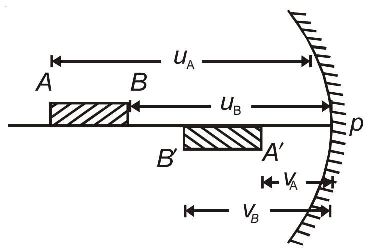
5. If an object moves along the axis of a convex lens from infinity towards its focus with a constant speed, then

$v_i = m^2 v_0$, for a lens $v_i = \dfrac{v^2}{u^2} v_0$

6. If an object moves perpendicular to axis of a convex lens with a velocity v_0 then $v_i = mv_0$

LATERAL MAGNIFICATION

$$m_L = \frac{A'B'}{AB} = \frac{v_B - v_A}{u_A - u_B}$$



For short object

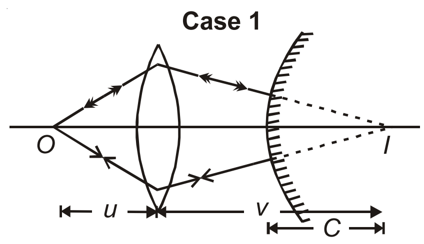
$$m_L = \frac{dv}{du}$$

For a concave mirror, $m_L = -\dfrac{v^2}{u^2}$

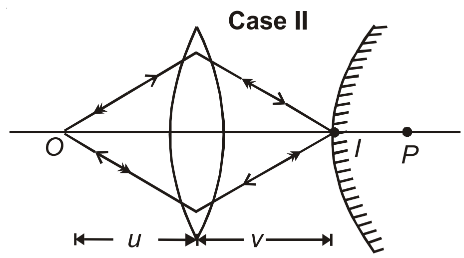
For a convex lens, $m_L = \dfrac{v^2}{u^2}$

Lens and Mirror

A point object is placed in front of a convex lens. A convex mirror is placed behind the lens, so that final image coincides with the object itself.



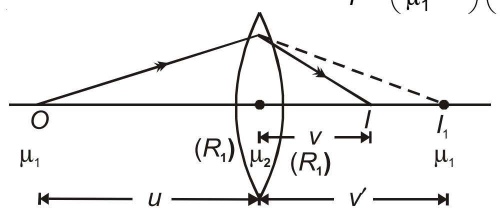
Final image is real and inverted and coincides with O



Final image is real and erect and coincides with O

Lens Maker's Formula (For thin lenses)

For a thin lens, $P(\text{Power}) = \frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

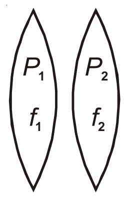


COMBINATION OF LENSES

1. In contact :

   $P = P_1 + P_2$, P is taken with sign

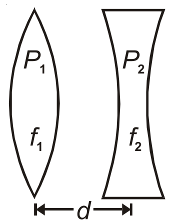
   $\Rightarrow \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$



2. $P = P_1 + P_2 - dP_1P_2$

   $\Rightarrow \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$ (Applicable only for parallel rays)

   where d is the separation between two lenses in air

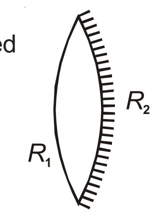


Silvering of Lenses

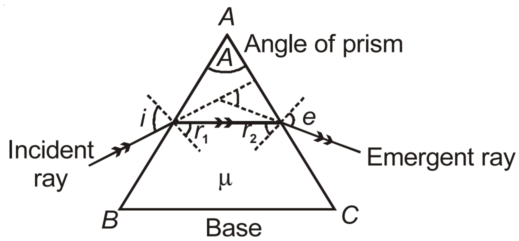
1. When a convex lens is silvered it will behave like a concave mirror.
2. When a concave lens is silvered it will behave like a convex mirror.
3. **Case-1 :** When one face (of radius of curvature R_2) of a double convex lens is silvered

   $P_{eq} = 2P_l + P_m$

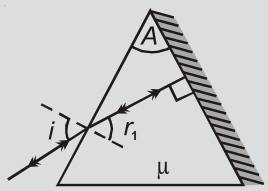
   $\Rightarrow \frac{1}{F_{eff}} = 2(\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] - \frac{2}{R_2}$



PRISM AND DISPERSION



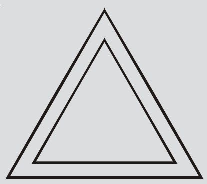
Note :

1. *The variation of   with i is unsymmetrical.*
2. *Under minimum deviation, ray passes symmetrically through the prism.*
3. *If prism is isosceles or equilateral, refracted ray is parallel to base of prism under minimum deviation.*
4. *If $A > 2C$ or μ cosec $\frac{A}{2}$, there will be no emergent light whatever may be the angle of incidence.*
5. *$A = 2C$ is called limiting value of angle of prism.*
6. *If $A < C$, total internal reflection at second face can never take place.*
7. 

   In case, one face is silvered, for incident ray to retrace its path after reflection from 2nd face.

   $r_2 = 0 \qquad r_1 = A$

   $\Rightarrow \qquad \mu = \dfrac{\sin i}{\sin A}$
8. *Angle of deviation is maximum when angle of incidence = 90º.*
9. *A thin hollow prism as shown produces zero deviation.*



Dispersion

$\mu = A + \dfrac{B}{\lambda^2} + \dfrac{C}{\lambda^4} + \dfrac{D}{\lambda^6} + \ldots$ [Cauchy's formula]

  $= (\mu - 1)A$

As, $\lambda_V < \lambda_R$

$\Rightarrow \mu_V > \mu_R$

$\Rightarrow \ _V < \ _R$

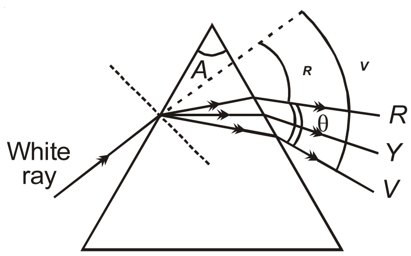
θ = angular dispersion = $\ _V - \ _R = (\mu_V - \mu_R)A$

  = Mean deviation = $(\mu - 1)A$

μ = mean refractive index = $\dfrac{\mu_V + \mu_R}{2}$

Dispersive power $\ = \dfrac{\theta}{\ } = \dfrac{(\mu_V - \mu_R)A}{(\mu - 1)A} = \dfrac{\mu_V - \mu_R}{\mu - 1} = \dfrac{d\mu}{\mu - 1}$

$d\mu$ = difference in refractive index



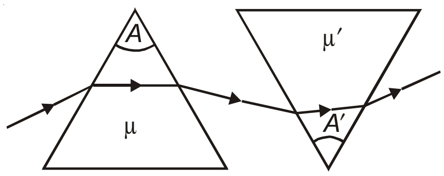
Combination of Prisms

$\theta = \theta_1 + \theta_2$, $= {}_1 + {}_2$

Case-1 : For dispersion without deviation

$= 0$

$\Rightarrow \quad (\mu - 1)A + (\mu' - 1)A' = 0$



Case-2 : For deviation without dispersion

$\theta = 0$

$\Rightarrow \quad (\mu_V - \mu_R)A + (\mu'_V - \mu'_R)A' = 0$

Chromatic Aberration

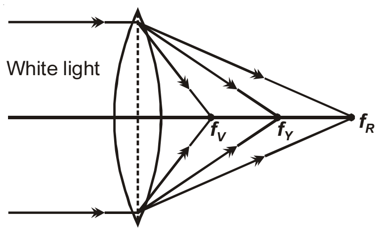
This is the phenomena observed when different colours come to focus at different points on the principal axis. This defect can be removed by using an achromatic lens combination.

As, $\mu = a + \dfrac{b}{\lambda^2} + \dfrac{c}{\lambda^4}$

and $\lambda_R > \lambda_Y > \lambda_v$

$\Rightarrow \mu_R < \mu_Y < \mu_v$

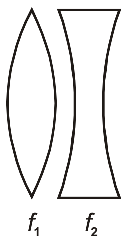
$\Rightarrow f_R > f_Y > f_v \qquad \left[\text{as } \dfrac{1}{f} = (\mu - 1)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)\right]$



Achromatic Combination

A combination free from chromatic aberration is achromatic combination.

1. Lenses in contact



Power $P = \dfrac{1}{f_1} + \dfrac{1}{f_2} = P_1 + P_2$

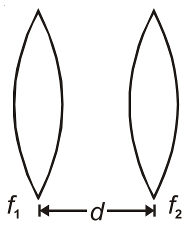
Condition for achromatism

$\dfrac{{}_1}{f_1} + \dfrac{{}_2}{f_2} = 0$ or ${}_1P_1 + {}_2P_2 = 0$

$\Rightarrow$ P_1 and P_2 or f_1 and f_2 should be opposite sign also ${}_1$ ${}_2$ as P will become zero.

2. $P = \dfrac{1}{f_1} + \dfrac{1}{f_2} - \dfrac{d}{f_1 f_2} = P_1 + P_2 - dP_1P_2$

Condition for achromatism $d = \dfrac{{}_1f_2 + {}_2f_1}{{}_1 + {}_2}$.



Note : *(a) f_1 and f_2 can be of same or opposite sign*

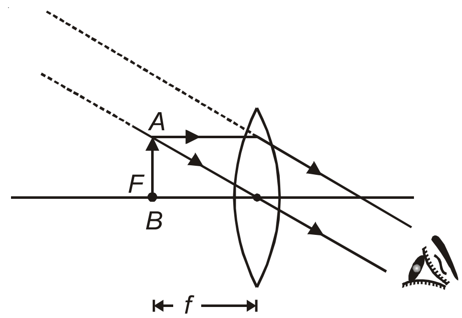
(b) ${}_1 = {}_2 \Rightarrow d = \dfrac{f_1 + f_2}{2}$

OPTICAL INSTRUMENTS

1. **Simple Microscope / Magnifying Glass :** It uses a single convex lens of focal length f

$m = \frac{D}{u}$, where u is distance of object

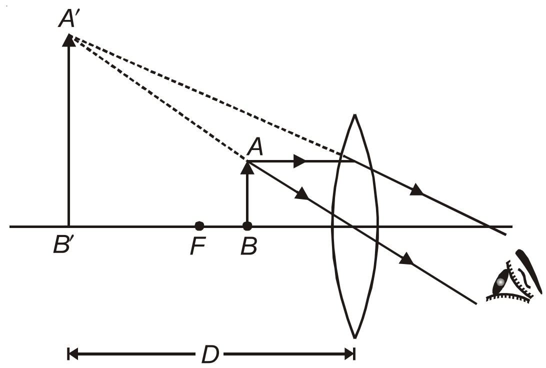
Case-1 :



(Final image at infinity)

For relaxed eye, $u = f$, image at $\Rightarrow m = \frac{D}{f} > 0$

Case-2 :



(Final image at least distance of distinct vision)

For strained eye, image is at $D \Rightarrow m = 1+\frac{D}{f} > 0$

2. **Compound Microscope**

It uses two convex lens objective (f_o) and eyepiece (f_e)

u_o = object distance from objective (u_o is close to f_o)

v_o = image distance from objective (close to length of tube)

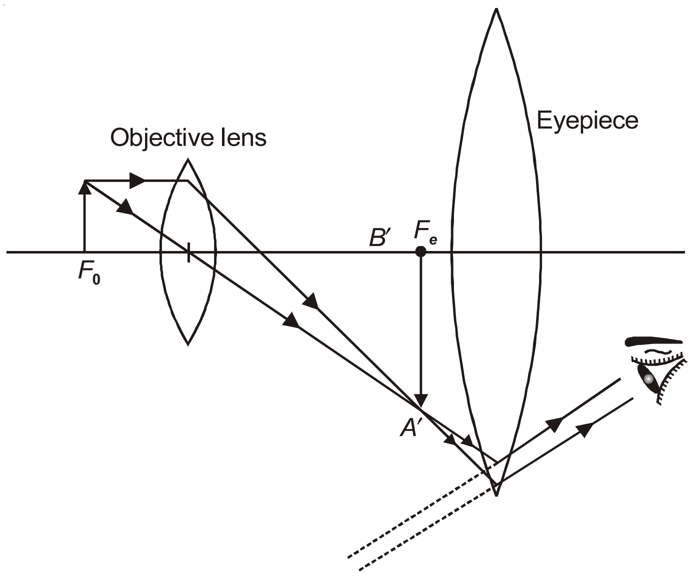
Magnification by objective $m_o = -\frac{v_o}{u_o}$ (–ve)

Magnification by eyepiece $m_e = \frac{D}{u_e}$

Magnification for microscope

$m = m_o \times m_e = \frac{-v_o}{u_o} \times \frac{D}{u_e}$

Case-1 :

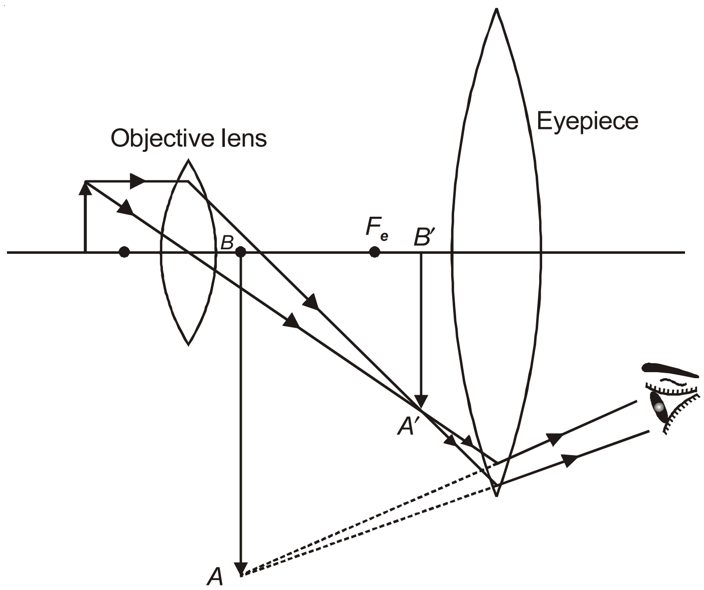


(Final image at infinity)

Relaxed eye : $m = \frac{-v_o}{uo} \times \frac{D}{f_e} \quad \frac{-L}{f_o}\left(\frac{D}{f_e}\right)$

Length of tube $L = V_0 + f_e$

Case-2 :



(Final image at least distance of distinct vision)

Strained eye : $m = \frac{-v_o}{u_o}\left(1 + \frac{D}{f_e}\right) \quad -\frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$

Length of tube $V_0 + \frac{f_e D}{f_e + D}$

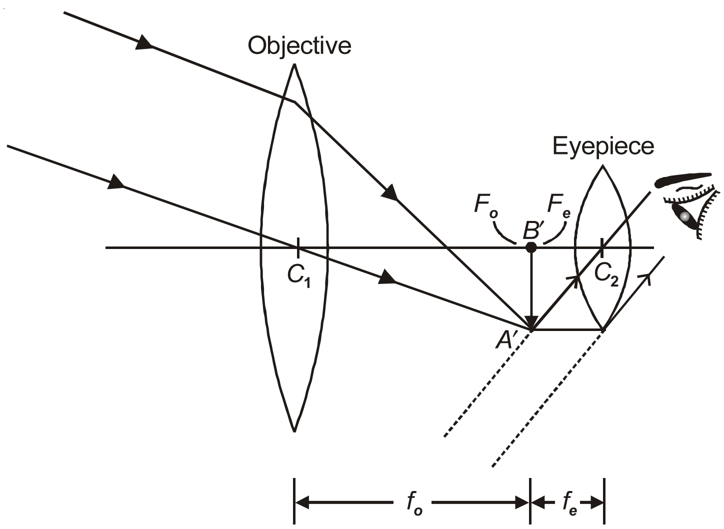
3. **Astronomical Telescope :** f_o is focal length of objective and f_e is focal length of eye-piece.

$m = m_o \times m_e$ here $m_o < 0$, $m_e > 0.$, $m < 0$

Case-1 :

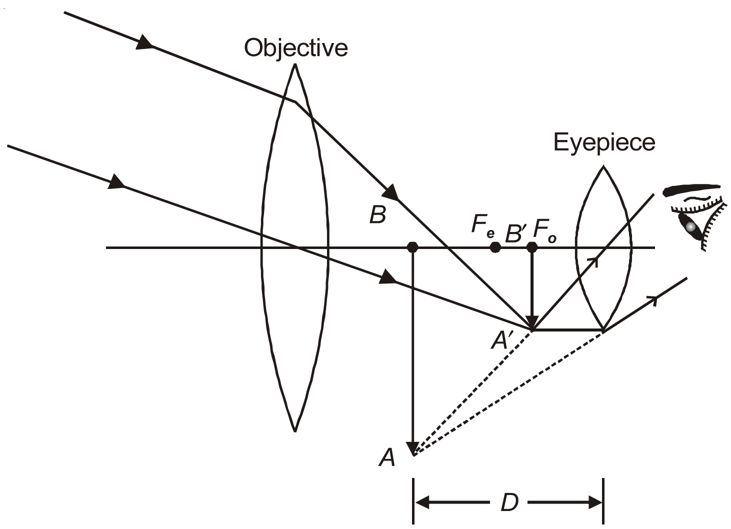
For relaxed eye *i.e.,* normal adjustment. $m = -\frac{f_o}{f_e}$

Length of tube $L = f_o + f_e$



(Final image at infinity)

Case-2 :



(Final image at least distance of distinct vision)

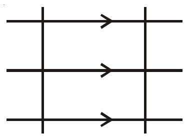
For strained eye : $m = \frac{-f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$

length of tube $L < f_o + f_e$

WAVE OPTICS

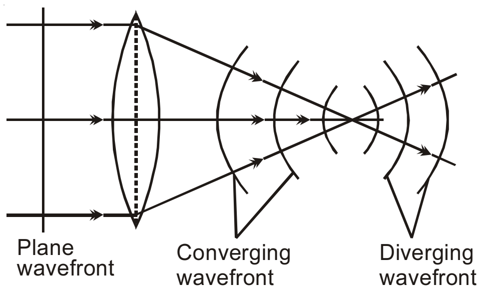
Plane Wavefront (Plane sheet source)

1. Amplitude = constant $\Rightarrow$ $A \propto r^0$
2. Intensity = constant $\Rightarrow$ $I \propto r^0$

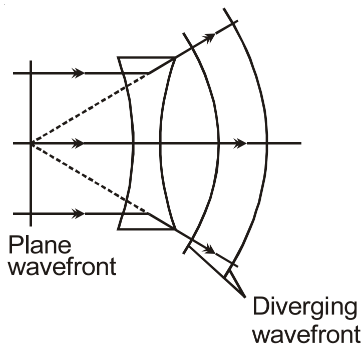


Refraction in form of Wavefronts

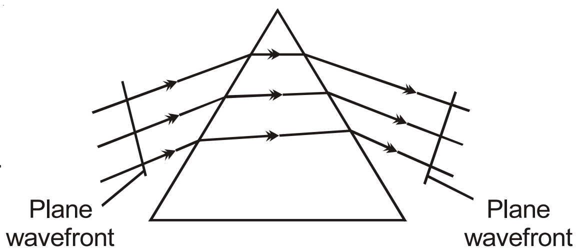
1.



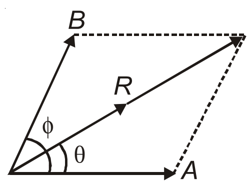
2.



3.



SUPERPOSITION OF WAVES



Wave-1 : $y_1 = A \sin \ t$

Wave-2 : $y_2 = B \sin (\ t + \phi)$

Resultant wave : $y = y_1 + y_2$ $\Rightarrow$ $y = A \sin \ t + B \sin (\ t + \phi)$

$\Rightarrow$ $y = R \sin (\ t + \theta)$

where $\tan\theta = \dfrac{B\sin\phi}{A + B\cos\phi}$

and $R = \sqrt{A^2 + B^2 + 2AB\cos\phi}$

Intensity $\propto$ (Amp)^2 $\therefore$ $I_1 \propto A^2$, $I_2 \propto B^2$, $I \propto R^2$

As, $R^2 = A^2 + B^2 + 2AB\cos\phi$ $\Rightarrow$ $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

https://neetmedicalacademy.blogspot.com

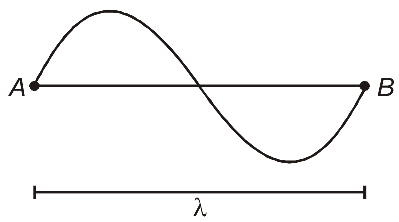
Relation between phase difference and path difference

For the two points A and B on a wave,

$\phi_{AB} = 2\pi$

$X_{AB} = \lambda$

$$\boxed{\phi = \frac{2\pi}{\lambda}\ x}$$



Condition for Maxima

When $\cos\phi = 1$ or $\phi = 2n\pi$ or $x = n\lambda$ (path difference)

$R_{max} = A + B$, $I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

Condition for Minima

When $\cos\phi = -1$ or $\phi = (2n-1)\pi$ or $x = (2n-1)\dfrac{\lambda}{2}$

$R_{min} = A - B$, $I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$

$$\Rightarrow \frac{R_{max}}{R_{min}} = \frac{A+B}{A-B} \qquad \frac{I_{max}}{I_{min}} = \left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2 = \left(\frac{\sqrt{\beta}+1}{\sqrt{\beta}-1}\right)^2, \text{ where } \beta = \frac{I_1}{I_2}$$

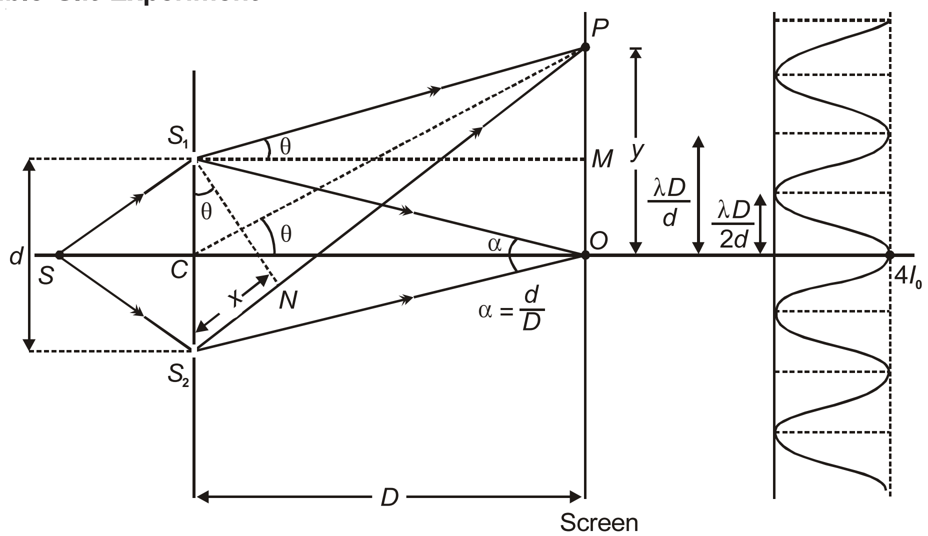
Interference

Phenomenon of redistribution of energy on account of superposition of waves is known as interference.

Coherent Sources

Condition for sustained interference-Sources must be coherent *i.e.*, phase difference between them must be constant.

Young's Double Slit Experiment



For waves reaching P from S_1 and S_2, path difference $x = d\sin\theta$.

For small θ $\quad x = \dfrac{yd}{D} \qquad \left[\because\ \sin\theta \quad \tan\theta = \dfrac{y}{D}\right]$

Condition for Maxima

For maxima $x = n\lambda \Rightarrow y = n\dfrac{\lambda D}{d}$ $(n = 0, 1, 2, \ldots)$

or $d \sin\theta = n\lambda$

$\Rightarrow \sin\theta = \dfrac{n\lambda}{d}$

Condition for Minima

For minima $x = (2n-1)\dfrac{\lambda}{2} \quad \Rightarrow \quad y = \dfrac{(2n-1)\lambda D}{2d}$ $(n = 1, 2, \ldots)$

or $d\sin\theta = \dfrac{(2n-1)\lambda}{2}$

$\Rightarrow \sin\theta = \dfrac{(2n-1)\lambda}{2d}$

If λ and d are comparable then, as $-1 \quad \sin\theta \quad 1$

$-1 \quad \dfrac{(2n-1)\lambda}{2d} \quad 1$

$\Rightarrow \dfrac{-2d+\lambda}{2\lambda} \quad n \quad \dfrac{2d+\lambda}{2\lambda}$

From here you can find maximum number of dark fringes observed on the screen.

For example, if $d = 2\lambda$, then

$\dfrac{-4\lambda+\lambda}{2\lambda} \quad n \quad \dfrac{4\lambda+\lambda}{2\lambda}$

$-\dfrac{3}{2} \quad n \quad \dfrac{5}{2}$ so, n can have four values *i.e.*, –1, 0, 1, 2.

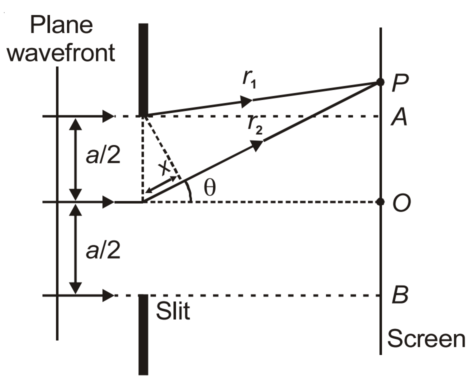
This means that only four minima are observed on the screen.

DIFFRACTION

Diffraction is the phenomenon of light observed due to superposition of secondary wavelets starting from different points of a wavefront which is not blocked by an obstacle or which are allowed by an aperture (of size comparable to the wavelength of light).

In other words you can say diffraction is the phenomena of entering of light in the region of geometrical shadow, due to bending around obstacle edges.

Diffraction by Single Slit



1. $x = \frac{a}{2}\sin\theta$
2. For '*O*' waves from all points in the slit travel about the same distance and are in phase.
3. At *P*, waves r_1 and r_2 have a phase difference $x = \frac{a}{2}\sin\theta$.
4. When $\frac{a}{2}\sin\theta_1 = \frac{\lambda}{2}$, there will be destructive interference.

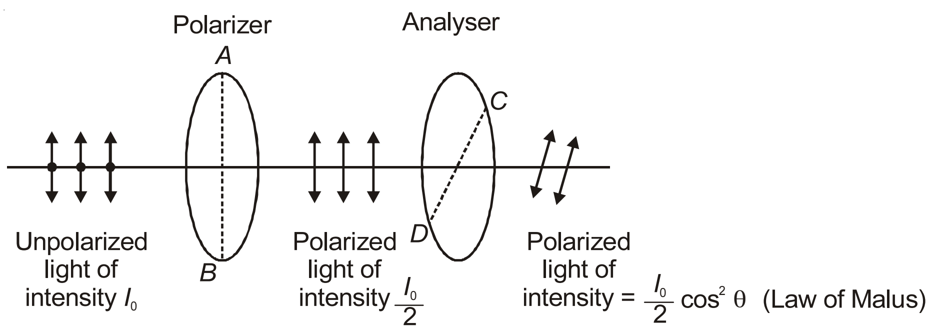
   $\therefore$ when $a \sin\theta_1 = \lambda$, first minima will be formed at *P*.
5. In general $a \sin\theta_n = n\lambda$ is position of n^{th} minima.
6. Angular position of first minima $\theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right)$
7. Angular spread of central maximum is $2\theta_1 = 2\sin^{-1}\left(\frac{\lambda}{a}\right)$. If $\lambda << a$, then angular spread = $\frac{2\lambda}{a}$.
8. When $\lambda > a \Rightarrow \sin\theta > 1$ which is not possible

   $\therefore$ diffraction cannot be observed.
9. $\lambda << a$, then sin θ $\theta = \frac{\lambda}{a}$ (in radians)
10. Width of central maximum = $\frac{2\lambda D}{a}$
11. Width of other fringes = $\frac{\lambda D}{a}$.
12. If I_0 is the intensity of central maximum, then intensity of n^{th} maxima is $I_n = \frac{4I_0}{(2n+1)^2\pi^2}$

    $\therefore$ $I_0 : I_1 : I_2 : : 1 : 0.045 : 0 : 016$.
13. The intensity of fringe goes on decreasing in case of diffraction while it remain nearly same in the interference.

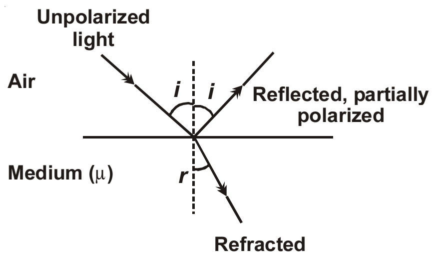
Polarization

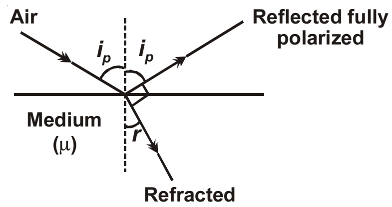
Polarization is a phenomenon exhibited by transverse waves only.



where θ = angle between the transmission (*i.e.*, *AB* and *CD*) axis of the two polaroids

Polarization by Reflection





SOME IMPORTANT POINTS :

1. Reflected light is partially polarized.
2. When $i = i_p$ (polarizing angle), reflected light is completely polarized, i_p is also called Brewster's angle.
3. When reflected light is completely polarized, reflected and refracted light are perpendicular to each other.
4. This was found experimentally by **Sir David Brewster**.

   At this situation $i_p + r = 90°$

   $\Rightarrow$ $\mu = \tan i_p$. This is called Brewster's law.

   $\Rightarrow$ $\tan i_p = \dfrac{1}{\sin i_c}$

   $\Rightarrow$ $\sin i_c . \tan i_p = 1$ [Here i_c is critical angle]

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Chapter 16

Dual Nature of Matter and Radiation, Atoms and Nuclei

DUAL NATURE OF MATTER AND RADIATION

WAVE NATURE OF PARTICLES

The following points should be kept in mind :

1. Any particle in motion can act like a wave. Wave associated with a particle is called Matter wave or de-Broglie wave.
2. de-Broglie wavelength of a particle $\lambda = \frac{h}{mv} = \frac{h}{p} = \frac{h}{\sqrt{2mE_k}}$.

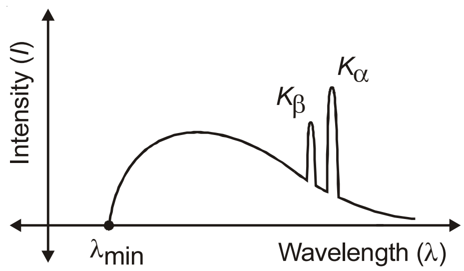
   where $p = mv$ is momentum of particle

   E_k = kinetic energy.
3. For an electron accelerated through V volts.

   $$E_k = eV \therefore \lambda = \frac{h}{\sqrt{2meV}} = \frac{12.27}{\sqrt{V}}\text{Å} \quad \text{or} \quad \lambda = \sqrt{\frac{150}{V}}\ \text{Å}$$
4. For a proton accelerated through V volts, $\lambda = \frac{0.286}{\sqrt{V}}$Å
5. For an α-particle accelerated through V volts, $\lambda = \frac{0.101}{\sqrt{V}}$Å
6. For an electron revolving in n^{th} orbit of Bohr's Hydrogen atom, $mvr = \frac{nh}{2\pi},\ \lambda = \frac{h}{mv} = \frac{2\pi r}{n}$.

X-RAYS

Variation of intensity (I) of X-rays with wavelength λ :



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Important points related to the above curve :

1. At certain sharply defined wavelength, the intensity of X-rays is very large as marked K_α and K_β. These are known as **characteristic X-rays**.
2. At other wavelengths intensity varies continuously. These are known as **continuous X-rays**.
3. Minimum wavelength or cut off wavelength or threshold wavelength of continuous X-rays,

   $\lambda_{\text{min}} = \dfrac{hc}{eV} = \dfrac{12400\text{Å}}{V}$, where V is applied voltage in volts.
4. The minimum wavelength does not depend on the material of target. It depends only on the accelerating potential.
5. Continuous X-rays are due to continuous loss of energy of electrons striking the target through successive collisions.
6. Characteristic X-rays are due to the transition of electrons from higher energy level to the vacant space present in the lower energy level.
7. Wavelength of K_α, $\lambda = \dfrac{hc}{E_L - E_K}$(transition from L to K)
8. Wavelength of K_β, $\lambda = \dfrac{hc}{E_M - E_K}$(transition from M to K)

Moseley's law : Applicable to characteristic X-rays only.

Mathematically $\boxed{\sqrt{\nu} = a(Z - b)}$ a and b are Moseley's constants, ν is frequency of X-rays.

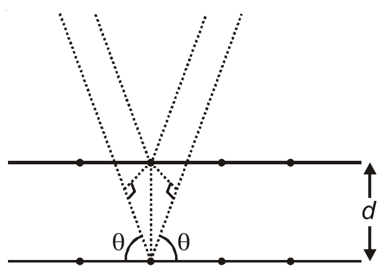
Z is atomic number of the target atom.

For K_α X-ray,

$$a = \sqrt{\frac{3Rc}{4}}$$

$$b = 1$$

Diffraction of X-Rays



Bragg's Law : $2d \sin\theta = n\lambda$ [condition for constructive interference]

where, λ = wavelength of X-ray.

d = separation between crystal planes.

θ = angle between X-ray beam and crystal plane.

$2d \sin\theta$: Path difference

Davisson and Germer's accidental discovery of the diffraction of electrons was the first direct evidence confirming de Broglie's hypothesis that particles have wave properties as well.

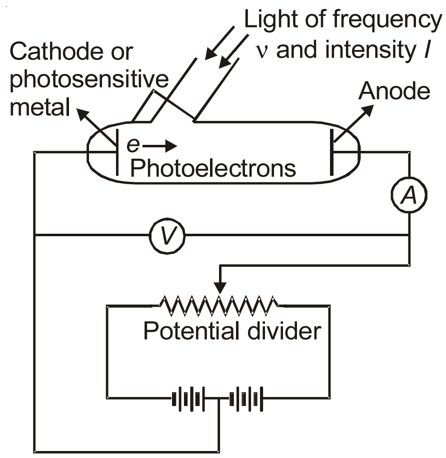
PHOTOELECTRIC EFFECT

The emission of electrons from a metallic surface when illuminated with light of appropriate wavelength (or frequency) is known as photoelectric effect. It was discovered by Hertz in 1887.

Einstein's Theory of Photoelectric Effect

Light of frequency ν consists of stream of packets or quanta of energy $E = h\nu$. These are called photons.

In the process of photoemission, a single photon gives up all its energy to a single electron. As a result, the electron can be ejected instantaneously.



Exp. set up photoelectric effect

Work Function (ϕ) : It is the minimum energy of photon required to liberate an electron from a metal surface.

Threshold Frequency (ν_0) : The frequency of incident radiation below which photoelectric effect does not take place. $h\nu_0 = \phi$.

Stopping Potential (V_0) : The smallest negative value of anode potential which just stops the photocurrent is called the stopping potential.

If the stopping potential is V_0 then $eV_0 = KE_{max}$ = Maximum kinetic energy of photoelectrons emitted.

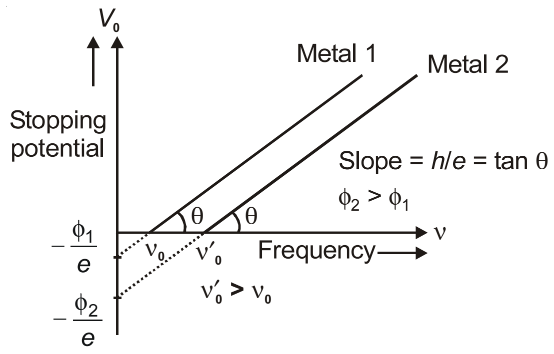
The following important points should be kept in mind :

1. The kinetic energy of photoelectrons varies between zero to KE_{max}.
2. If ν (> ν_0) is frequency of incident photon, $h\nu_0$ is work function then $h\nu - h\nu_0 = KE_{max}$. This is Einstein's photoelectric equation. Here h is Planck's constant.
3. Efficiency of photoelectric emission is less than 1%. It means it is not necessary that if the energy of incident photon is greater than work function electrons will definitely be ejected out.
4. If frequency of incident radiation (ν) is doubled, stopping potential (V_0) or kinetic energy maximum ($K.E._{max}$) gets more than doubled.
5. If on a neutral ball made up of metal of work function ϕ, radiation of frequency ν (greater than threshold frequency) is incident, number of photoelectrons emitted from the ball before the photoelectric emission stops is given by $n = \dfrac{(h\nu - \phi)4\pi\varepsilon_0 R}{e^2}$.
6. Saturation current depends upon intensity of incident light whereas stopping potential depends upon frequency of light as mentioned in graphs also.

Graphs for Photoelectric Effect (Lenard's Observations)

Following graphs are important :

1.



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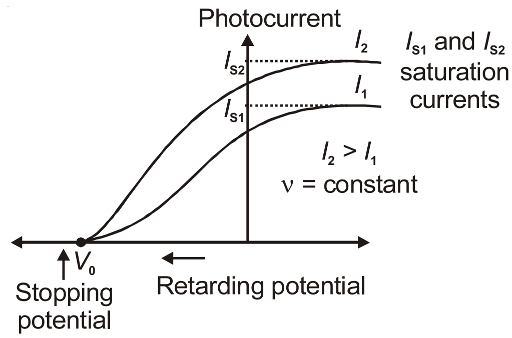

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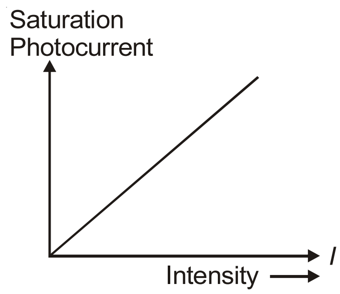

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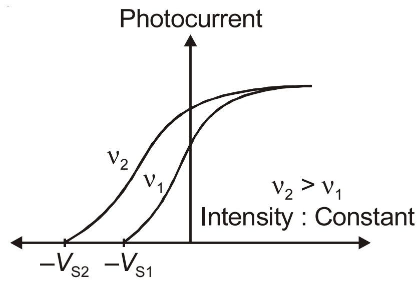


ATOMS AND NUCLEI

BOHR'S ATOMIC MODEL

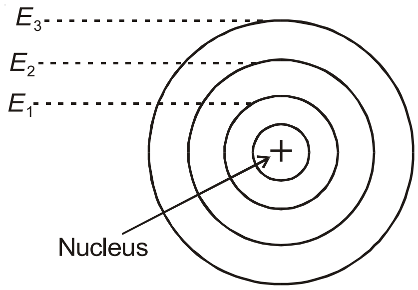
In 1913 Niels Bohr, a Danish physicist, introduced a revolutionary concept *i.e.*, the quantum concept to explain the stability of an atom. He made a simple but bold statement that "The old classical laws which are applicable to bigger bodies cannot be directly applied to the sub-atomic particles such as electrons or protons".

Postulates of Bohr's Theory

1. Electron revolves round the nucleus in circular orbits.
2. Electron can revolve only in those orbits in which angular momentum of the electron about the nucleus is an integral multiple of $\frac{h}{2\pi}$

   i.e., $mvr = \frac{nh}{2\pi}$

   n = principal quantum number of the orbit in which electron is revolving.
3. Electrons in an atom can revolve only in discrete circular orbits called stationary energy levels (shells). An electron in such a shell is characterised by a definite energy, angular momentum and orbit number. While an electron is in any of these orbits it does not radiate energy although it is accelerated.
4. Electrons can jump from one stationary orbit to another stationary orbit. Electrons in outer orbits have greater energy than those in inner orbits. The orbiting electron emits energy when it jumps from a higher energy state to a lower energy state and absorbs energy when it makes a jump from lower orbits to higher orbits. This energy (emitted or absorbed) is in form of photons.



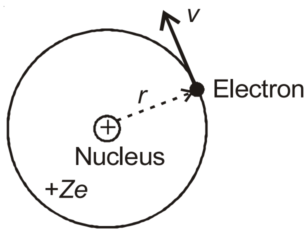
$E_2 - E_1 = h\nu$

where, E_2 = higher energy state

E_1 = lower energy state

and ν = frequency of photons of radiation/emitted absorbed.

Mathematical Analysis of Bohr's Theory



Electric force of attraction provides the centripetal force

$$\Rightarrow \quad \frac{1}{4\pi\varepsilon_0}\frac{(Ze)e}{r^2} = \frac{mv^2}{r} \qquad ...(i)$$

where, m = mass of electron

v = velocity (linear) of electron

r = radius of the orbit in which electron is revolving

Z = atomic number of hydrogen like atom

Angular momentum about the nucleus, $mvr = \frac{nh}{2\pi}$ ...(ii)

(a) Velocity of electron in *n*th orbit

Putting value of *mvr* from equation (ii) into equation (i),

$$\frac{1}{4\pi\varepsilon_0}Ze^2 = \left(\frac{nh}{2\pi}\right)v$$

$$\Rightarrow \quad v = \frac{Z}{n}\left[\frac{e^2}{2\varepsilon_0 h}\right] = \frac{Z}{n}.v_0 \qquad ...(iii)$$

where,

$$v_0 = \frac{c}{137} = 2.2 \times 10^6 \text{ m/s}$$

where $c = 3 \times 10^8$ m/s = speed of light in vacuum, $\frac{v_0}{c} = \frac{1}{137}$ = fine structure constant

(b) Radius of the *n*th orbit

Putting value of v from equation (iii) in equation (ii), we get,

$$m\left(\frac{Z}{n}\times\frac{e^2}{2\varepsilon_0 h}\right)r=\frac{nh}{2\pi}$$

$$\Rightarrow \quad r=\frac{n^2}{Z}\left[\frac{\varepsilon_0 h^2}{\pi m e^2}\right]=\frac{n^2}{Z}.r_0 \qquad \text{...(iv)}$$

where,

$r_0 = 0.53$ Å.

(c) Total energy of electron in n^{th} orbit

From equation (i)

$$\text{K.E.}=\frac{1}{2}mv^2=\frac{Ze^2}{8\pi\varepsilon_0 r}$$

$$\text{and P.E.}=\frac{1}{4\pi\varepsilon_0}\frac{(Ze)(-e)}{r}=-2\text{K.E.}$$

P.E. = – 2 K.E.

Total energy, E = K.E. + P.E. = –K.E.

$$E=\frac{Z^2}{n^2}\left(-\frac{me^4}{8\varepsilon_0^2 h^2}\right)=\frac{Z^2}{n^2}.E_0$$

where, $E_0 = -13.6$ eV.

(d) Time period of revolution of electron in *n*th orbit

$$T=\frac{2\pi r}{v}=\frac{n^3}{Z^2}.T_0$$

where,

$T_0 = 1.51 \times 10^{-16}$ s.

(e) Frequency of revolution in *n*th orbit

$$f=\frac{1}{T}=\frac{Z^2}{n^3}.f_0$$

where,

$f_0 = 6.6 \times 10^{15}$ Hz.

(f) Magnetic field at the centre due to revolution of electron

$$B=\frac{\mu_0 I}{2r}=\frac{\mu_0 e}{2r\,T}=\frac{\mu_0 e}{2r}\times\frac{v}{2\pi r}$$

$$B\propto\frac{v}{r^2} \quad\Rightarrow\quad B\propto\frac{Z}{n}\times\left(\frac{Z}{n^2}\right)^2 \quad\Rightarrow\quad B\propto\frac{Z^3}{n^5}$$

(g) Wavelength of photon

$$\frac{1}{\lambda} = \bar{\nu} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]Z^2$$

where,

$\bar{\nu}$ **is called wave number.**

R = Rydberg constant

= $1.09677 \times 10^7\ \text{m}^{-1}$

= $1.09677 \times 10^{-3}\ \text{Å}^{-1}$ (for stationary nucleus)

$$= \left(\frac{1}{912}\right)\text{Å}^{-1}$$

BINDING ENERGY

The amount of energy needed to separate the constituent nucleons to large distances is called binding energy.

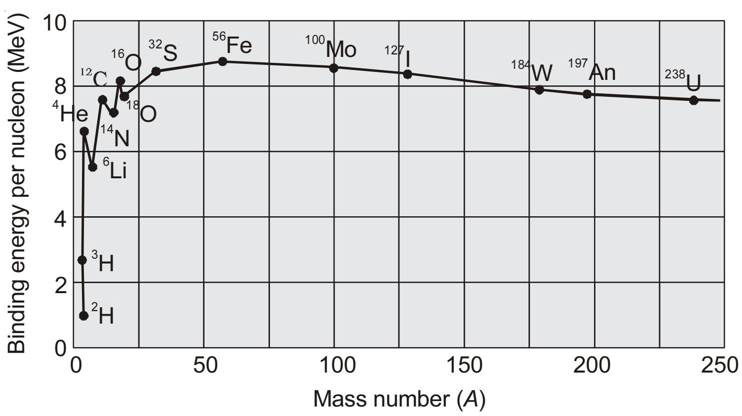
If the nucleons are initially well separated and are brought to form the nucleus, this much energy is released.

$\boxed{\text{BE} = (ZM_p + NM_n - M)c^2}$ (Where M = mass of nucleus and $N = A - Z$)

M_p = Mass of proton, M_n = Mass of neutron.

Binding Energy Curve

B.E./nucleon is very low for light nuclei. This means energy will be released if two nuclei combine to form a single middle mass nucleus. The release of energy in a fusion process is based on this fact.



Likewise, the low B.E. per nucleon for heavy nuclei indicates that if a single heavy nucleus breaks up into middle mass nuclei, energy will be released. Release of energy in fission process is based on this fact.

Note : *1. Binding energy per nucleon is practically constant for 30 < A < 170.*

2. B.E. per nucleon is lower for both light nuclei (A < 30) and heavy nuclei (A > 170).

RADIOACTIVITY

Law of Radioactive Disintegration

$$-\frac{dN}{dt} \propto N$$

$$-\frac{dN}{dt} = \lambda N \quad (\lambda \text{ is decay constant})$$

$$\Rightarrow \quad \boxed{N = N_0 e^{-\lambda t}}$$

Activity $A = -\dfrac{dN}{dt} = \lambda N_0 e^{-\lambda t}$

$$\boxed{A = A_0 e^{-\lambda t}}$$

Half Life ($T_{1/2}$)

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

Let N_0 be the initial number of active nuclei and N be the number of active nuclei remaining after n half lives then $N = \dfrac{\mu_0}{2^n}$.

Application : Let R_1 be activity of radioactive substance at $t = T_1$ and R_2 be the activity at $t = T_2$, then

$R_1 = \lambda N_1$ and $R_2 = \lambda N_2$

Number of nuclei disintegrated in $(T_2 - T_1)$ is

$$N_1 - N_2 = \frac{R_1 - R_2}{\lambda} = \frac{(R_1 - R_2)T}{\ln 2}$$

where T is the half life of radioactive substance.

Average Life (T_{av})

$$\boxed{T_{av} = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2}} \quad \text{or} \quad T_{av} = 1.44\ T_{1/2} \quad \text{or} \quad T_{1/2} = 0.693\ T_{av}$$

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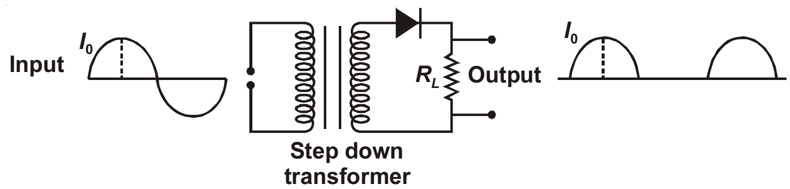
Chapter 17

Electronic Devices and Communication Systems

DIODE AS RECTIFIER

A device to convert ac into dc.

1. **Half Wave Rectifier**



$I_{rms} = \frac{I_0}{2}$

$I_{mean} = \frac{I_0}{\pi}$

form factor $= \frac{I_{rms}}{I_{mean}} = 1.57$

Important points :

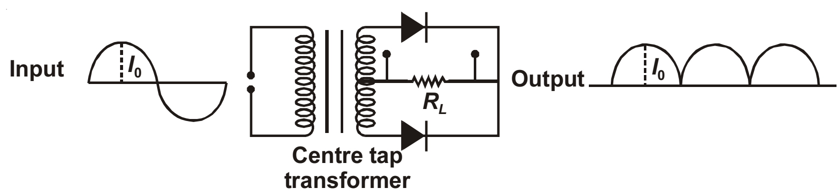
a. Input frequency = Out put frequency.

b. Maximum Efficiency = 40.6%

c. Ripple factor $r = \dfrac{\text{ac component}}{\text{dc component}} = \sqrt{\dfrac{I_{rms}^2}{I_{mean}^2} - 1} = 1.21$.

d. Efficiency of half wave rectifier $= \dfrac{0.406 R_L}{r_f + R_L}$

2. **Full Wave Rectifier**



$I_{rms} = \frac{I_0}{\sqrt{2}}$

$I_{mean} = \frac{2I_0}{\pi}$

a. Out put frequency = 2 × input frequency

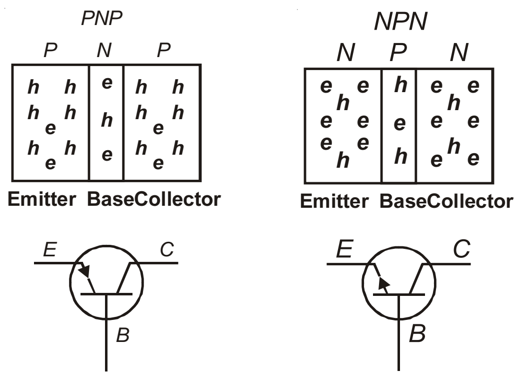
b. Maximum Efficiency = 81.2%.

c. Ripple factor r = 0.48.

d. Efficiency of full wave rectifier $= \dfrac{0.812 R_L}{r_f + R_L}$

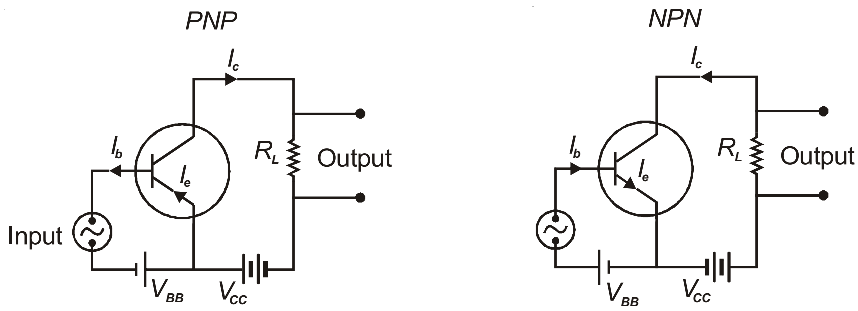
JUNCTION TRANSISTOR

Symbols used for transistors are shown here :



Common-Emitter (CE) Amplifier

$$\boxed{I_e = I_b + I_c}$$



Input $v_i = V_i \sin\ t$ Output $v_0 = V_0 \sin(\ t \pm \pi)$

Important Points related to CE-Amplifier

1. There is phase reversal of 180°.
2. DC current gain $\beta = \frac{I_c}{I_b}$.
3. AC current gain $\beta_{ac} = \frac{I_c}{I_b}$.
4. AC voltage gain $A_v = \frac{V_0}{V_i}$.
5. AC power gain = $\beta_{ac} \times A_v$ [maximum in *CE* mode]

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6. Resistance gain $R_g = \dfrac{R_L}{R_i}$ (R_i = input resistance).

7. $\dfrac{1}{\alpha} - \dfrac{1}{\beta} = 1$

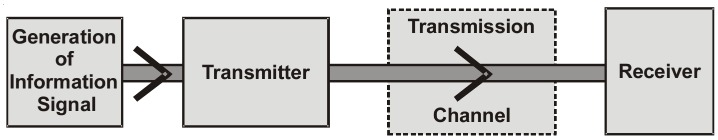
8. $\alpha < 1$, $\beta > 1$

9. Transconductance $g_m = \dfrac{I_c}{V_i}$

COMMUNICATION SYSTEMS

Communication of information to each other is a basic human activity. For example, one person wishes to tell something or give a message to another person sitting near him. Then he speaks and transmits sound waves through air medium or channel. The other person receives the message by listening through his/her ears. In modern communication systems the information is first converted into electrical signals or electromagnetic waves and then sent electronically. This has the advantage of speed, reliability and possibility of communicating over long distances.

The key to communication system is to obtain an electrical signal voltage or current which contains the information. For example, a microphone can convert speech signals into electrical signals. Similarly, pressure can be sensed by **piezoelectric** sensor which gives pressure in terms of electrical signal. A signal is defined as **a single-valued function of time (that conveys the information) and which, at every instant of time has a unique value.**



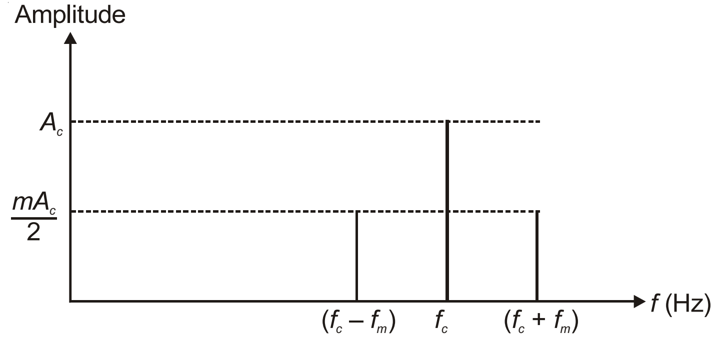
Basic units of all communication systems

1. An AM-wave is equivalent to the summation of three sinusoidal waves whose frequencies are $_c$, ($_c$ + $_m$) and ($_c$ – $_m$) and the amplitudes are respectively V_c, $\dfrac{mV_c}{2}$ and $\dfrac{mV_c}{2}$

2. The frequency f_c is known as **carrier frequency** and $(f_c + f_m)$ and $(f_c - f_m)$ are known as **Upper side band** and **Lower side band** respectively.

> ***Note :*** *In an AM-wave the difference between upper side band and lower side band (which is equal to $2f_m$) is known as the **band width**.*

Graphically, the carrier frequency and the side bands can be shown as :



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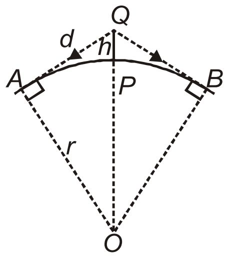
Space Wave Propagation or Tropospheric Wave Propagation

The transmitted waves, travelling in a straight line, directly reach the receiver end and are then picked up by the receiving antenna. This mode of communication is termed as **Line of sight communication.**



Ray path of transmitted waves following space-wave (or line of sight) mode of propagation. The transmitter is located at the ground on a tall lower.

Range

$d = \sqrt{2rh}$ $\left[\begin{array}{l} r : \text{Radius of earth} \\ h : \text{Height of transmitting antenna} \end{array}\right]$

This distance is of the order of 40 km.

❑ ❑ ❑